A COMPREHENSIVE COMPUTER SCIENCE TEXTBOOK

FOUNDATIONS OF

COMPUTER SCIENCE

FROM BASICS TO ARTIFICIAL INTELLIGENCE

19 COMPREHENSIVE CHAPTERS

Hardware · Software · OS · MS Office · Internet · Cyber Safety
Algorithms · Flowcharts · Scratch · Python · AI · Ethics of AI

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Beginner to Advanced · Theory & Practice · Hands-On Projects

“ Designed for First-Time Learners — No Prior Knowledge Required ”

— BY —

The Monotheist

✦ ✦ ✦

Computer Science · Beginner Level · 19 Chapters

From Introduction to Computers → All the way through to → Artificial Intelligence & Ethics

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PREFACE

By The Monotheist

> *" In the name of God, the Most Gracious, the Most Merciful. "*
> *Every journey of knowledge begins with a single step. This book is that step.*

Why This Book Was Written

We live in an age where computers are no longer a luxury or a mystery reserved for scientists and engineers. They are the air we breathe — present in our schools, our hospitals, our markets, our hands, and increasingly, in the very decisions that shape our lives. Yet for millions of students, the door to understanding this world remains firmly closed, not because they lack ability, but because they have never been given the right key.

This book was written with one purpose in mind: to be that key. It is designed for the student who has never touched a keyboard in a classroom, for the young mind full of curiosity but without guidance, and for the teacher who wants a single, complete, honest resource they can place in front of a beginner and say: "Start here."

I am not a publisher. I am not a corporation. I am a teacher — a person who believes that knowledge is a trust, that education is an act of worship, and that every child, no matter where they come from, deserves to understand the world they are growing up in.

What You Will Find in This Book

This textbook covers nineteen carefully sequenced chapters, beginning with the most fundamental question — What is a computer? — and building steadily, step by step, all the way to Artificial Intelligence and the ethical questions it raises for humanity.

You will learn how computers are built and how they think. You will learn to work with the world's most used software tools: Microsoft Word, PowerPoint, and Excel. You will learn how the internet works, how to stay safe online, and how to communicate professionally in the digital world. You will learn to think logically, to design algorithms, to draw flowcharts, and to write your first real programs in Scratch and Python.

And in the final chapters, you will confront one of the most important conversations of our time: what Artificial Intelligence is, how it works, and perhaps most importantly, what responsibilities come with building and using it.

How to Use This Book

Each chapter is designed to stand on its own while also building on what came before. Every chapter opens with Learning Objectives so you always know exactly what you are about to learn. The main content is explained in plain, clear language with real-world examples, illustrated tables, and highlighted concept boxes.

Every chapter closes with a full set of review questions — short answers, fill-in-the-blanks, true or false, and applied exercises. These are not tricks. They are tools. Use them

honestly, and they will tell you exactly what you have learned and what still needs attention. A glossary of key terms is included in every chapter so that no word is left unexplained.

Whether you are a student reading alone, a teacher guiding a class, or a parent supporting a child at home — this book is structured to work for you. Read it in order if you are a complete beginner. Dip into specific chapters if you need to brush up on a particular topic. The knowledge belongs to you.

A Note on Artificial Intelligence

You will notice that this book ends with two chapters on Artificial Intelligence — one on how it works, and one on its ethics. This is intentional and important.

AI is not a distant future technology. It is already making decisions about what you see online, whether you get a loan, how doctors diagnose illness, and how wars may one day be fought. Understanding AI — and being able to think critically about it — is not optional for the generation growing up today. It is a civic responsibility.

I have written these chapters not to frighten, but to empower. The student who understands how AI works, who recognises its biases, who can ask hard questions about fairness, privacy, and accountability — that student is not just computer literate. That student is ready for the world.

A Personal Word

I chose to write this book under the name The Monotheist as a reminder to myself — and to every reader — that all knowledge ultimately points to something greater than ourselves. The ability to reason, to create tools, to teach and to learn: these are gifts. And gifts, in the truest sense, carry obligations.

The obligation here is simple: use this knowledge well. Be honest. Be curious. Be kind to others in digital spaces as you would be in physical ones. Build things that help people. Question things that harm them. And never stop learning.

To every student opening this book for the first time: I wrote it for you. I hope it opens doors.

With gratitude and sincerity,

The Monotheist

Compiler · Educator · Lifelong Student

CHAPTER 1
Introduction to Computers

Learning Objectives

By the end of this chapter, students will be able to:

- Define a computer and explain what it does
- Identify and describe the main parts of a computer
- Understand hardware and software with examples
- Explain how data is stored and measured
- Describe different types and generations of computers
- Understand the basic concept of networking and the internet
- Identify real-world uses of computers

1.1 What is a Computer?

A computer is an electronic device that accepts data as input, processes it according to a set of instructions (called a program), and produces useful information as output. It can store data for future use and perform tasks very quickly and accurately.

The word "computer" originally referred to a person who performed calculations. Over time, machines were built to do this work, and today a computer can do millions of tasks in just one second.

Did You Know?

The first general-purpose electronic computer, ENIAC, was built in 1945. It weighed about 30 tons and filled an entire room. Today, your smartphone is millions of times more powerful!

Key Characteristics of a Computer

Characteristic	Description
Speed	Computers perform millions of operations per second.
Accuracy	They produce correct results if given correct instructions.
Storage	They can store large amounts of data permanently or temporarily.
Automation	Once programmed, they work without human intervention.

Characteristic	Description
Diligence	They never get tired and work consistently around the clock.
Versatility	They can perform many different kinds of tasks.

1.2 The IPO Cycle (Input–Process–Output)

Every computer works on the IPO cycle. This is the most fundamental concept in computing. Every single thing a computer does follows this pattern:

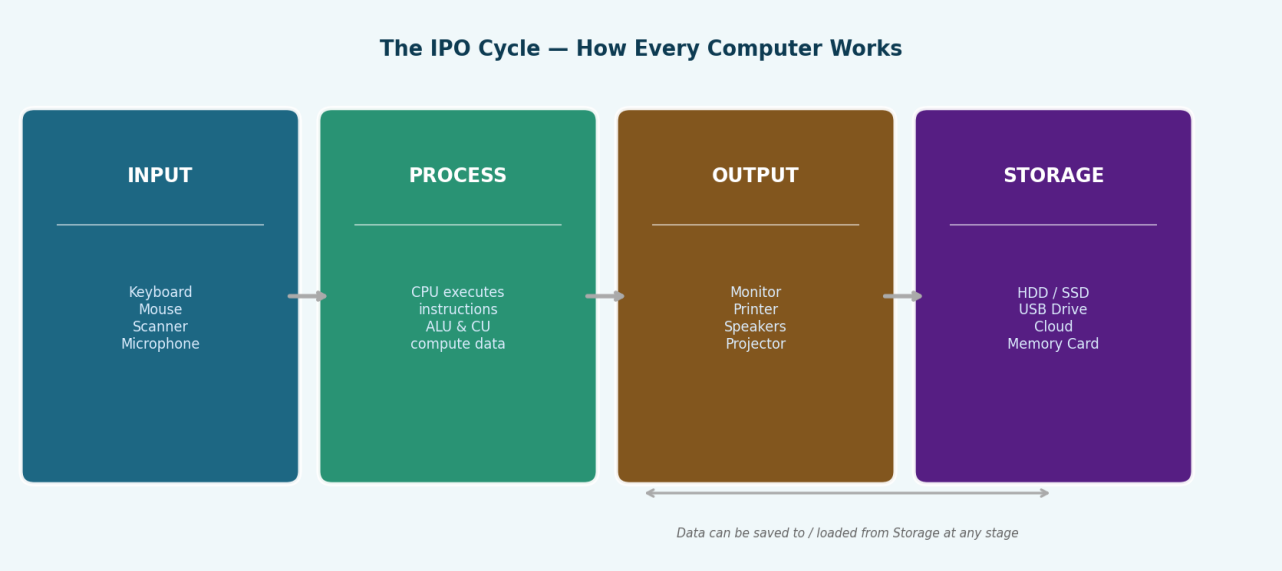


Figure 1.1 — The IPO Cycle: How Every Computer Processes Information

Stage	What Happens	Example
INPUT	Data is entered into the computer	Typing on a keyboard, clicking a mouse
PROCESS	The CPU works on the data using instructions	CPU calculates, compares, or transforms data
OUTPUT	Results are shown to the user	Displaying text on screen, printing a document
STORAGE	Data is saved for later use	Saving a file to the hard disk

1.3 Hardware

Hardware refers to all the physical parts of a computer — the parts you can see and touch. Hardware is divided into four main categories: Input Devices, Output Devices, Processing Unit, and Memory.

1.3.1 Input Devices

Input devices allow users to enter data or commands into the computer:

Device	Description
Keyboard	Used to type letters, numbers, and symbols.

Device	Description
	The most common input device.
Mouse	A pointing device used to click, drag, and select items on screen.
Scanner	Converts printed documents or images into digital form.
Microphone	Captures sound and converts it to digital audio.
Webcam	Captures live video and sends it to the computer.
Touchscreen	Allows input by touching the screen directly.
Joystick	Used mainly for games to control movement.
Barcode Reader	Scans barcodes to read product information.

1.3.2 Output Devices

Device	Description
Monitor	The screen that displays text, images, and video.
Printer	Produces a paper (hard) copy of documents or images.
Speakers	Produce sound — music, speech, or alerts.
Projector	Projects the screen image onto a large surface or wall.
Headphones	Personal audio output worn on or over the ears.
Plotter	Draws large-scale diagrams or blueprints (used in engineering).

1.3.3 Processing Unit — The CPU

The Central Processing Unit (CPU) is often called the "brain" of the computer. It carries out all instructions given to the computer. The CPU has two main parts:

- Arithmetic and Logic Unit (ALU): Performs all mathematical calculations (addition, subtraction, etc.) and logical comparisons (greater than, equal to, etc.).
- Control Unit (CU): Directs and coordinates all activities of the computer. It fetches instructions, interprets them, and controls data flow.

CPU speed is measured in GHz (Gigahertz). A 3.0 GHz processor can perform 3 billion cycles per second. Modern computers may also have multiple cores allowing them to work on several tasks simultaneously.

1.3.4 Memory

Primary Memory (Main Memory)

Type	Description	Key Feature
RAM (Random Access Memory)	Temporary storage used while the computer is running.	Volatile — data lost when power is off. More RAM = smoother multitasking.
ROM (Read-Only Memory)	Permanent memory containing the computer's startup instructions.	Non-volatile — keeps data even when power is off.

Type	Description	Key Feature
		Cannot be changed.

Secondary Memory (Storage)

Storage Device	Description
Hard Disk Drive (HDD)	Magnetic storage inside the computer. Common in desktops and laptops. Capacity: 500 GB to several TB.
Solid State Drive (SSD)	Faster and more durable than HDD. Uses flash memory chips. No moving parts.
USB Flash Drive	Small, portable storage device. Easy to carry and use.
CD / DVD	Optical discs. CDs hold ~700 MB; DVDs hold 4.7 GB to 17 GB.
Memory Card	Small cards used in cameras, phones, and tablets (SD card, microSD).
Cloud Storage	Storing files online on remote servers (e.g., Google Drive, OneDrive).

1.4 Data Storage Units

Data in a computer is stored and measured using specific units. The smallest unit is a bit — a single binary digit (0 or 1).

Unit	Size / Description	Real-World Example
Bit (b)	Smallest unit — a single 0 or 1	A single binary switch
Nibble	4 bits	Half a byte
Byte (B)	8 bits	One character, e.g., the letter "A"
Kilobyte (KB)	1,024 Bytes	About one page of text
Megabyte (MB)	1,024 KB	About one small digital photo
Gigabyte (GB)	1,024 MB	About 1,000 large photos
Terabyte (TB)	1,024 GB	About 500 hours of HD video
Petabyte (PB)	1,024 TB	Used in large data centres

Real-Life Comparison

A typical song is about 4 MB. | A full HD movie is about 4 GB. | A computer with 1 TB storage can hold about 250,000 songs!

1.5 Software

Software is the set of instructions (programs) that tells the hardware what to do. Unlike hardware, you cannot touch software. Software is divided into two major categories: System Software and Application Software.

1.5.1 System Software

- Operating System (OS): Controls all hardware and software resources. Examples: Windows 11, macOS Sonoma, Linux, Android, iOS.
- Device Drivers: Programs that allow the OS to communicate with hardware (e.g., printer driver, graphics driver).
- Utility Programs: Tools that maintain and optimise the computer.
  disk cleaner, file manager.

1.5.2 Application Software

Category	Examples
Word Processing	Microsoft Word, Google Docs, LibreOffice Writer
Spreadsheet	Microsoft Excel, Google Sheets
Presentation	Microsoft PowerPoint, Google Slides
Web Browser	Google Chrome, Mozilla Firefox, Microsoft Edge
Multimedia	VLC Media Player, Adobe Photoshop, Audacity
Communication	WhatsApp, Zoom, Microsoft Teams, Gmail
Education	Khan Academy, Duolingo, Google Classroom
Games	Chess, Minecraft, FIFA (gaming software)

1.5.3 Programming Languages

- Low-level languages: Machine language (binary) and Assembly language — very close to hardware.
- High-level languages: Python, Java, C++, Scratch — easier for humans to write and understand.

1.6 Types of Computers

Computers come in many shapes and sizes, each designed for different purposes — from room-sized supercomputers to tiny embedded chips inside everyday devices.

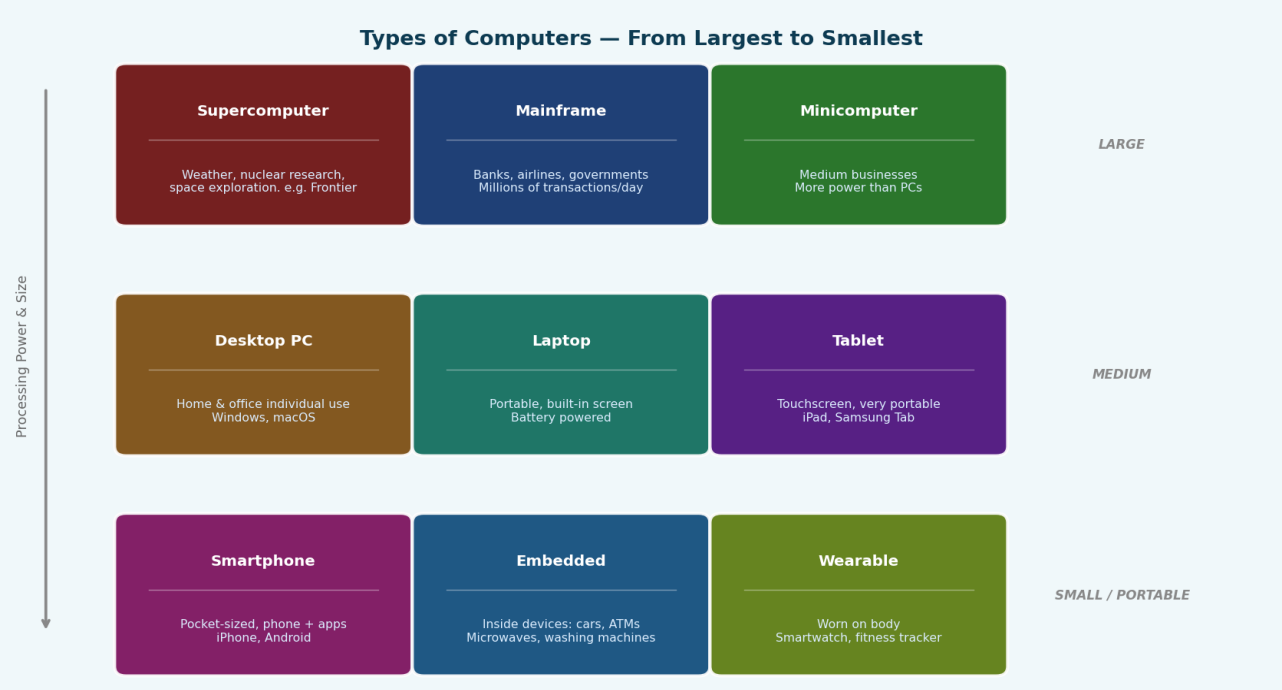


Figure 1.2 — Types of Computers: From Largest to Smallest

Type	Description
Supercomputer	Fastest and most powerful. Used for weather forecasting, nuclear research, and space exploration. Example: Frontier (USA).
Mainframe	Very large computers used by banks, airlines, and governments to process millions of transactions daily.
Minicomputer	Medium-sized computers for medium-scale businesses. Less powerful than mainframes but more than PCs.
Desktop PC	Designed for individual use at home or office. Runs Windows or macOS.
Laptop / Notebook	A portable PC with built-in keyboard, screen, and battery.
Tablet	A flat touchscreen device. More portable than a laptop. Examples: iPad, Samsung Galaxy Tab.
Smartphone	A pocket-sized computer with phone capabilities. Runs mobile apps.
Embedded Computer	Built inside other devices. Examples: washing machines, microwaves, cars, ATMs.
Wearable Computer	Worn on the body. Examples: smartwatch (Apple Watch), fitness tracker.

1.7 Generations of Computers

The history of computers is divided into five generations, each marked by a major technological breakthrough that made computers smaller, faster, and more powerful.

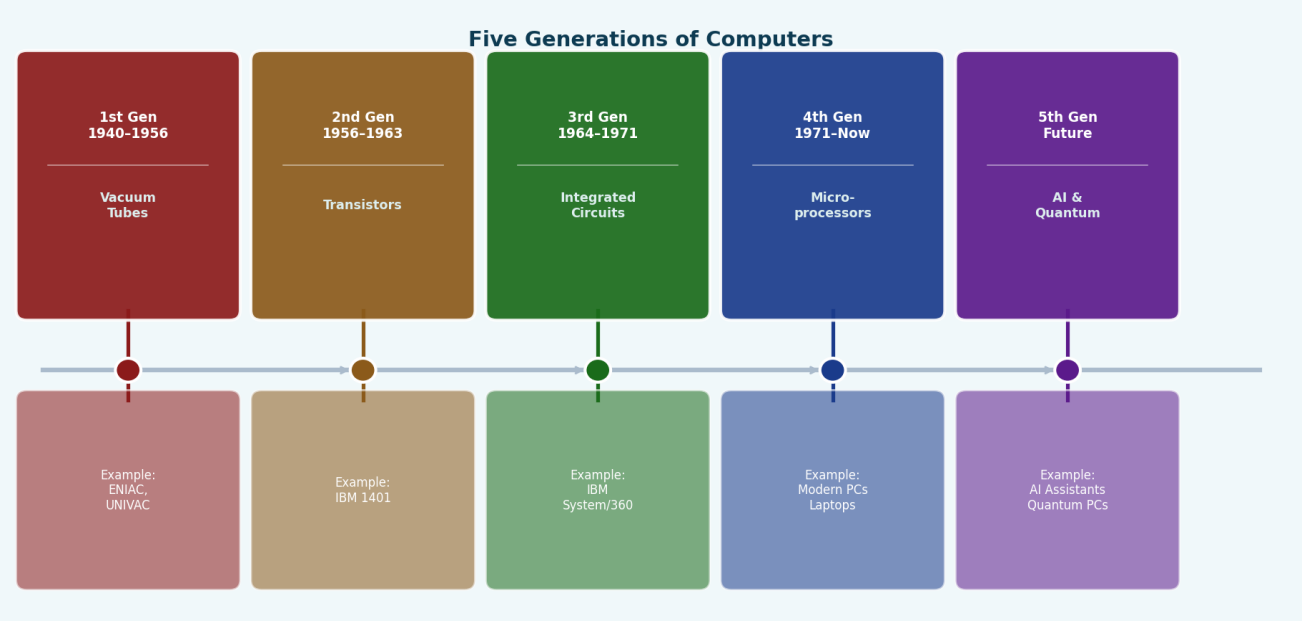


Figure 1.3 — Five Generations of Computers: Technology and Timeline

Generation	Period & Technology	Key Features
1st Generation	1940–1956 | Vacuum Tubes	Very large, consumed huge power, produced a lot of heat, used machine language. Example: ENIAC, UNIVAC.
2nd Generation	1956–1963 | Transistors	Smaller, faster, and more reliable. Used assembly language. Less heat. Example: IBM 1401.
3rd Generation	1964–1971 | Integrated Circuits	Even smaller and faster. Multiple transistors on a chip. Used high-level languages. Example: IBM System/360.
4th Generation	1971–Present | Microprocessors	Entire CPU on a single chip. Birth of PCs. Very fast and affordable. Example: Intel processors, modern laptops.
5th Generation	Present & Future | AI & Quantum	Based on Artificial Intelligence. Machines that can think and learn. Uses voice recognition and machine learning.

1.8 Computer Networks and the Internet

A computer network is a group of computers connected together to share resources such as files, printers, and internet access.

1.8.1 Types of Networks

Network Type	Full Name	Coverage & Use
LAN	Local Area Network	Connects computers in a small area — classroom, school, or office building.
WAN	Wide Area Network	Connects computers across cities or countries. The Internet is the largest WAN.
MAN	Metropolitan Area Network	Covers a city or large campus.

Network Type	Full Name	Coverage & Use
WLAN	Wireless LAN	A LAN that uses Wi-Fi instead of cables.

1.8.2 The Internet

The Internet is a global network of billions of computers and devices all connected together. It allows users to communicate, share information, and access services worldwide.

- World Wide Web (WWW): A system of websites and web pages accessed through a browser. The Web runs on the Internet.
- Email: Electronic messages sent and received over the internet.
- Cloud Computing: Using remote servers on the internet to store, manage, and process data instead of a local computer.

Web vs. Internet — An Important Distinction

The Internet is the physical network of cables and connections. The World Wide Web (WWW) is a service that runs ON the Internet. Not all internet services are the web — email and online gaming also use the internet.

1.9 Uses of Computers

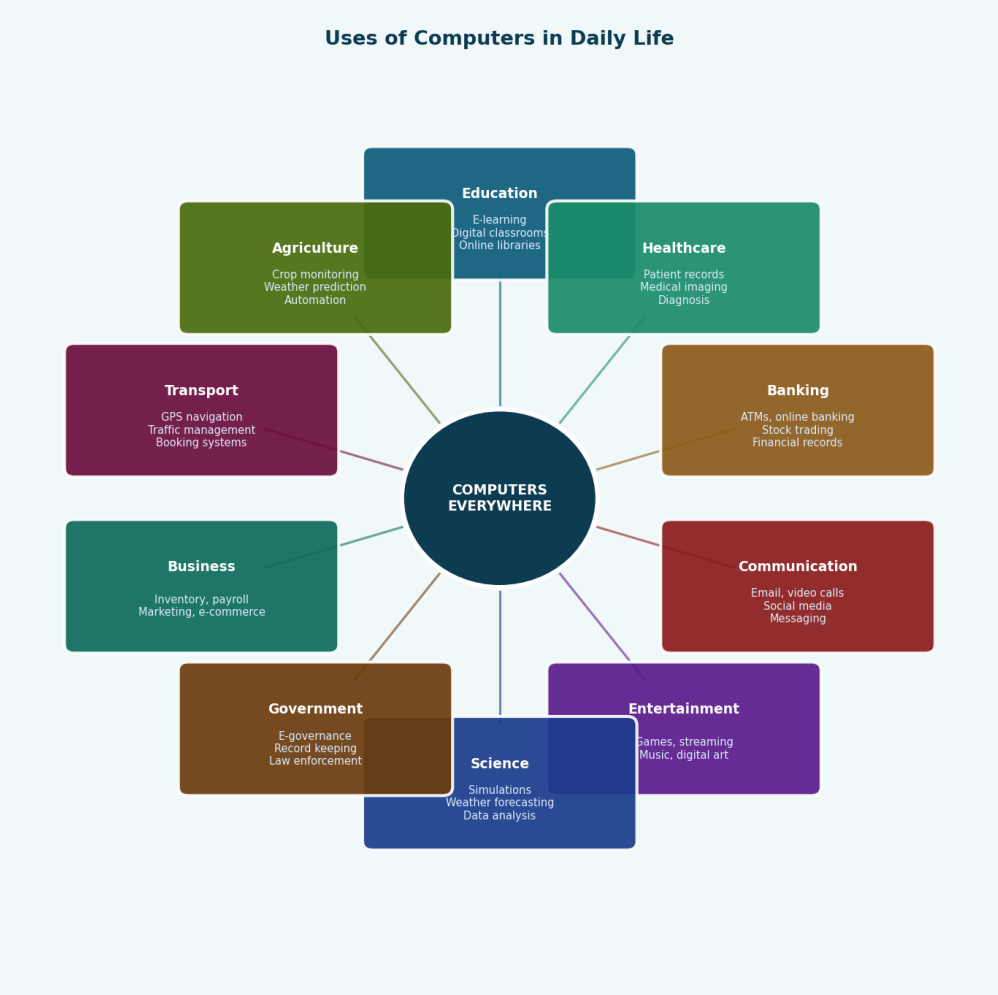


Figure 1.4 — Computers in Every Field of Modern Life

Field	How Computers Are Used
Education	E-learning, digital classrooms, research, online libraries
Healthcare	Patient records, diagnosing diseases, medical imaging (X-ray, MRI)
Banking & Finance	Online banking, ATMs, stock trading, financial records
Communication	Email, video calls, social media, instant messaging
Entertainment	Games, streaming movies and music, digital art
Science & Research	Simulations, data analysis, weather forecasting
Government	Record keeping, e-governance, law enforcement
Business	Inventory management, payroll, marketing, e-commerce
Agriculture	Crop monitoring, weather prediction, automated farming
Transportation	GPS navigation, traffic management, booking systems

1.10 Advantages and Disadvantages of Computers

ADVANTAGES	DISADVANTAGES
Very fast at processing large amounts of data.	Requires electricity — useless during power cuts without backup.
High accuracy with no mistakes when given correct input.	Can malfunction due to viruses, software bugs, or hardware failure.
Can store enormous amounts of data in a small space.	Over-reliance can reduce critical thinking skills.
Capable of multitasking — running many programs at once.	Privacy and security risks — hacking, data theft.
Available 24/7 without fatigue.	Health issues from prolonged use — eye strain, poor posture.
Enables global communication and access to information.	Expensive to purchase, maintain, and upgrade.

Chapter Review Questions

A. Short Answer Questions

1. What is a computer? List any three characteristics.
2. Explain the IPO cycle with an example.
3. What is the difference between RAM and ROM?
4. Name four input devices and four output devices.
5. What is the difference between hardware and software?
6. Name the five generations of computers and the technology used in each.
7. What is a LAN? How is it different from a WAN?
8. List any five uses of computers in daily life.

B. Fill in the Blanks

1. The _____ is called the brain of the computer.
2. 8 bits make one _____.
3. _____ is temporary memory that is lost when the computer is turned off.
4. The _____ is the largest WAN in the world.
5. The first generation of computers used _____ as their main technology.

C. True or False

1. A printer is an input device. (True / False)
2. ROM retains data even when the computer is off. (True / False)
3. A Supercomputer is the slowest type of computer. (True / False)
4. The Internet and the World Wide Web are exactly the same. (True / False)
5. Software can be physically touched. (True / False)

D. Key Terms Glossary

Term	Definition
Computer	An electronic device that inputs, processes, stores, and outputs data.
Hardware	Physical components of a computer that can be seen and touched.
Software	Programs and instructions that tell hardware what to do.
CPU	Central Processing Unit — the brain of the computer.
RAM	Random Access Memory — fast, temporary memory used while the computer is on.
ROM	Read-Only Memory — permanent memory storing startup instructions.
Input Device	Device used to enter data into the computer (e.g., keyboard, mouse).
Output Device	Device that delivers processed results to the user (e.g., monitor, printer).
Operating System	System software that manages hardware and runs applications.
Network	A group of computers connected to share

Term	Definition
	resources and information.
Internet	A global network connecting billions of computers worldwide.
Bit	The smallest unit of data; a single 0 or 1.
Byte	8 bits; the basic unit used to measure data storage.

Chapter 1 · Overview Complete

Units 1·6 build systematically on everything you have just read.

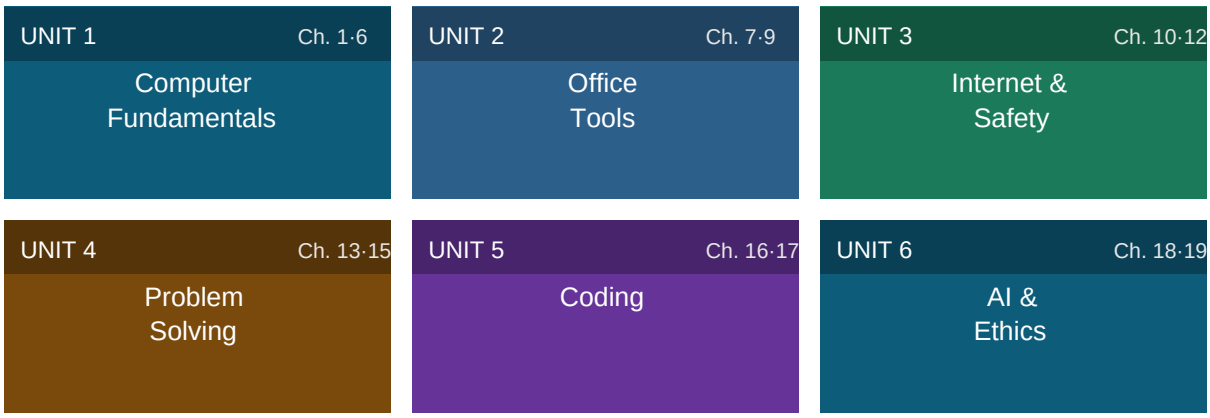
Chapter 1 has given you a broad map of the entire field of Computer Science · hardware, software, storage, networking, and more. You may have noticed that several topics were introduced briefly. This is intentional. Each of the following chapters takes one of those topics and explores it in much greater depth with detailed explanations, worked examples, and review questions.

Chapter	How It Deepens Chapter 1
Ch. 2 Hardware	Explores every hardware component in detail: motherboard, CPU specs, RAM types, HDD vs SSD vs NVMe, GPU, PSU, ports, and peripherals.
Ch. 3 Software	Deep-dives into OS functions, device drivers, utility software, programming language levels, compilers vs interpreters, and software licensing.
Ch. 4 Storage & Memory	Explains primary vs secondary storage, cache levels, binary representation, ASCII vs Unicode, and cloud storage trade-offs.
Ch. 5 Operating System	Covers process management, multitasking, file systems, GUI vs CLI, booting, virtual memory, and OS comparison.
Ch. 6 File Management	Teaches how to organise, navigate, copy, compress, and search files effectively using File Explorer.

How to Use This Book

If you are a complete beginner: read every chapter in order · each one builds on the last. If you need to brush up on a specific topic: jump directly to that chapter. The knowledge belongs to you. There is no wrong way to learn.

Visual Roadmap · All 6 Units at a Glance



CHAPTER 2
Computer Hardware

Learning Objectives

By the end of this chapter, students will be able to:

- Identify and describe internal and external hardware components
- Explain the function of the CPU, RAM, ROM, and storage devices
- Distinguish between input, output, processing, and storage hardware
- Compare different ports and connectors
- Understand how hardware components work together

2.1 What is Computer Hardware?

Computer hardware refers to all the physical, tangible components that make up a computer system. These are the parts you can see, touch, and handle. Hardware can be broadly divided into internal components (inside the computer case) and external components (outside the case, also called peripherals).

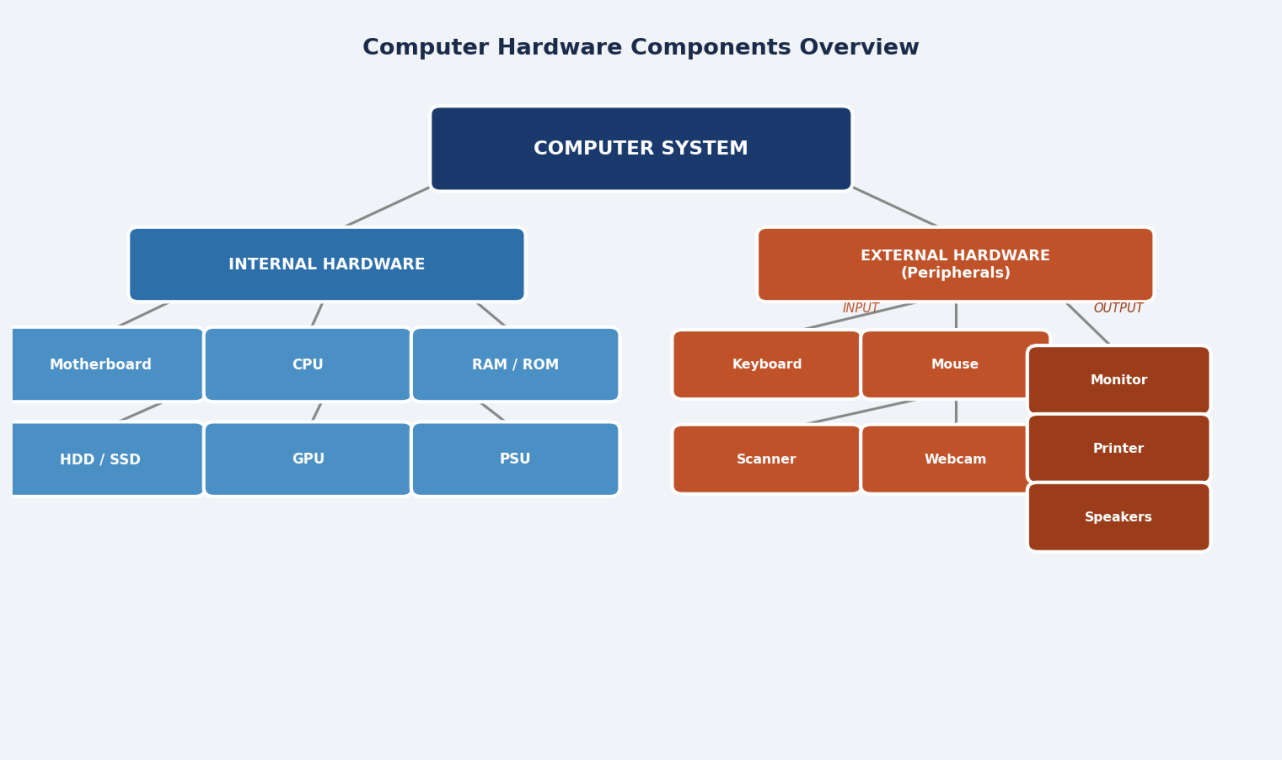


Figure 2.1 — Overview of Computer Hardware Components

2.2 Internal Hardware Components

2.2.1 Motherboard

The motherboard is the main circuit board of the computer. It connects and allows communication between all other components. Every part of the computer — CPU, RAM, storage — connects to the motherboard directly or through cables.

- Also called the mainboard or system board.
- Contains slots for RAM, CPU, and expansion cards.
- Has ports and connectors for storage devices and peripherals.

2.2.2 Central Processing Unit (CPU)

The CPU is the most important processing chip in the computer. It executes instructions from programs and controls all other hardware. Key CPU specifications include:

Specification	Description
Clock Speed	Measured in GHz. Higher = faster processing. Example: 3.5 GHz.
Cores	Multiple cores allow simultaneous task processing. Dual-core, quad-core, octa-core.
Cache Memory	Ultra-fast memory built into the CPU for

Specification	Description
	frequently used instructions.
Manufacturer	Intel (Core i3/i5/i7/i9) and AMD (Ryzen series) are the main CPU makers.

2.2.3 RAM (Random Access Memory)

RAM is the computer's working memory. It temporarily stores data that the CPU is actively using. The more RAM a computer has, the more programs it can run smoothly at the same time.

- Volatile memory — data is lost when power is turned off.
- Common sizes: 4 GB, 8 GB, 16 GB, 32 GB.
- Types: DDR3, DDR4, DDR5 (newer = faster).

2.2.4 ROM (Read-Only Memory)

ROM stores permanent instructions used to start up the computer (called the BIOS/UEFI). Unlike RAM, ROM retains data without power and cannot normally be changed by the user.

Key Distinction: RAM vs ROM vs Storage

RAM is like your work desk — you keep only what you are currently using there.

Storage (HDD/SSD) is like a filing cabinet — everything is stored there long-term.

ROM is like a laminated instruction card — fixed, always there, never changes.

The CPU is the person doing the work.

2.2.5 Hard Disk Drive (HDD) vs Solid State Drive (SSD)

SSDs are increasingly replacing HDDs in modern computers due to their superior speed and durability. The chart below compares their key characteristics:

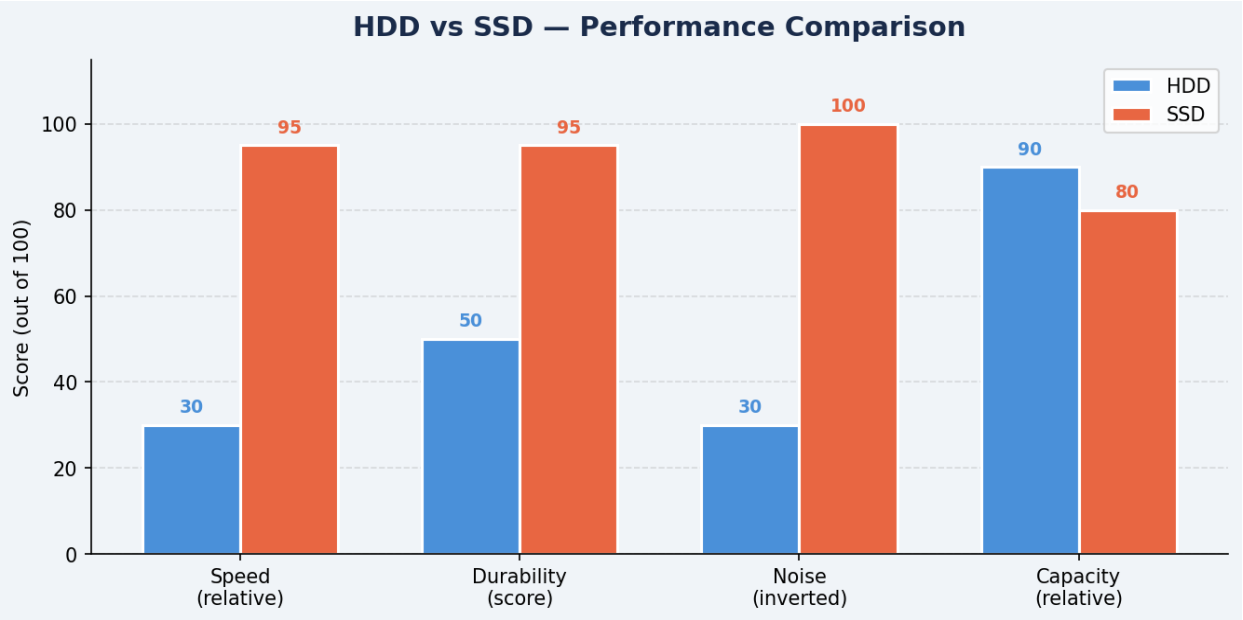


Figure 2.2 — HDD vs SSD Performance Comparison

Feature	HDD	SSD
Technology	Spinning magnetic disks	Flash memory chips
Speed	Standard	5–10x faster than HDD
Durability	Has moving parts — fragile	No moving parts — shock resistant
Cost per GB	Cheaper per GB	More expensive but falling
Noise	Audible spinning noise	Completely silent
Capacity	500 GB to several TB	256 GB to several TB

2.2.6 Power Supply Unit (PSU)

The PSU converts mains electricity (AC) into the lower voltages (DC) used by computer components. It distributes power to the motherboard, CPU, drives, and fans. Rated in Watts (e.g., 500W, 750W).

2.2.7 Graphics Processing Unit (GPU)

The GPU handles all graphical calculations — rendering images, video, and animations. Integrated GPUs are built into the CPU; dedicated GPUs are separate cards (e.g., NVIDIA GeForce, AMD Radeon) for gaming or design work.

2.2.8 Cooling System

Computers generate heat during operation. Cooling systems prevent overheating:

- CPU Fan: Blows air over the CPU heat sink.
- Case Fans: Circulate air inside the computer case.
- Liquid Cooling: Uses water loops for high-performance PCs.
- Thermal Paste: Applied between CPU and cooler for better heat transfer.

2.3 External Hardware (Peripherals)

2.3.1 Input Peripherals

Input devices allow users to send data and commands to the computer:

Device	Description
Keyboard	Standard typing interface. Layouts: QWERTY (most common), AZERTY, DVORAK.
Mouse	Pointing device. Types: optical, laser, wireless. Has left/right buttons and scroll wheel.
Trackpad / Touchpad	Built-in mouse replacement on laptops — senses finger movement.
Scanner	Converts physical documents or photos into digital files. Measured in DPI.
Webcam	Captures video for video calls, streaming, or security.
Microphone	Records audio input. Built-in in laptops; standalone for better quality.
Graphics Tablet	Artists draw directly with a stylus on a pressure-sensitive surface.

2.2.9 NVMe and M.2 Drives

Section 2.2.5 compared HDDs and SATA SSDs. However, modern computers increasingly use a newer, much faster type of SSD: the NVMe (Non-Volatile Memory Express) drive, typically in an M.2 form factor. This technology is now standard in most new laptops and desktop PCs and is important to understand when reading hardware specifications.

Feature	SATA SSD	NVMe M.2 SSD
Protocol	SATA (shared with HDD)	NVMe · direct to CPU via PCIe
Read Speed	Up to ~550 MB/s	3,000·7,000 MB/s
Write Speed	Up to ~520 MB/s	2,500·6,500 MB/s
Form Factor	2.5" drive or M.2	M.2 slot on motherboard
Size	Varies	Very compact · ~22mm × 80mm
Typical Price	Moderate	Slightly higher, rapidly falling
Best For	Budget upgrades	New builds and modern laptops

NVMe drives connect directly to the CPU through PCIe lanes rather than through the slower SATA controller. This eliminates a major bottleneck and allows sequential read speeds of 3,000 MB/s on standard NVMe drives, up to 7,000 MB/s on high-end PCIe 4.0 and 5.0 models · compared to 550 MB/s maximum for SATA SSDs.

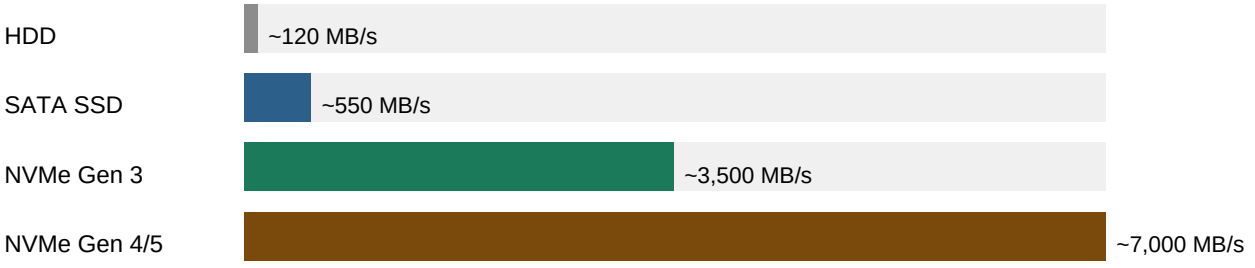
Buyer's Guide · How to Identify NVMe Storage

When buying a new laptop or PC, always check whether the SSD is SATA or NVMe. Look for terms like: 'PCIe NVMe', 'M.2 NVMe', or a speed listed above 1,000 MB/s. A SATA SSD listed as 'M.2' is still limited to ~550 MB/s · the M.2 slot is a physical format; NVMe is the protocol that makes it fast.

Key Terms · Added

Term	Definition
NVMe	Non-Volatile Memory Express · a fast SSD protocol communicating directly with the CPU via PCIe.
M.2	A compact card-sized SSD form factor that plugs into a slot on the motherboard.
PCIe	Peripheral Component Interconnect Express · high-bandwidth bus for connecting fast hardware.

Storage Speed Comparison · Visual Reference



* Typical sequential read speeds. Actual speeds vary by model.

Device	Description
Biometric Scanner	Reads fingerprints or retinas for security access.

2.3.2 Output Peripherals

Device	Description
Monitor / Display	Shows visual output. Types: LCD, LED, OLED. Key specs: size, resolution (1080p, 4K), refresh rate.
Printer	Inkjet: sprays ink — good for photos. Laser: uses toner — fast for text. 3D Printer: builds objects layer by layer.
Speakers	Produce audio output. 2.0 (stereo), 2.1 (with subwoofer), 5.1 (surround sound).
Projector	Displays enlarged images onto a screen or wall. Used in classrooms and boardrooms.
Headphones	Personal audio output. Wired or wireless (Bluetooth).

2.4 Ports and Connectors

Ports are sockets on the computer where you plug in peripherals. Knowing common ports helps you connect devices correctly.

Port / Connector	Use
USB-A	Standard rectangular USB port for flash drives, keyboards, mice.
USB-C	Newer reversible connector — for charging, data, and video. Faster than USB-A.
HDMI	High Definition Multimedia Interface — connects monitor or TV for HD video and audio.
DisplayPort	Similar to HDMI but supports higher refresh rates — used in gaming monitors.
3.5mm Audio Jack	For headphones or speakers (green = output, pink = microphone).
Ethernet (RJ-45)	Connects computer to a wired network using a network cable.
SD Card Slot	Accepts memory cards from cameras or other devices.
Thunderbolt	Very high-speed port (Intel/Apple) supporting data, video, and power.

2.5 How Hardware Components Work Together

When you turn on a computer and open a program, here is what happens behind the scenes:

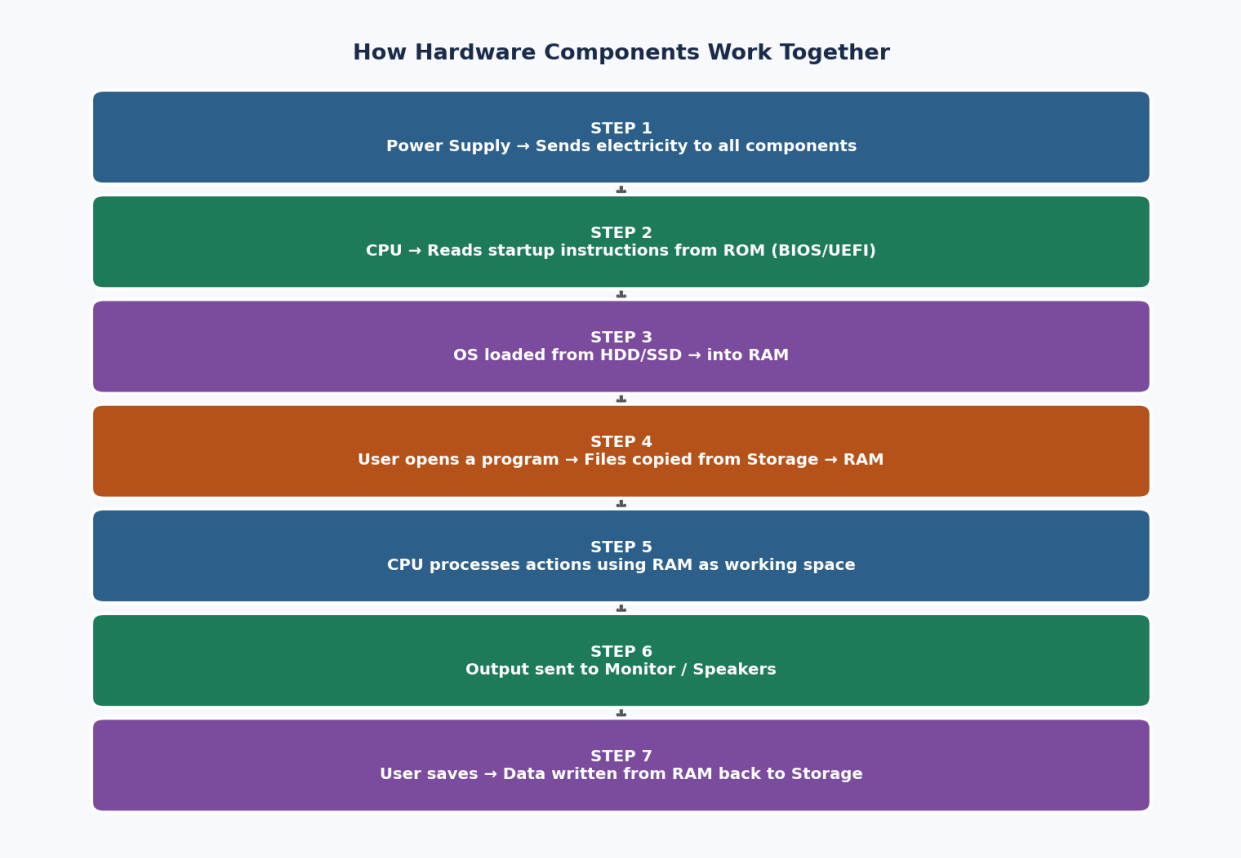


Figure 2.3 — Step-by-Step: How Computer Hardware Works Together

Chapter Review Questions

A. Short Answer

1. What is the role of the motherboard in a computer?
2. What does CPU stand for? Name two things it does.
3. What is the difference between RAM and ROM?
4. List three differences between HDD and SSD.
5. Name four types of ports and what each is used for.

B. Fill in the Blanks

1. The _____ is the main circuit board that connects all computer components.
2. CPU speed is measured in _____.
3. A/An _____ port is used to connect a monitor for HD video output.
4. _____ memory is lost when the computer is switched off.
5. The _____ unit converts mains electricity for use by internal components.

C. True or False

1. The GPU is responsible for handling audio output. (True / False)
2. SSD has moving parts inside. (True / False)
3. A webcam is an input device. (True / False)
4. USB-C is a newer and more versatile connector than USB-A. (True / False)
5. RAM retains data when the computer is switched off. (True / False)

D. Key Terms Glossary

Term	Definition
Motherboard	The main circuit board connecting all computer components.
CPU	The chip that processes all instructions in a computer.
RAM	Temporary, fast memory used while the computer is running.
GPU	Chip dedicated to processing graphics and video.
PSU	Converts mains electricity into voltages used by the computer.
Peripheral	An external device connected to the computer (e.g., printer, scanner).
Port	A socket on the computer used to connect external devices.
HDMI	A port and cable standard for transmitting high-definition video and audio.

Additional Review Questions · NVMe & M.2 (§2.2.9)

A. Short Answer

6. What is the difference between an NVMe M.2 SSD and a SATA SSD?

7. What does M.2 refer to? Why does M.2 not always mean NVMe?

8. A PC has an 'M.2 NVMe SSD'. What does this combination tell you about its speed?

9. Why are NVMe drives replacing SATA SSDs as the standard in modern laptops?

B. Fill in the Blanks

1. NVMe drives connect to the CPU via _____ lanes rather than the SATA controller.

2. A standard SATA SSD's top speed is around _____ MB/s.

3. NVMe drives can achieve read speeds of _____ to _____ MB/s.

4. The M.2 slot is a physical _____ factor · the drive inside can be SATA or NVMe.

5. When buying a laptop, look for the phrase '_____ NVMe' to confirm fast storage.

C. True or False

NVMe is a physical form factor like M.2. (True / False)

All NVMe drives are faster than all SATA SSDs. (True / False)

An M.2 slot can be used for both SATA and NVMe drives. (True / False)

NVMe stands for Non-Volatile Memory Express. (True / False)

HDDs are generally faster than NVMe SSDs. (True / False)

Quick Reference · NVMe & M.2 Key Facts

Term / Concept	Key Fact
NVMe full name	Non-Volatile Memory Express
NVMe protocol	PCIe (Peripheral Component Interconnect Express)
HDD speed	~80·160 MB/s sequential read
SATA SSD speed	~500·550 MB/s sequential read
NVMe Gen 3 speed	~3,000·3,500 MB/s sequential read
NVMe Gen 4 speed	~5,000·7,000 MB/s sequential read
M.2 form factor	~22mm × 80mm card plugging into motherboard slot
M.2 + SATA	Physical M.2 slot + SATA protocol = ~550 MB/s max
M.2 + NVMe	Physical M.2 slot + NVMe protocol = 3,000·7,000 MB/s
Key buyer tip	Look for 'PCIe NVMe' or speeds >1,000 MB/s in specs

True / False Answer Key

1. NVMe is a physical form factor like M.2. · FALSE (NVMe is a protocol, M.2 is physical)
2. All NVMe drives are faster than all SATA SSDs. · TRUE (NVMe always beats SATA SSD)
3. M.2 slot can be used for both SATA and NVMe drives. · TRUE
4. NVMe stands for Non-Volatile Memory Express. · TRUE
5. HDDs are generally faster than NVMe SSDs. · FALSE (HDDs are ~40× slower)

CHAPTER 3
Computer Software

Learning Objectives

By the end of this chapter, students will be able to:

- Define software and explain why it is needed
- Distinguish between system software and application software
- Identify common examples of each type of software
- Understand what programming languages are and how they work
- Explain the software development process

3.1 What is Software?

Software is a collection of programs, instructions, and data that tell a computer what to do. Unlike hardware, software cannot be physically touched — it exists as digital code stored on a computer. Without software, a computer is just a collection of electronic parts with no purpose.

Hardware vs Software — An Analogy

Hardware is the physical parts (keyboard, CPU, monitor).

Software is the set of instructions that runs on that hardware.

Think of a piano (hardware) and the sheet music (software) — one without the other is not very useful.

3.2 Types of Software

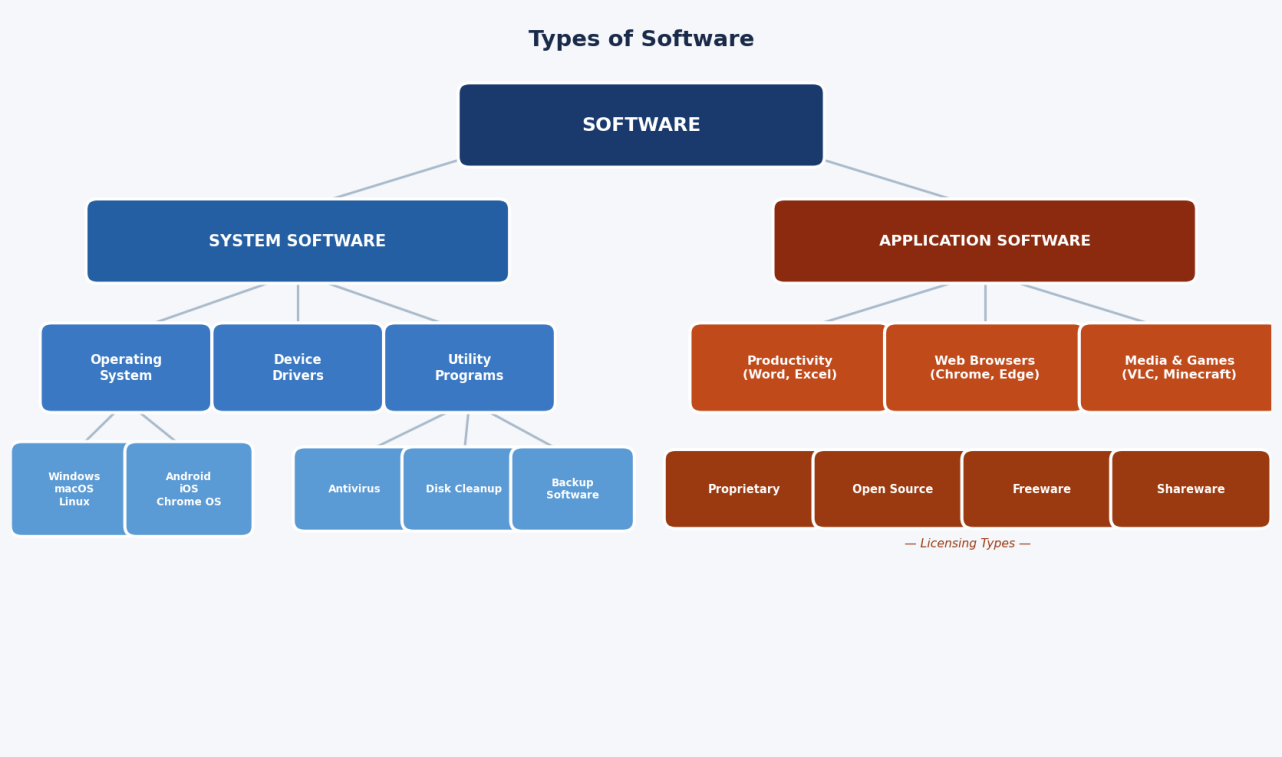


Figure 3.1 — Software Classification Tree

3.2.1 System Software

System software manages and controls the hardware, and provides a foundation on which other software can run. Users interact with it mostly without realising it.

Operating System (OS)

The OS is the most critical piece of system software. It manages all hardware resources and provides an interface between the user and the hardware.



Figure 3.2 — Operating System Core Functions

Operating System	Used On
Microsoft Windows 11	Personal computers, laptops (most common worldwide)
macOS Sonoma	Apple Mac computers and MacBooks
Linux (Ubuntu, Fedora)	Servers, developers, and advanced users (free and open-source)
Android	Smartphones and tablets (most popular mobile OS)
iOS / iPadOS	Apple iPhones and iPads
Chrome OS	Google Chromebooks (cloud-based)

Key OS Functions:

- Process Management: Manages which programs run and shares CPU time between them.
- Memory Management: Allocates RAM to running programs and frees it when they close.
- File Management: Organises files in folders and controls read/write access.
- Device Management: Communicates with hardware via device drivers.
- Security: Manages user accounts, passwords, and permissions.

Device Drivers

Device drivers are small programs that allow the OS to communicate with specific hardware devices. For example, a printer driver tells the OS exactly how to send data to a particular printer model. Without the correct driver, the device will not work.

Utility Programs

Utility Program	Purpose
Antivirus / Anti-malware	Detects and removes viruses, spyware, and malicious software.
Disk Defragmenter	Rearranges fragmented data on an HDD for faster access. · Never run on SSDs · causes wear.
Disk Cleanup	Removes temporary files and frees up storage space.
Backup Software	Creates copies of important files in case of data loss.
File Compression (ZIP)	Reduces file sizes for easier storage and sharing.
System Monitor / Task Manager	Shows CPU, RAM, and disk usage; lets you close frozen programs.

3.2.2 Application Software

Application software (apps) are programs designed for end-users to accomplish specific tasks. They run on top of the operating system.

Category	Examples and Uses
Word Processing	MS Word, Google Docs — create and edit documents, letters, and reports.
Spreadsheet	MS Excel, Google Sheets — organise data, perform calculations, create charts.
Presentation	MS PowerPoint, Google Slides — create slideshows for presentations.
Web Browser	Google Chrome, Firefox, Edge — browse the internet.
Email Client	Microsoft Outlook, Thunderbird — send, receive, and organise emails.
Media Player	VLC, Windows Media Player — play audio and video files.
Image Editing	Adobe Photoshop, GIMP — edit and create digital images.
Video Editing	Adobe Premiere, DaVinci Resolve — edit and produce videos.
Database	Microsoft Access, MySQL — store and query structured data.
Accounting	QuickBooks, Sage — manage business finances and payroll.
Games	Minecraft, FIFA, Chess — entertainment and

Category	Examples and Uses
	recreation.

3.2.3 Open Source vs Proprietary Software

Type	Description
Proprietary Software	Owned by a company. Source code is secret. Must purchase a licence. Examples: Microsoft Office, Adobe Photoshop.
Open Source Software	Source code is freely available. Anyone can modify and distribute it. Examples: LibreOffice, VLC, Linux, Firefox.
Freeware	Free to use but source code is not available. Examples: Adobe Acrobat Reader, Skype.
Shareware	Free trial — limited time or features. Must pay to unlock full version.

3.3 Programming Languages

Programming languages allow humans to write instructions that a computer can execute. Computers only understand binary (0s and 1s — called machine code). Programming languages act as a bridge between human thinking and machine code.

3.3.1 Levels of Programming Languages

Level	Description
Machine Language (1st gen)	Pure binary code (0s and 1s). Directly understood by CPU. Extremely difficult for humans to write.
Assembly Language (2nd gen)	Uses short codes (mnemonics) like MOV, ADD. Converted to machine code by an assembler.
High-Level Languages (3rd gen)	n, Java (bytecode), C++. Need a compiler or interpreter.
4th Generation (4GL)	Even closer to natural language. Often used for databases. Example: SQL.
5th Generation (5GL)	Based on AI and natural language. Examples: Prolog, Mercury.

3.3.2 Compiler vs Interpreter

High-level code must be translated to machine code before the CPU can run it. Two methods are used:

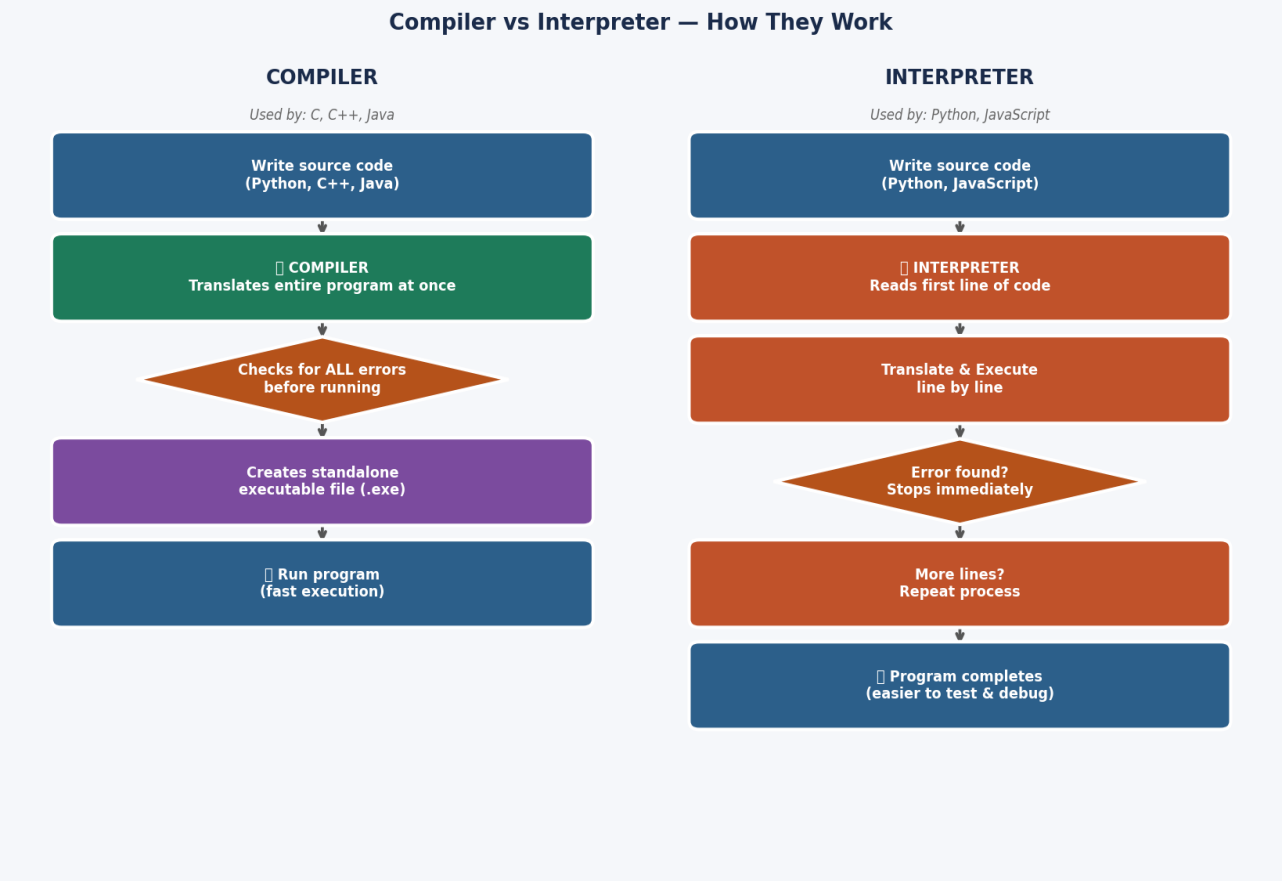


Figure 3.3 — Compiler vs Interpreter: How They Process Code

- Compiler: Translates the entire program at once into an executable file. Faster to run. Used by C and Used by C and C++. (Java compiles to bytecode, interpreted by JVM.)
- Interpreter: Translates and executes the program line by line. Easier to test. Used by Python, JavaScript.

3.3.3 Popular Programming Languages

Language	Common Uses
Python	Beginner-friendly. Used in data science, AI, automation, and web development.
JavaScript	Makes websites interactive. Runs in every web browser.
Java	Used for Android apps, enterprise software, and cross-platform programs.
C / C++	Fast, low-level. Used in operating systems, games, and embedded systems.
HTML / CSS	Markup and style languages for building web pages.
SQL	Used to query and manage data in databases.
Scratch	Visual block-based language for beginners and children.
Swift / Kotlin	Used for iOS (Swift) and Android (Kotlin) mobile app development.

3.4 Software Licensing and Updates

When you buy or download software, you receive a software licence — a legal agreement that defines how you may use the software. Important concepts:

- Single-user licence: One person on one computer.
- Multi-user / site licence: Multiple users, e.g., all computers in a school.
- Subscription licence: Pay monthly or yearly (e.g., Microsoft 365, Adobe Creative Cloud).
- Software Update / Patch: A small update that fixes bugs or security vulnerabilities.
- Software Upgrade: A major new version with new features.

Chapter Review Questions

A. Short Answer

1. What is software? How is it different from hardware?
2. What is the role of an operating system? Name three functions.
3. What is a device driver? Give one example.
4. What is the difference between a compiler and an interpreter?
5. Name four categories of application software with an example of each.

B. Fill in the Blanks

1. _____ software manages hardware and provides a platform for other software.
2. Microsoft Windows, macOS, and Linux are examples of _____.
3. Software whose source code is freely available is called _____ software.
4. Python uses an _____ to run code line by line.
5. A _____ is a small update released to fix bugs or security problems.

C. True or False

1. Software can be physically touched and handled. (True / False)
2. An antivirus program is a utility program. (True / False)
3. Machine language uses words like ADD and MOV. (True / False)
4. on. (True / False)
5. A compiler translates code line by line as it runs. (True / False)

D. Key Terms Glossary

Term	Definition
Software	Programs and instructions that tell hardware what to do.
Operating System	System software managing hardware, memory, files, and processes.
Device Driver	Software allowing the OS to communicate with a specific hardware device.
Utility Program	Software that maintains and optimises a computer.
Application Software	Programs designed for specific user tasks.
Compiler	Translates an entire high-level program into machine code at once.
Interpreter	Translates and runs a program one line at a time.
Open Source	Software with freely available and modifiable source code.

CHAPTER 4
Data Storage & Memory

Learning Objectives

By the end of this chapter, students will be able to:

- Explain the difference between primary and secondary storage
- Compare different storage devices and their characteristics
- Understand data storage units from bit to petabyte
- Describe how data is represented in binary
- Evaluate appropriate storage media for different situations

4.1 Why Do Computers Need Storage?

A computer needs to store two things: the programs it runs and the data those programs create. Without storage, a computer could not remember anything — not your files, your photos, or even how to start up. Storage keeps data safe for as long as needed, from a few seconds to many years.

4.2 Primary Memory

Primary memory is directly accessible by the CPU and is used for active, real-time operations. It is fast but limited in capacity. There are three main types: RAM, ROM, and Cache Memory.

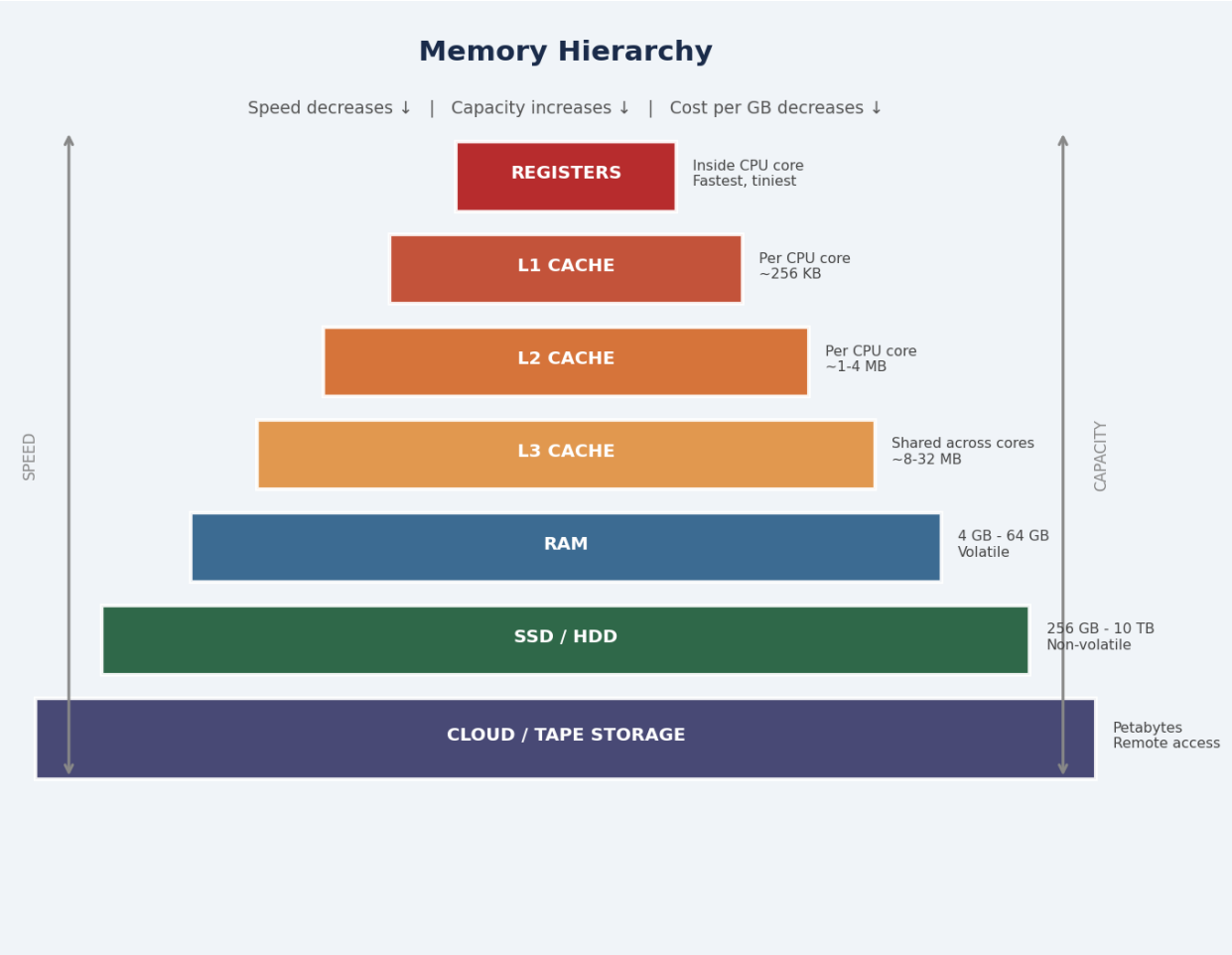


Figure 4.1 — Memory Hierarchy: Speed vs Capacity vs Cost

4.2.1 RAM (Random Access Memory)

RAM is the working memory of a computer. All programs and data currently in use are loaded into RAM. When you close a program or turn off the computer, the RAM contents are erased.

Property	Detail
Type	Volatile — loses all data without power
Speed	Very fast — measured in MHz/GHz and bandwidth (GB/s)
Capacity	Typically 4 GB – 64 GB in modern computers

Property	Detail
Purpose	Holds the OS, open applications, and active data

4.2.2 ROM (Read-Only Memory)

ROM is a small chip on the motherboard containing the firmware — the initial instructions the computer needs to start up (BIOS or UEFI). It is permanent and cannot be changed by normal user operations.

- Type: Non-volatile — retains data without power.
- Content: Boot instructions, hardware checks (POST — Power-On Self-Test).
- Capacity: Very small (a few megabytes).

4.2.3 Cache Memory

Cache memory is ultra-fast memory built directly into the CPU. It stores the most frequently used instructions and data so the CPU does not have to fetch them from the slower RAM. There are three levels:

Cache Level	Size	Speed	Location
L1 Cache	Smallest (~256 KB)	Fastest	Built into each CPU core
L2 Cache	Medium (~1–4 MB)	Very fast	Per core or shared
L3 Cache	Largest (~8–32 MB)	Fast	Shared between all CPU cores

4.3 Secondary Storage

Secondary storage is used for long-term, permanent storage. Data remains stored even when the computer is turned off. It is slower than RAM but far larger in capacity.

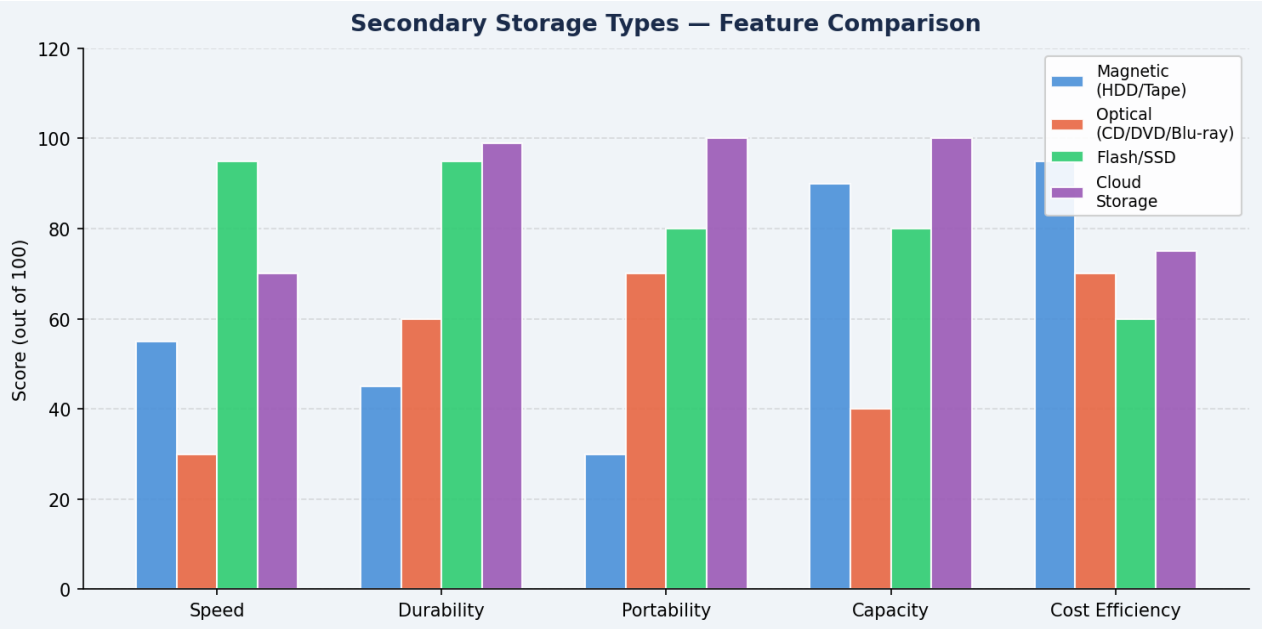


Figure 4.2 — Secondary Storage Types: Feature Comparison

4.3.1 Magnetic Storage

Data is stored by magnetising tiny particles on a surface. Reading and writing is done by a magnetic head.

- Hard Disk Drive (HDD): The most common internal storage. Spinning platters coated with magnetic material. Capacity: 500 GB – 10+ TB. Affordable but fragile.
- Magnetic Tape: Used for backing up very large amounts of data. Sequential access only. Used in data centres.

4.3.2 Optical Storage

Data is stored as microscopic pits and lands on a shiny disc surface, read by a laser beam.

Disc Type	Capacity	Common Use
CD (Compact Disc)	~700 MB	Music, software distribution
DVD (Digital Versatile Disc)	4.7 GB – 17 GB	Movies, software
Blu-ray Disc	25 GB – 100 GB	HD movies and large data backups

4.3.3 Flash / Solid-State Storage

Flash memory stores data in electronic cells with no moving parts. It is fast, silent, and durable.

Device	Capacity	Use Case
SSD (Solid State Drive)	256 GB – 4 TB	Main internal storage in modern laptops and PCs
USB Flash Drive	4 GB – 1 TB	Portable, plugs via USB port
SD Card / MicroSD	4 GB – 1 TB	Cameras, phones, tablets
eMMC	16 GB – 256 GB	Soldered into budget laptops and phones

4.3.4 Cloud Storage

Cloud storage saves data on remote servers maintained by companies and accessed via the internet. There is no physical storage device the user manages.

Service	Provider	Free Storage
Google Drive	Google	15 GB free
OneDrive	Microsoft	5 GB free
iCloud	Apple	5 GB free
Dropbox	Dropbox Inc.	2 GB free

Cloud Storage — Advantages & Disadvantages

Advantages: Access from anywhere, automatic backup, easy sharing.

Disadvantages: Requires internet, monthly costs for large storage, privacy concerns.

4.4 Data Representation: Binary

4.3.3a NVMe and M.2 Solid-State Drives

While the flash storage section above covers SSDs in general, modern computers use a specific type of SSD that is significantly faster: the NVMe drive in an M.2 form factor. This is now the standard internal storage in most laptops and desktop computers sold since 2019 and is important to understand when comparing storage specifications.

What is NVMe?

NVMe (Non-Volatile Memory Express) is a communication protocol designed specifically for flash storage. Traditional SSDs used the SATA protocol, originally designed for spinning hard disks. NVMe bypasses these limitations by connecting directly to the CPU via PCIe lanes, achieving dramatically higher speeds.

What is M.2?

M.2 is a compact physical form factor (a small card approximately 22mm × 80mm) that plugs directly into a slot on the motherboard. An M.2 drive can use either the SATA or NVMe protocol · NVMe M.2 drives are far faster.

Storage Type	Protocol	Typical Speed	Use Case
HDD	SATA	80·160 MB/s	Budget desktops
SATA SSD	SATA	500·550 MB/s	Older laptops/PCs
NVMe M.2 SSD	NVMe	3,000·7,000 MB/s	Modern laptops, PCs

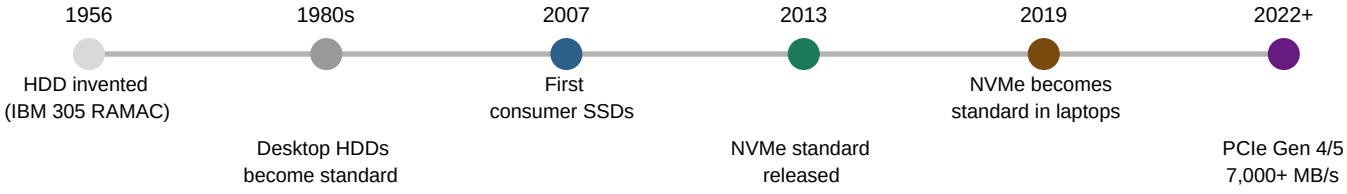
Did You Know? · NVMe Speed

A modern NVMe SSD can read data at 5,000·7,000 MB/s · that is over 40x faster than a traditional hard disk drive. When buying a laptop, look for 'NVMe SSD' rather than just 'SSD' to ensure maximum performance.

Key Terms · Added

Term	Definition
NVMe	Non-Volatile Memory Express · a high-speed protocol for SSDs connecting via PCIe
M.2	A compact SSD form factor that slots directly into the motherboard. Supports SATA or NVMe
PCIe	Peripheral Component Interconnect Express · high-speed bus connecting components to CPU.

Storage Technology Evolution Timeline



Key Insight: NVMe drives are now standard in new laptops (since ~2019).

When buying storage, always check whether the device uses SATA or NVMe protocol.

All data in a computer is ultimately represented as binary — sequences of 0s and 1s. This is because electronic circuits have two states: off (0) and on (1).

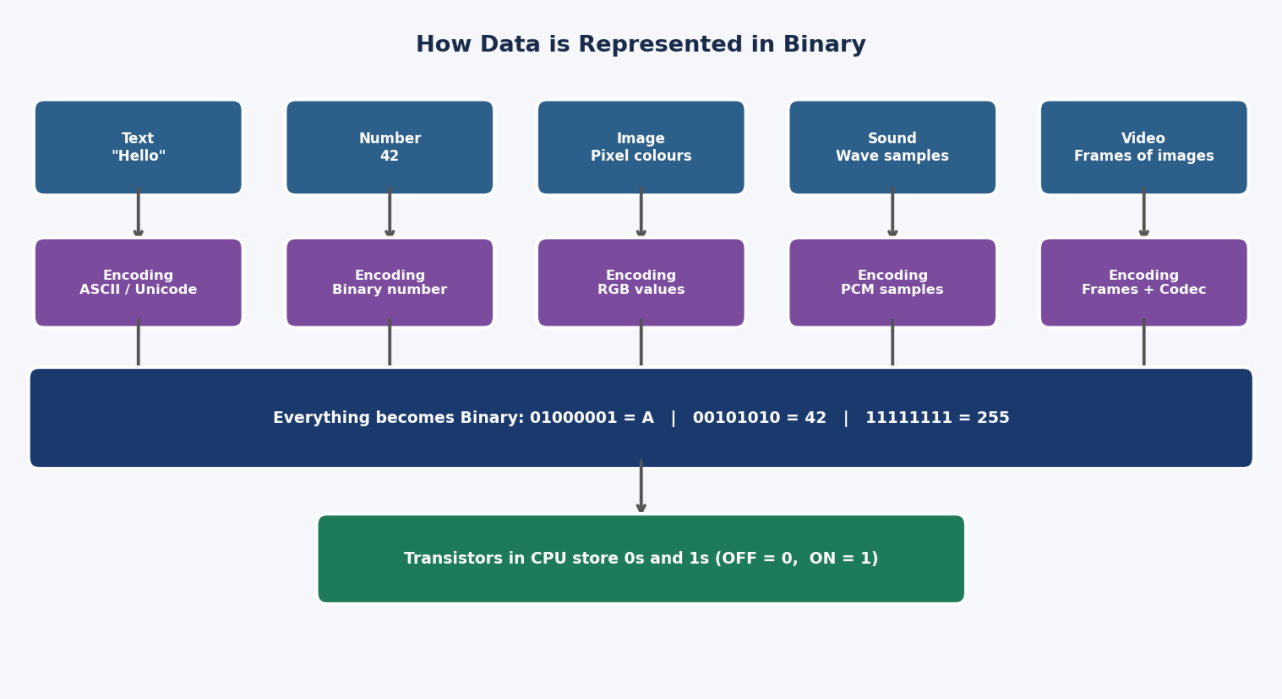


Figure 4.3 — How All Data Types Are Converted to Binary

4.4.1 Why Binary?

Binary (base-2) uses only two digits. It perfectly matches the on/off nature of transistors. Every number, letter, image, or sound is eventually stored as a pattern of 0s and 1s.

4.4.2 Binary Numbers

Each binary digit is called a bit. A group of 8 bits is a byte. Here is how binary counting works:

Binary	Decimal	Binary	Decimal
0000	0	1000	8
0001	1	1001	9
0010	2	1010	10
0011	3	1011	11
0100	4	1100	12
0101	5	1101	13
0110	6	1110	14
0111	7	1111	15

Converting Binary to Decimal

Each bit position represents a power of 2 (from right: 1, 2, 4, 8, 16...).

Example: 1011 = (1×8) + (0×4) + (1×2) + (1×1) = 8 + 0 + 2 + 1 = 11

4.4.3 Character Encoding

Letters, symbols, and punctuation are stored as numbers using encoding standards:

- ASCII (American Standard Code for Information Interchange): 7-bit code for 128 characters. 'A' = 65, 'a' = 97, '0' = 48.
- Unicode (UTF-8): Modern standard supporting over 154,000 characters · all world languages, emojis, mathematical symbols.

4.5 Storage Units — Full Breakdown

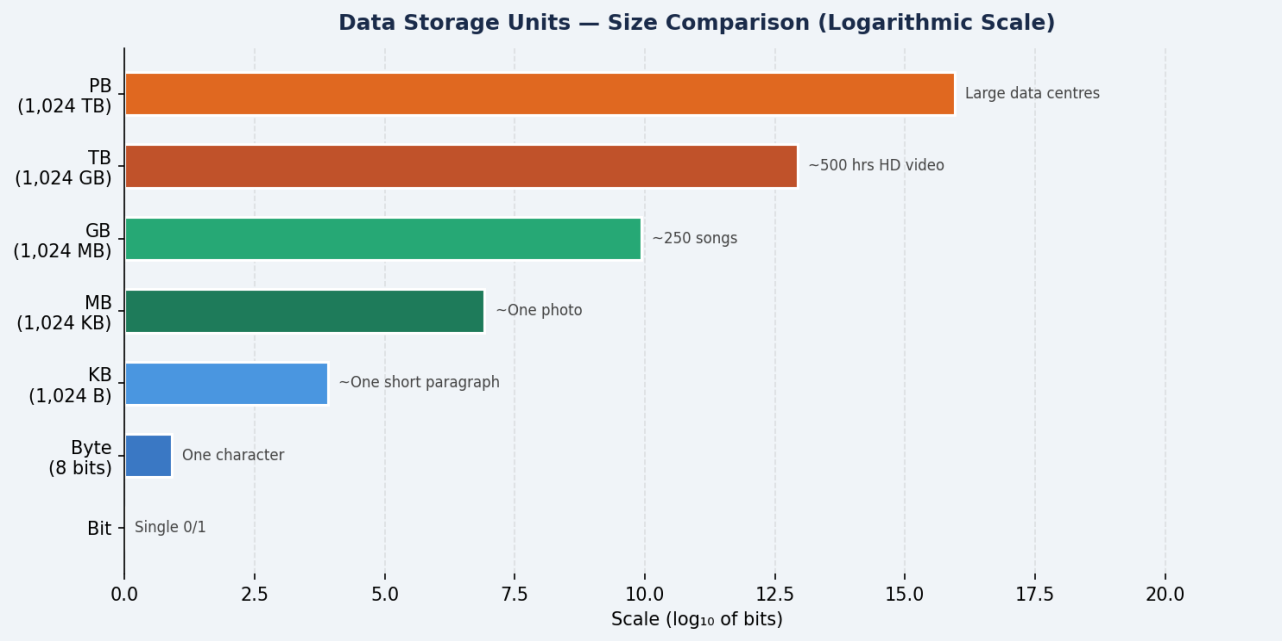


Figure 4.4 — Data Storage Units: Logarithmic Size Comparison

Unit	Size	Real-World Example
Bit (b)	A single 0 or 1	The smallest unit of data
Nibble	4 bits	Half a byte
Byte (B)	8 bits	One character, e.g. 'A'
Kilobyte (KB)	1,024 bytes	Approx. one short paragraph of text
Megabyte (MB)	1,024 KB	Approx. one digital photo
Gigabyte (GB)	1,024 MB	Approx. 250 songs
Terabyte (TB)	1,024 GB	Approx. 500 hours of HD video
Petabyte (PB)	1,024 TB	Used in large data centres
Exabyte (EB)	1,024 PB	Entire internet traffic per day

Chapter Review Questions

A. Short Answer

1. What is the difference between primary and secondary storage?
2. Why is cache memory important? Name the three levels of cache.
3. Compare HDDs and SSDs in terms of speed, durability, and cost.
4. What is binary? Why do computers use binary?
5. What is the difference between ASCII and Unicode?

B. Fill in the Blanks

1. _____ memory is lost when the computer loses power.
2. A group of 8 bits is called a _____.
3. The binary number 1010 equals _____ in decimal.
4. _____ storage saves data on remote servers accessed over the internet.
5. ROM stores the _____ that starts up the computer.

C. True or False

1. RAM is non-volatile memory. (True / False)
2. A USB flash drive uses flash memory with no moving parts. (True / False)
3. Blu-ray discs can store more data than DVDs. (True / False)
4. Unicode only supports the English alphabet. (True / False)
5. L1 cache is the largest and slowest cache level. (True / False)

D. Key Terms Glossary

Term	Definition
Volatile Memory	Memory that loses its data when power is removed (e.g., RAM).
Non-volatile Memory	Memory that retains data without power (e.g., ROM, SSD, HDD).
Cache Memory	Ultra-fast memory built into the CPU for frequently used data.
Binary	A number system using only 0 and 1 — the language of computers.
ASCII	A character encoding standard representing English letters as numbers.
Unicode	A modern encoding standard for all world languages and symbols.
Cloud Storage	Storing data on remote internet servers for access from anywhere.
Flash Memory	Non-volatile storage using electronic cells — used in SSDs and USB drives.

— End of Chapter 4: Data Storage and Memory —

Additional Review Questions · NVMe & M.2 (§4.3.3a)

A. Short Answer

6. What is NVMe? Why was it developed instead of using the existing SATA protocol?

7. What is an M.2 drive? Can an M.2 slot contain both SATA and NVMe drives?

8. A laptop is advertised with '512 GB M.2 SSD'. Does this guarantee NVMe speeds? Explain.

9. Compare HDD, SATA SSD, and NVMe SSD in terms of speed, durability, and use case.

B. Fill in the Blanks

1. NVMe stands for _____ _____ Memory _____.

2. NVMe drives connect directly to the CPU via _____ lanes.

3. A SATA SSD has a maximum read speed of approximately _____ MB/s.

4. A high-end NVMe SSD can reach read speeds of up to _____ MB/s.

5. M.2 refers to a physical _____ factor, not the speed protocol.

C. True or False

All M.2 SSDs use the NVMe protocol. (True / False)

NVMe SSDs are generally faster than SATA SSDs. (True / False)

NVMe bypasses the SATA controller to communicate directly with the CPU. (True / False)

An HDD is faster than an NVMe SSD. (True / False)

PCIe is the bus NVMe drives use to connect to the CPU. (True / False)

Fill in the Blanks · Answer Key

1.	NVMe stands for Non-Volatile Memory Express.
2.	NVMe drives connect directly to the CPU via PCIe lanes.
3.	A SATA SSD's top speed is around 550 MB/s.
4.	NVMe drives can reach speeds of 3,000 to 7,000 MB/s.
5.	M.2 refers to a physical form factor · the drive inside can be SATA or NVMe.

True / False Answer Key

1.	FALSE	All M.2 SSDs use NVMe protocol.· | M.2 can be SATA or NVMe
2.	TRUE	NVMe SSDs are generally faster than· | NVMe uses PCIe, SATA is bottlenecked
3.	TRUE	NVMe bypasses SATA controller to co· | Via PCIe lanes directly
4.	FALSE	An HDD is faster than an NVMe SSD.· | HDD ~150 MB/s vs NVMe 3,000-7,000 MB/s
5.	TRUE	PCIe is the bus NVMe drives use to · | PCIe = high-bandwidth connection

Quick Storage Speed Reference · For Exam Use

HDD	~80·160 MB/s	Spinning magnetic disk, slowest
SATA SSD	~500·550 MB/s	Flash memory, SATA protocol

CHAPTER 5
Operating System Basics

Learning Objectives

By the end of this chapter, students will be able to:

- Define an operating system and explain its purpose
- Describe the main functions of an OS
- Navigate a graphical user interface (GUI)
- Compare different types of operating systems
- Understand basic OS processes: booting, multitasking, and shutdown

5.1 What is an Operating System?

An Operating System (OS) is the most important software on a computer. It acts as the manager and coordinator between the hardware and all other software. Without an OS, you could not run any program, access any file, or use any device.

The Orchestra Analogy

Think of the OS as the conductor of an orchestra — it does not play an instrument itself, but it makes sure every instrument (component) plays at the right time, at the right volume, in perfect coordination.

5.2 Functions of an Operating System

5.2.1 Process Management

A process is a program that is currently running. The OS manages multiple processes at the same time by allocating CPU time to each. This is called multitasking.

- Scheduler: Decides which process gets CPU time and for how long.
- Context Switching: Saves a process's state when it pauses and restores it when it resumes.
- Process States: New → Ready → Running → Waiting → Terminated.

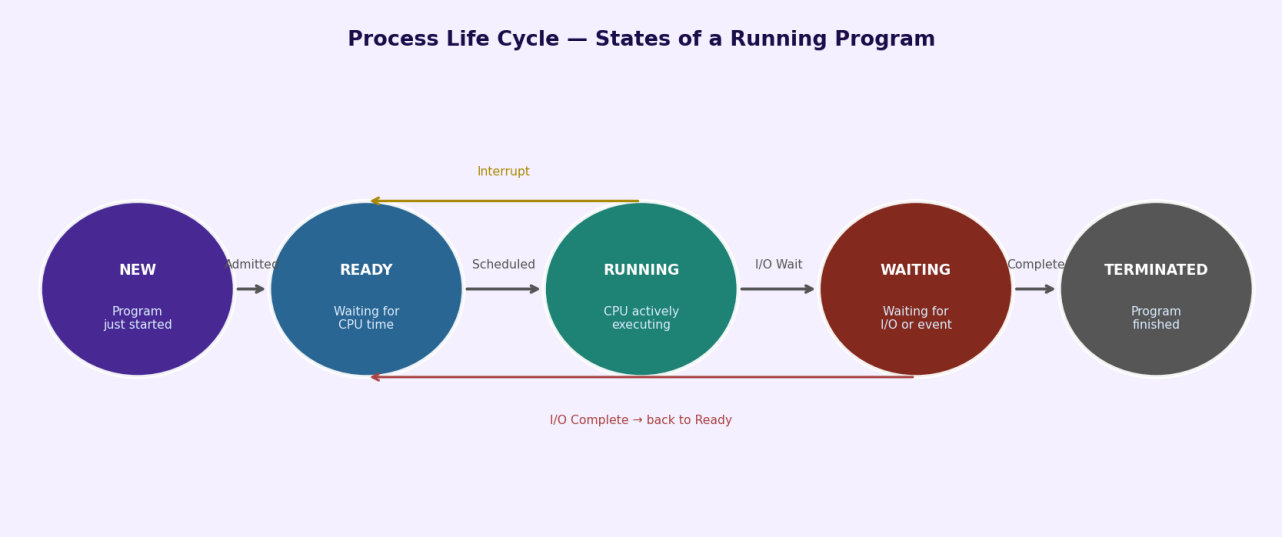


Figure 5.1 — Process Life Cycle: States of a Running Program

5.2.2 Memory Management

The OS decides how RAM is allocated to running programs. It ensures no program takes memory belonging to another, and reclaims memory when programs close.

- Virtual Memory: When RAM is full, the OS uses part of the hard disk as temporary RAM. Slower but prevents crashes.
- Memory Protection: Prevents one program from corrupting another's memory space.

5.2.3 File System Management

The OS organises data on storage devices using a file system. It tracks where every file is stored, controls who can access it, and manages reading and writing.

File System	Used By
NTFS (New Technology File System)	Windows — supports large files, permissions, encryption.
FAT32 (File Allocation Table)	USB drives, older systems — widely compatible.
exFAT	USB drives and SD cards — supports files larger than 4 GB.
ext4	Linux — standard Linux file system.
APFS (Apple File System)	macOS and iOS — optimised for flash storage.

5.2.4 Device Management

The OS manages communication with all connected hardware via device drivers. When you plug in a USB device, the OS detects it, loads the appropriate driver, and makes the device available for use.

5.2.5 User Interface

The OS provides an interface for users to interact with the computer. There are two main types: GUI and CLI.

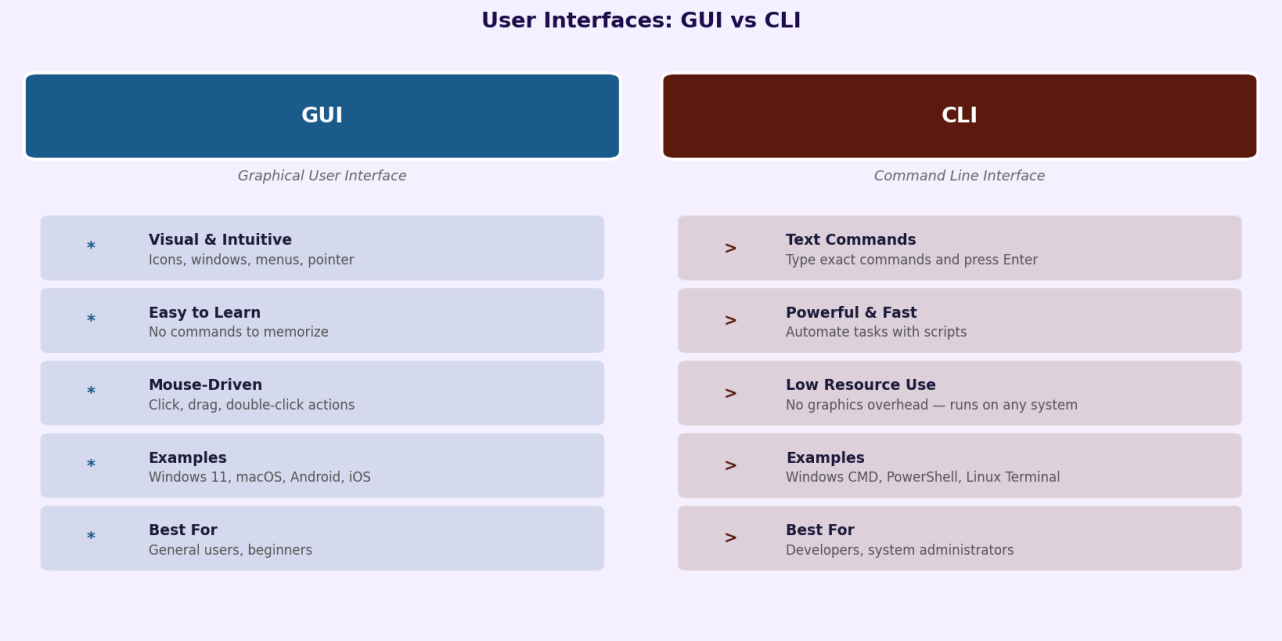


Figure 5.2 — GUI vs CLI: Two Ways to Interact with the Operating System

- GUI (Graphical User Interface): Uses windows, icons, menus, and a pointer. Easy and intuitive. Examples: Windows, macOS.
- CLI (Command Line Interface): User types text commands. Powerful and efficient. Examples: Windows Command Prompt, Linux Terminal.

5.2.6 Security and Access Control

The OS manages user accounts and controls access to the system and files. Features include:

Security Feature	Description
User Accounts	Each user has a separate profile and permissions.

Security Feature	Description
Password Protection	Controls who can log in to the computer.
File Permissions	Controls who can read, write, or execute files.
Firewall	Built-in network security to block unauthorised access.

5.3 Booting the Computer

Booting is the process of starting up a computer. When you press the power button a precise sequence of events occurs before you can use the computer.



Figure 5.3 — The Boot Process: Step-by-Step from Power On to Ready

Cold Boot vs Warm Boot

Cold Boot: Starting a computer that is completely off — goes through the full POST and OS load.

Warm Boot (Restart): Restarting a computer that is already on — used to apply updates or fix

problems without a full power-off.

5.4 The Desktop Environment

When you log into a GUI operating system, you see the desktop. Key elements of the desktop include:

Element	Description
Desktop	The main work area — shows icons and wallpaper.
Taskbar / Dock	Bar at the bottom (Windows) or side (macOS) for running apps and notifications.
Start Menu / Launcher	Opens programs, settings, and file explorer.
Icons	Small pictures that represent programs, files, or folders.
Window	A rectangular area showing a running program. Can be resized, minimised, or closed.
System Tray	Bottom-right area showing clock, volume, Wi-Fi, and background programs.
File Explorer / Finder	Tool to browse and manage files and folders.

5.5 Multitasking and Task Management

Modern operating systems allow multitasking — running several programs simultaneously. The CPU switches between tasks so quickly that it appears everything runs at the same time.

Task Manager (Windows) / Activity Monitor (macOS) shows all running processes and their resource usage. You can use it to:

- See CPU and RAM usage in real time.
- End (force quit) a frozen or unresponsive program.
- Monitor startup programs and network activity.

5.6 Comparing Operating Systems

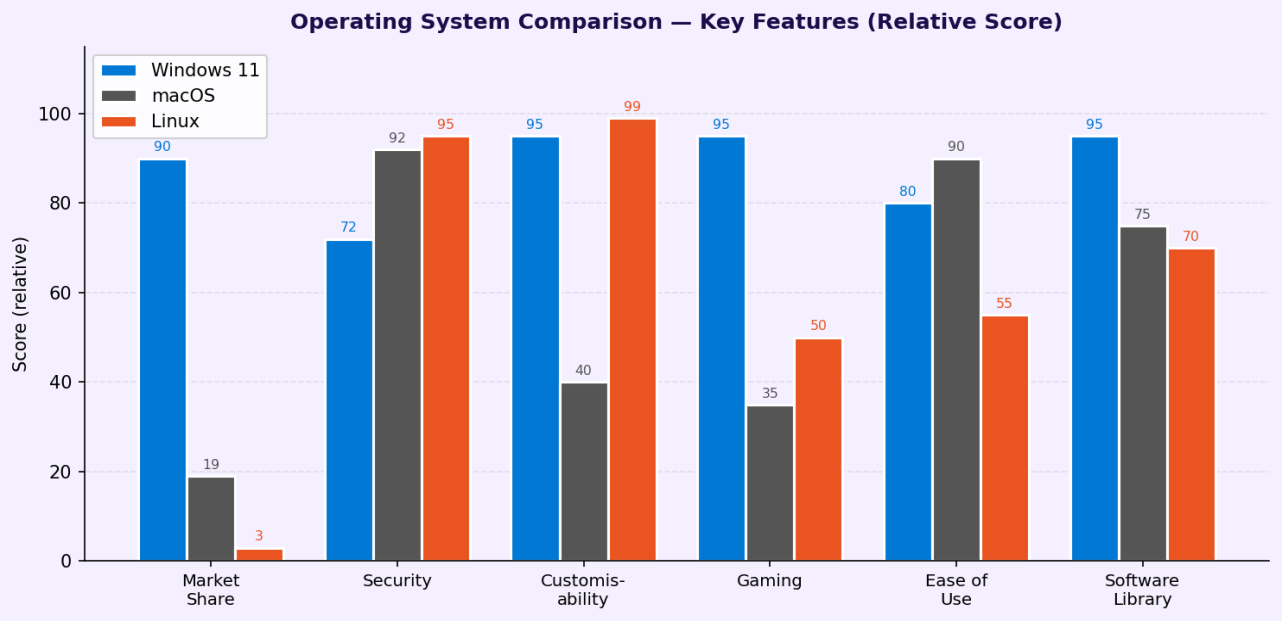


Figure 5.4 — Operating System Feature Comparison: Windows, macOS, and Linux

Feature	Windows 11	macOS	Linux
Developer	Microsoft	Apple	Open Source Community
Cost	Paid (included with PC)	Free (Mac hardware only)	Free
Hardware	Works on most PCs	Only on Apple hardware	Works on most hardware
Market Share	~70% of desktops	~15% of desktops	~5% of desktops
Gaming	Excellent	Limited	Growing
Security	Good (with updates)	Excellent	Very strong
Customisability	Very high	Limited	Fully customisable

Chapter Review Questions

A. Short Answer

1. What is an operating system? Why is it important?
2. Explain what multitasking means and how the OS enables it.
3. What happens during the boot process? Describe at least four steps.
4. What is the difference between a GUI and a CLI?
5. What is virtual memory and why is it needed?

B. Fill in the Blanks

1. POST stands for _____.
2. The _____ file system is used by Windows and supports encryption.
3. A _____ boot starts a computer that is completely turned off.
4. The OS uses _____ memory when RAM is full, using part of the hard disk.
5. Task Manager shows all _____ currently running on the computer.

C. True or False

1. CLI is easier to learn than GUI for beginners. (True / False)
2. APFS is the file system used by macOS and iOS. (True / False)
3. An OS does not manage file access permissions. (True / False)
4. Virtual memory is slower than RAM. (True / False)
5. The OS is loaded into RAM from the hard disk during booting. (True / False)

D. Key Terms Glossary

Term	Definition
Operating System	Software that manages hardware, memory, files, and provides a user interface.
Process	A program that is currently executing.
Multitasking	The ability to run multiple programs at the same time.
GUI	Graphical User Interface — uses windows, icons, and menus.
CLI	Command Line Interface — user types text commands.
Boot	The process of starting up a computer.
File System	The method an OS uses to organise and track files on storage.
Virtual Memory	Disk space used as extra RAM when physical RAM is full.

— End of Chapter 5: Operating System Basics —

CHAPTER 6
File Management

Learning Objectives

By the end of this chapter, students will be able to:

- Understand how files and folders are organised on a computer
- Navigate and use File Explorer / Finder effectively
- Create, rename, move, copy, and delete files and folders
- Understand file types and extensions
- Use search tools to find files
- Maintain a clean and organised file system

6.1 What are Files and Folders?

A file is a named collection of data stored on a computer. Every document you create, every photo you save, every program you install — all are stored as files. Folders (also called directories) are containers that hold files and other folders, allowing you to organise your data in a structured way.

Real-Life Analogy: The Filing Cabinet

Think of your computer's storage like a filing cabinet. The cabinet = the hard drive. The drawers = top-level folders. The dividers inside each drawer = sub-folders. The individual papers = files.

6.2 File Extensions and Types

A file extension is the letters after the dot in a filename (e.g., report.docx — the extension is .docx). The extension tells the OS what type of file it is and which program should open it.



Figure 6.1 — Common File Extensions Grouped by Category

Extension	File Type and Program
.docx	Microsoft Word document
.xlsx	Microsoft Excel spreadsheet
.pptx	Microsoft PowerPoint presentation
.pdf	PDF document (Adobe Reader, browser)
.txt	Plain text file (Notepad)
.jpg / .jpeg	JPEG image file (photos)
.png	PNG image file (transparent backgrounds)
.mp3	MP3 audio file (music)
.mp4	MP4 video file

Extension	File Type and Program
.zip	Compressed archive of multiple files
.exe	Executable program file (Windows)
.py	Python source code file

6.3 Folder Structure and Paths

Files on a computer are organised in a hierarchical tree structure. At the top is the root (C:\ on Windows, / on Linux/macOS), then folders branch out from there.

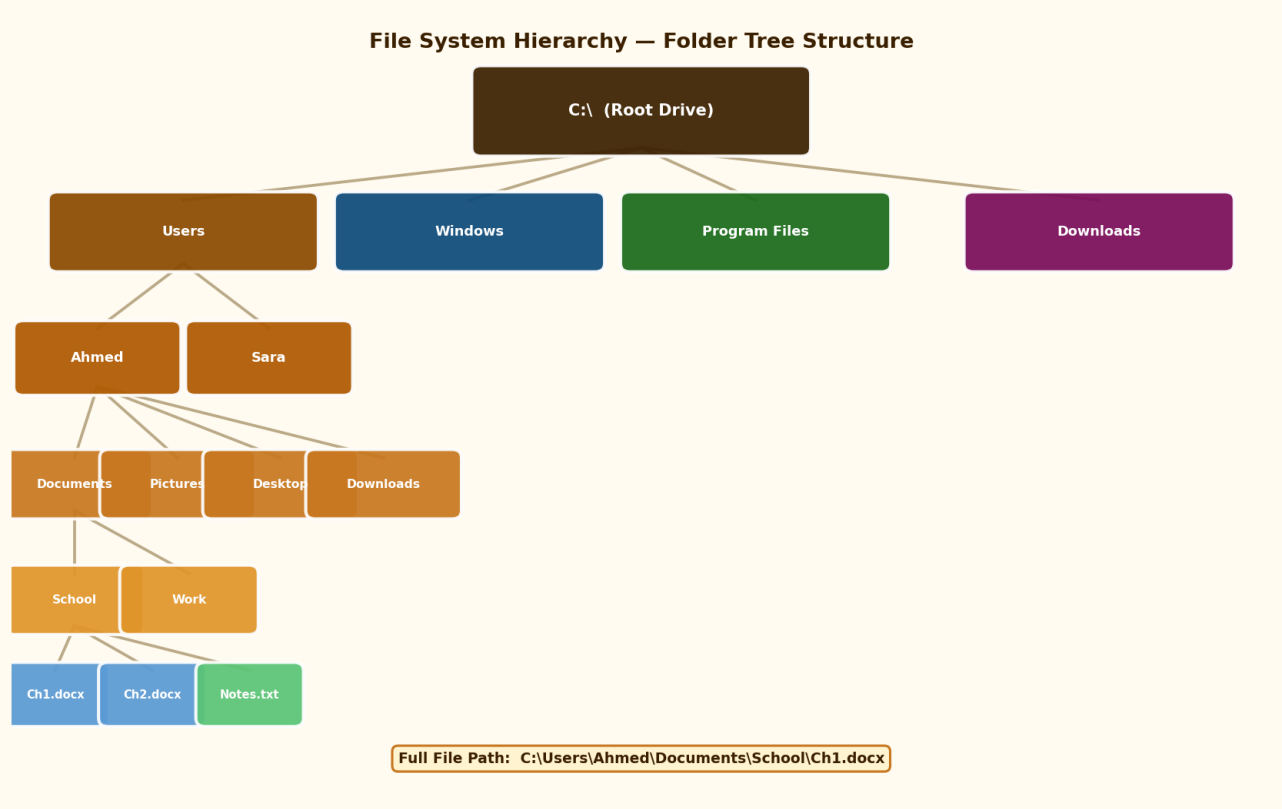


Figure 6.2 — File System Hierarchy: Folder Tree and Path Structure

Example File Path:

`C:\Users\Ahmed\Documents\School\Chapter1.docx`

This means: On drive C, inside Users, inside Ahmed, inside Documents, inside School, there is a file called Chapter1.docx.

Understanding paths is important because:

- Every file has a unique path (location address).
- Programs use paths to find files automatically.
- Shortcuts use paths to link to files or programs.

6.4 Using File Explorer (Windows)

File Explorer is the built-in file management tool in Windows. You can open it by pressing the Windows key + E or clicking the folder icon on the taskbar.

Area	Description
Navigation Pane (left)	Shows drives, Quick Access, and folder tree for fast navigation.
Address Bar (top)	Shows the current folder path. Click to type a path directly.
Search Box (top right)	Type a file name to search within the current folder.
Main Window	Shows the contents of the selected folder (files and sub-folders).
Ribbon / Toolbar	Buttons for common actions: New, Copy, Paste, Delete, Rename.
Status Bar (bottom)	Shows how many items are in the folder and how many are selected.

6.5 Essential File Operations

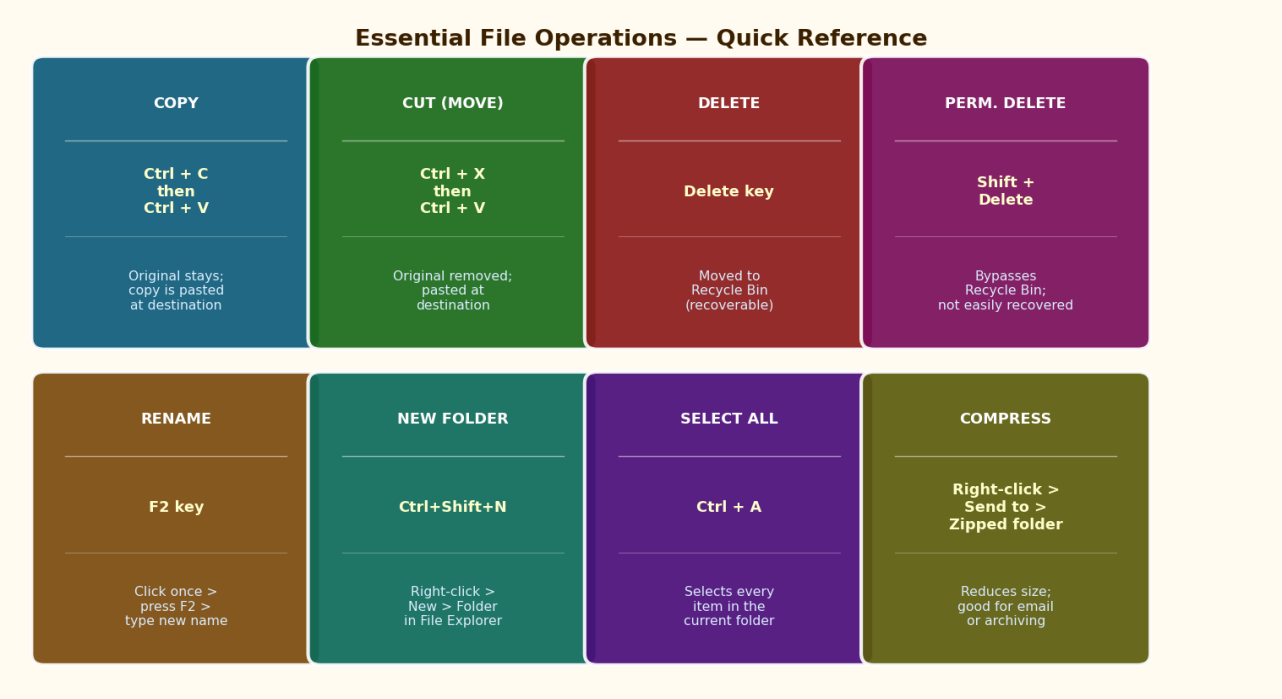


Figure 6.3 — Essential File Operations: Shortcuts and Actions

Creating Files and Folders

- New Folder: Right-click in File Explorer > New > Folder. Or press Ctrl+Shift+N.
- New File: Right-click > New > select file type (e.g., Text Document).
- From an App: Open Word, Excel, etc., and use File > Save As to choose location.

Selecting Files

Selection Type	Method
Single file	Click once on the file.
Multiple files (contiguous)	Click first file, hold Shift, click last file.
Multiple files (non-contiguous)	Hold Ctrl while clicking each individual file.
All files in folder	Press Ctrl+A.

Copying and Moving Files

Action	Methods
Copy	Ctrl+C — puts a copy in clipboard. Ctrl+V to paste in the new location.
Cut (Move)	Ctrl+X — removes from original. Ctrl+V to paste at destination.
Drag and Drop Copy	Hold Ctrl while dragging to a new folder on the same drive.
Drag and Drop Move	Drag without Ctrl to move (within the same drive).
Right-click Menu	Right-click > Copy or Cut > navigate to destination > Paste.

Renaming Files

- Click once to select, then press F2. Type the new name and press Enter.
- Or: Right-click the file > Rename.
- Important: Do not change the file extension when renaming (e.g., keep .docx).

Deleting Files

- Select the file > press Delete key. This moves it to the Recycle Bin.
- Permanent delete: Hold Shift and press Delete (bypasses Recycle Bin — cannot be recovered easily).
- Restore from Recycle Bin: Open Recycle Bin > right-click file > Restore.

6.6 Sorting and Viewing Files

You can control how files appear in File Explorer to make finding files easier:

View / Sort Option	Description
Large Icons	Shows big thumbnail previews — good for photos and images.
List	Shows file names in a compact vertical list.
Details	Shows name, date modified, type, and size — most useful for managing files.
Sort by Name	Alphabetical order A–Z or Z–A.
Sort by Date Modified	Most recently changed files first — useful for finding recent work.

View / Sort Option	Description
Sort by Size	Largest files first — useful for finding files taking up space.

6.7 Searching for Files

When you cannot remember where a file is saved, use the search function:

- In File Explorer: Click the search box (top right) and type the file name or part of it.
- Windows Search: Press the Windows key and start typing.
- Search by type: Type *.docx to find all Word documents in a folder.
- Search by date: Use the Date Modified filter in the search options.

6.8 Compressing and Extracting Files

Compression reduces file size by encoding data more efficiently. This is useful for sending files by email or saving space.

- To compress: Select files > right-click > Send to > Compressed (zipped) folder.
- To extract: Right-click the .zip file > Extract All > choose destination folder.
- Common formats: .zip (universal), .rar (more compression), .7z (open-source).

6.9 Good File Organisation Habits

Keeping your files well-organised saves time and prevents data loss. Follow these best practices:

Best Practices for File Organisation

✓ Use clear, descriptive file names (e.g., Science_Assignment_March.docx, not doc1.docx).

✓ Create a logical folder structure: School > Subjects > Year > Term.

✓ Save files in the correct folder immediately — do not leave everything in Downloads.

✓ Back up important files regularly to an external drive or cloud storage.

✓ Delete files you no longer need to free up storage space.

✓ Avoid special characters (?, *, :, /, \) in file names — they cause errors.

6.4a Using Finder (macOS)

Section 6.4 covers Windows File Explorer in detail. macOS users have an equivalent tool called Finder. Most concepts are identical · files, folders, paths, and operations all work the same way · but the interface and keyboard shortcuts differ.

The Finder Interface

Area	Description
Sidebar (left)	Shows Favourites, iCloud Drive, devices, and recent locations. Click to navigate quickly.
Toolbar (top)	Back/forward navigation, view options (Icons, List, Columns, Gallery), and search box.
Path Bar (bottom)	Shows the full folder path of the current location. Enable: View > Show Path Bar.
Status Bar (bottom)	Shows item count and available disk space. Enable: View > Show Status Bar.
Column View	Unique to macOS · shows folder hierarchy across columns. Excellent for navigating deep structures.

Windows vs macOS · Equivalent Operations

Operation	Windows	macOS
Open File Manager	Windows + E	Click Finder in Dock
Copy	Ctrl+C	Cmd+C
Cut (Move)	Ctrl+X then Ctrl+V	Cmd+C then Cmd+Option+V
Paste	Ctrl+V	Cmd+V
Rename	F2	Enter (press once)
Delete to Bin	Delete key	Cmd+Delete
Permanent delete	Shift+Delete	Cmd+Shift+Delete
Select all	Ctrl+A	Cmd+A
New Folder	Ctrl+Shift+N	Cmd+Shift+N
Search	Windows key + type	Cmd+F or Spotlight: Cmd+Space
File Properties	Right-click > Properties	Cmd+I (Get Info)
Show hidden files	View > Hidden items	Cmd+Shift+. (dot)

macOS Path · Understanding Finder Locations

macOS uses / as the path separator (same as Linux): /Users/Ahmed/Documents/School/
The macOS root is / (not C:\ like Windows). Your home folder is /Users/YourName/
In Finder, press Cmd+Shift+G to type a path directly and jump to any folder.

Chapter Review Questions

A. Short Answer

1. What is a file extension? Give three examples with their associated programs.
2. What is a file path? Write an example path and explain each part.
3. What is the difference between Cut and Copy?
4. What happens when you delete a file with the Delete key? How is this different from Shift+Delete?
5. Name four good file organisation habits.

B. Fill in the Blanks

1. The letters after the dot in a file name are called the file _____.
2. To select all files in a folder, press _____.
3. Pressing _____ and Delete permanently deletes a file without going to the Recycle Bin.
4. A _____ file reduces the size of multiple files into one archive.
5. To open File Explorer quickly, press _____.

C. True or False

1. Folders can contain other folders. (True / False)
2. You should change the extension when renaming a file to a new name. (True / False)
3. The .mp3 extension is used for video files. (True / False)
4. Sorting by "Date Modified" shows the most recently changed files first. (True / False)
5. Files deleted with Shift+Delete can be recovered from the Recycle Bin. (True / False)

D. Key Terms Glossary

Term	Definition
File	A named collection of data stored on a computer.
Folder / Directory	A container that organises files and other folders.
File Extension	Letters after the dot in a filename indicating the file type.
File Path	The full address showing where a file is stored (e.g., C:\Docs\file.docx).
Recycle Bin	A temporary holding area for deleted files before permanent removal.
File Compression	Reducing file size by encoding data more efficiently (e.g., .zip).
Root	The top level of a file system (C:\ on Windows, / on Linux).
Shortcut	A link that points to a file, folder, or program in another location.

— End of Chapter 6: File Management —

CHAPTER 7
Microsoft Word

Learning Objectives

By the end of this chapter, students will be able to:

- Navigate the Microsoft Word interface confidently
- Create, format, save, and print documents
- Use formatting tools: fonts, alignment, spacing, lists
- Insert tables, images, and headers/footers
- Use proofing tools: spell check and find & replace
- Apply styles and page formatting

7.1 What is Microsoft Word?

Microsoft Word is the world's most widely used word processing software, developed by Microsoft Corporation. It allows users to create, edit, format, and print text documents. Word documents are saved with the .docx file extension.

Word is used to create letters, reports, assignments, resumes, books, newsletters, and much more. It is part of the Microsoft Office / Microsoft 365 suite.

7.2 The Word Interface

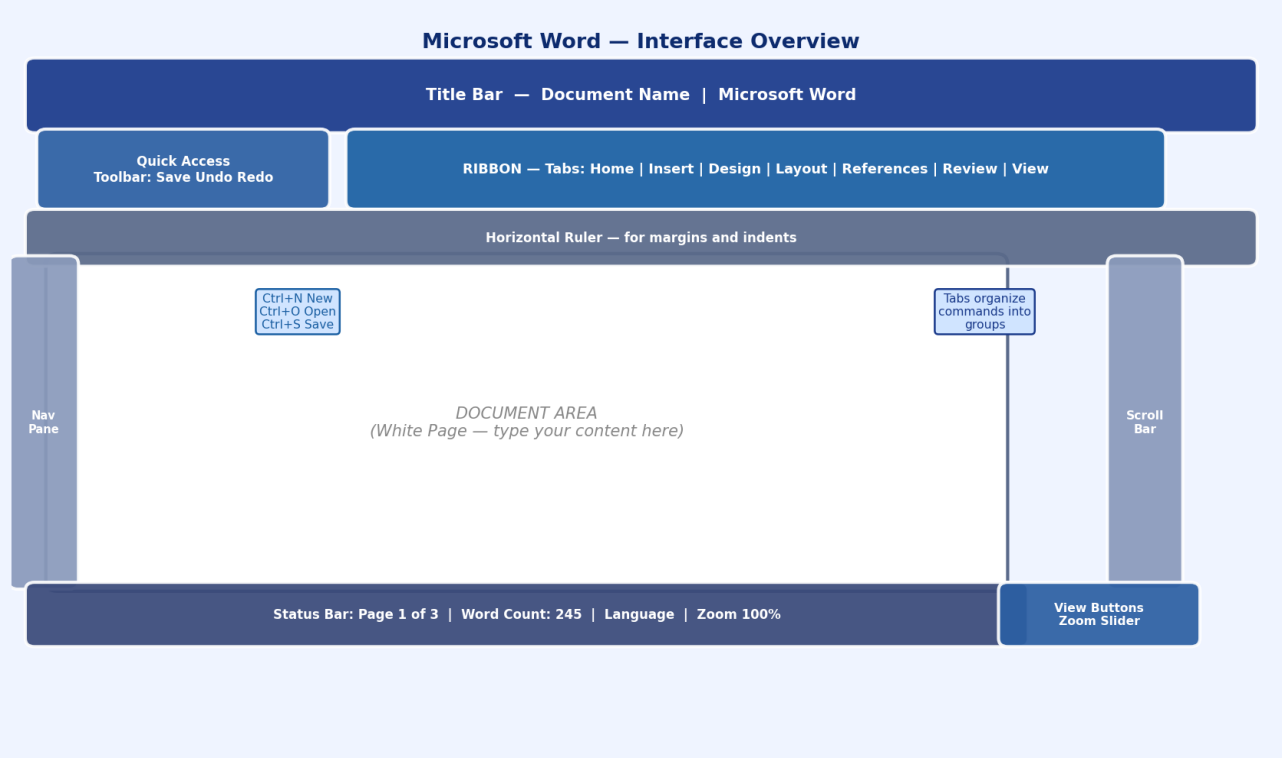


Figure 7.1 — Microsoft Word Interface Overview

Interface Element	Description
Title Bar	Shows the document name and application name (Microsoft Word).
Quick Access Toolbar	Top-left: customisable icons for Save, Undo, Redo.
Ribbon	The main toolbar divided into tabs with command groups.
Tabs	Home, Insert, Draw, Design, Layout, References, Mailings, Review, View.
Ruler	Horizontal and vertical rulers for setting margins and indents.
Document Area	The white page where you type your document.
Status Bar	Bottom: page number, word count, language, zoom level.
View Buttons	Bottom-right: Print Layout, Read Mode, Web Layout, Zoom slider.

7.3 Creating and Saving Documents

Creating a New Document

- Open Word — it shows the Start Screen with templates.
- Click Blank Document, or choose a template (letter, resume, etc.).
- Keyboard shortcut for new document: Ctrl+N.

Saving a Document

- First save: Press Ctrl+S or File > Save As. Choose location, enter file name, select file type (.docx).
- Save again: Ctrl+S (saves without asking). Save regularly to avoid data loss!
- Save as PDF: File > Save As > change file type to PDF.
- AutoSave: Available in Microsoft 365 when using OneDrive — saves every few seconds.

Opening an Existing Document

- Ctrl+O, or File > Open > browse to file location.
- Recent files appear in File > Home for quick access.

7.4 Basic Text Entry and Editing

Action	Method
Type text	Click in the document and start typing.
Select word	Double-click on the word.
Select paragraph	Triple-click in the paragraph.
Select all text	Ctrl+A
Undo / Redo	Ctrl+Z / Ctrl+Y
Cut / Copy / Paste	Ctrl+X / Ctrl+C / Ctrl+V
Find text	Ctrl+F
Find and Replace	Ctrl+H
Delete left / right	Backspace / Delete key

7.5 Formatting Text

7.5.1 Font Formatting (Home Tab)



Figure 7.2 — Essential MS Word Keyboard Shortcuts

Format	Shortcut / Location
Bold	Ctrl+B — makes text thick and dark.
Italic	Ctrl+I — slants text for emphasis.
Underline	Ctrl+U — adds a line under text.
Strikethrough	Home tab > Strikethrough button — draws a line through text.
Font Type	Home tab > Font dropdown (e.g., Arial, Times New Roman, Calibri).
Font Size	Home tab > Font Size box (e.g., 12, 14, 24 pt).
Font Colour	Home tab > Font Color button (A with colour bar).
Superscript	Ctrl+Shift+= (e.g., x^2)
Subscript	Ctrl+= (e.g., H_2O)
Clear Formatting	Ctrl+Space, or Home > Clear All Formatting button.

7.5.2 Paragraph Formatting

Format	Description
Align Left	Ctrl+L — text lines up with the left margin (default).
Centre	Ctrl+E — text is centred on the page.
Align Right	Ctrl+R — text lines up with the right margin.
Justify	Ctrl+J — text stretches to fill both margins evenly (like a newspaper).
Line Spacing	Home > Line and Paragraph Spacing. Common: 1.0, 1.5, 2.0.
Indentation	Tab key indents. Home > Increase/Decrease Indent buttons.

Format	Description
Bullets	Home > Bullets button — creates an unordered (dot) list.
Numbering	Home > Numbering button — creates a numbered list.

7.5.3 Styles

Styles are pre-set combinations of font, size, colour, and spacing. They make formatting consistent and fast. Use styles for headings to enable the automatic Table of Contents.

- Apply a style: Select text > Home > Styles gallery > click a style (Heading 1, Heading 2, Normal, etc.).
- Modify a style: Right-click a style > Modify > change the formatting.

7.6 Page Formatting

Page Layout (Layout Tab)

Setting	Description
Margins	Layout > Margins. Presets: Normal (1 inch all sides), Narrow, Wide.
Orientation	Layout > Orientation. Portrait (tall) or Landscape (wide).
Paper Size	Layout > Size. A4 (most countries), Letter (USA).
Columns	Layout > Columns — divide page into 2 or 3 columns (for newsletters).
Page Break	Insert > Page Break (Ctrl+Enter) — starts a new page.
Watermark	Design > Watermark — adds background text like DRAFT or CONFIDENTIAL.

Headers and Footers (Insert Tab)

Headers appear at the top of every page; footers at the bottom. They are used for titles, page numbers, dates, and company names.

- Insert > Header or Footer > choose a style.
- Page Number: Insert > Page Number > choose position.
- Edit header/footer: Double-click the header or footer area.
- Close header/footer: Double-click back in the main document.

7.7 Inserting Content

Tables

- Insert > Table > drag to select rows and columns, or Insert Table for exact numbers.
- Add row: Click in last cell > press Tab.
- Table Design tab: Change colours, borders, and shading.
- Table Layout tab: Add/delete rows and columns, merge cells, set cell size.

Images and Shapes

- Insert > Pictures > This Device (from your computer) or Online Pictures.
- Resize: Click image > drag corner handles.
- Text Wrapping: Right-click image > Wrap Text > choose In Line, Around, Behind, In Front.
- Shapes: Insert > Shapes > select shape, draw on document. Format with Shape Styles.

Feature	Description
SmartArt	Insert > SmartArt — creates visual diagrams, process flows, and org charts.
Chart	Insert > Chart — inserts bar, pie, line charts linked to data.
Symbols	Insert > Symbol — for special characters: ©, ™, °, Greek letters.
Hyperlink	Ctrl+K — links selected text to a website, email, or file.

7.8 Proofing Tools

Tool	How to Use
Spell Check	Review > Spelling & Grammar (F7). Red wavy underline = spelling error. Right-click for suggestions.
Grammar Check	Blue wavy underline = grammar suggestion. Right-click to accept or dismiss.
Word Count	Review > Word Count, or check status bar at bottom of screen.
Find & Replace	Ctrl+H. Type word to find, type replacement. Replace, Replace All, or Find Next.
Thesaurus	Right-click a word > Synonyms, or Review > Thesaurus.
Track Changes	Review > Track Changes — marks all edits in coloured text for collaboration.

7.9 Printing a Document

- File > Print (Ctrl+P).
- Check Print Preview to see how the document will look on paper.
- Select printer, number of copies, page range, and orientation.
- Click Print to send to printer.
- Print to PDF: Choose "Microsoft Print to PDF" as the printer — saves as a PDF file.

Chapter Review Questions

A. Short Answer

1. What is Microsoft Word used for? Name five types of documents you can create.
2. What is the Ribbon in Word? Name any four tabs on the Ribbon.
3. What is the difference between Save and Save As?
4. Explain the four paragraph alignment options: Left, Centre, Right, Justify.
5. What is a Header and Footer? When would you use them?

B. Keyboard Shortcut Match — Write the Action

Shortcut	Action
Ctrl+B	
Ctrl+Z	
Ctrl+P	
Ctrl+H	
Ctrl+A	
Ctrl+Enter	
F7	

C. True or False

1. Word documents are saved with the .docx extension by default. (True / False)
2. Justify alignment makes text align only to the left margin. (True / False)
3. Double-clicking a word selects the entire paragraph. (True / False)
4. AutoSave is only available when using OneDrive. (True / False)
5. Track Changes records edits so a collaborator can review them. (True / False)

D. Key Terms Glossary

Term	Definition
Word Processing	Creating, editing, and formatting text documents using software.
Ribbon	The tabbed toolbar in MS Office applications containing commands.
Style	A pre-set combination of font, size, and spacing applied to text.
Alignment	How text is positioned horizontally on the page.
Header / Footer	Text printed at the top/bottom of every page in a document.
Track Changes	A feature that records edits so they can be reviewed and accepted/rejected.
Find & Replace	A tool to locate specific text and optionally replace it with different text.
Watermark	Background text or image on a page (e.g., DRAFT, CONFIDENTIAL).

CHAPTER 8
Microsoft PowerPoint

Learning Objectives

By the end of this chapter, students will be able to:

- Understand the purpose of presentation software
- Navigate the PowerPoint interface
- Create and design slides with text, images, and shapes
- Add transitions and animations
- Use Slide Master for consistent design
- Deliver a presentation in Slide Show mode

8.1 What is Microsoft PowerPoint?

Microsoft PowerPoint is a presentation software that allows users to create slideshows made up of individual slides. Each slide can contain text, images, charts, videos, and animations. Presentations are used in classrooms, businesses, conferences, and events to communicate ideas visually.

PowerPoint files are saved with the .pptx extension. The software is part of the Microsoft Office / Microsoft 365 suite.

8.2 The PowerPoint Interface

Interface Element	Description
Title Bar	Shows presentation name and the PowerPoint application name.
Ribbon & Tabs	Home, Insert, Draw, Design, Transitions, Animations, Slide Show, Review, View.
Slide Panel (left)	Thumbnail view of all slides — click to navigate, drag to reorder.
Main Editing Area	The current slide — click to edit content.
Notes Panel (bottom)	Add speaker notes below each slide — visible to presenter only.
Status Bar	Shows slide number, theme, language, and view buttons.
View Options	Normal, Outline, Slide Sorter, Notes Page, Reading View, Slide Show.

8.3 Creating a New Presentation

- Open PowerPoint > New > Blank Presentation, or choose a theme/template.
- The first slide is a Title Slide — click placeholders to type title and subtitle.
- Add a new slide: Home > New Slide (Ctrl+M), or right-click in Slide Panel > New Slide.
- Choose a layout: Right-click slide > Layout (Title Slide, Title and Content, Two Content, Blank, etc.).

8.4 Working with Slides

Slide Layouts

Layout	Best Used For
Title Slide	Opening slide — big title and subtitle.
Title and Content	Most common — title at top, large content area below.
Two Content	Compare two items side by side.
Blank	Complete freedom — add your own content anywhere.

Layout	Best Used For
Picture with Caption	Large photo with a description below.
Section Header	Divides the presentation into labelled sections.

Adding, Deleting, and Reordering Slides

- Add: Ctrl+M or Home > New Slide.
- Duplicate: Right-click slide > Duplicate Slide.
- Delete: Right-click slide > Delete Slide, or select + press Delete key.
- Reorder: Drag slides in the Slide Panel to change their order.

8.5 Adding and Formatting Text

- Click inside any text placeholder and start typing.
- To add a new text box: Insert > Text Box > draw on the slide.
- Format text: Use Home tab — Font, Size, Bold, Italic, Colour, Alignment.
- Use bullet points for lists: Home > Bullets or Numbering.

The 6x6 Presentation Rule

Keep text minimal on slides — use large fonts (minimum 24pt). Slides should support what you say, not replace your speech.

Use the 6x6 Rule: no more than 6 bullet points per slide, and no more than 6 words per bullet.

8.6 Inserting Objects

Object	How to Insert
Image	Insert > Pictures > from computer or online. Resize and position freely.
Shape	Insert > Shapes > draw on slide. Right-click to add text inside.
SmartArt	Insert > SmartArt — visual diagrams, timelines, cycles, and hierarchy charts.
Chart	Insert > Chart — choose type (bar, pie, line). Edit data in the built-in spreadsheet.
Table	Insert > Table — select rows and columns. Format with Table Design tab.
Video	Insert > Video — embed local video or link to online video.
Audio	Insert > Audio — embed sound clips or record audio directly.
Icons	Insert > Icons — choose from a large library of vector icons.

8.7 Design and Themes

Applying a Theme

Themes apply a coordinated set of colours, fonts, and background to all slides instantly.

- Design tab > Themes — hover to preview, click to apply.
- Variants: Each theme has colour variants — shown on the right of the Themes gallery.
- Background: Design > Format Background — solid colour, gradient, picture, or texture.

Slide Master

The Slide Master controls the design of all slides in the presentation. Changes made in the Slide Master automatically apply to every slide, ensuring a consistent look.

- View > Slide Master to open.
- Edit the master slide to change fonts, logos, and backgrounds for the whole presentation.
- Close Master View when done.

8.8 Transitions & 8.9 Animations

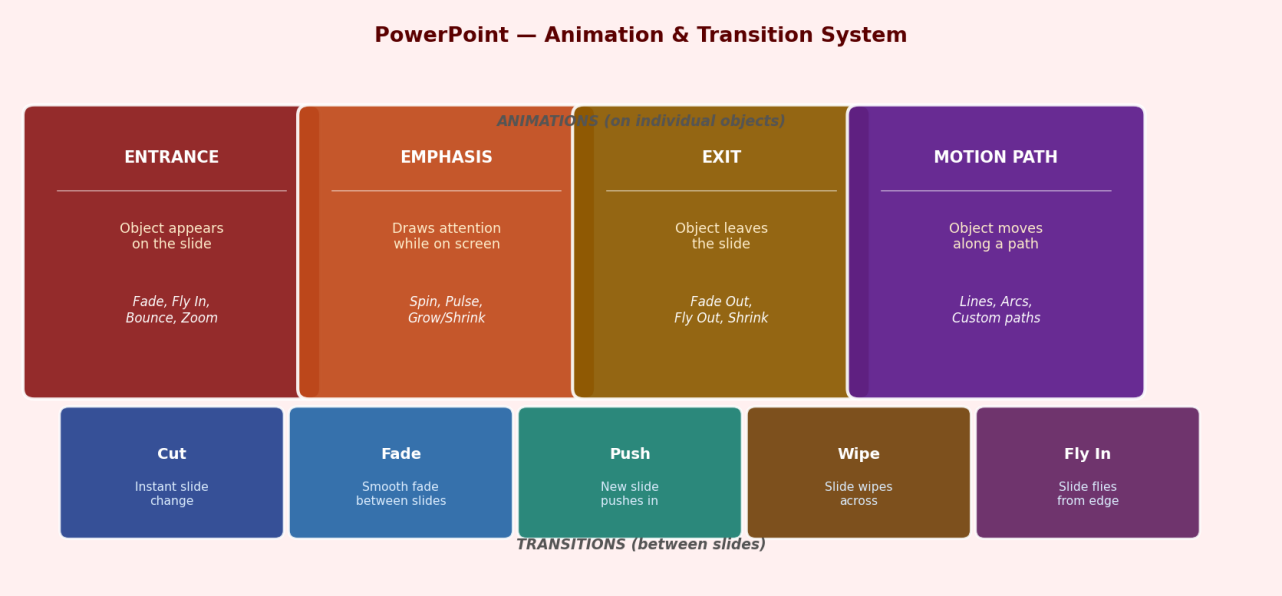


Figure 8.1 — Animation Types and Common Transitions

Transitions

Transitions are visual effects that occur when moving from one slide to the next.

- Select a slide > Transitions tab > choose an effect (Fade, Push, Wipe, Fly In, etc.).
- Duration: Set how long the transition takes. Apply to All: same transition for every slide.
- Advance Slide: Set to On Mouse Click, or automatically after a set time.

Animations

Animation Category	Description
Entrance	Object appears on the slide (e.g., Fade, Fly In, Bounce).

Animation Category	Description
Emphasis	Object draws attention while on screen (e.g., Spin, Pulse, Grow).
Exit	Object leaves the slide (e.g., Fade Out, Fly Out).
Motion Path	Object moves along a drawn path.

- Select object > Animations tab > choose animation.
- Animation Pane: Shows all animations on the slide, their order, and timing.
- Start options: On Click, With Previous, After Previous.

8.10 Slide Show — Presenting

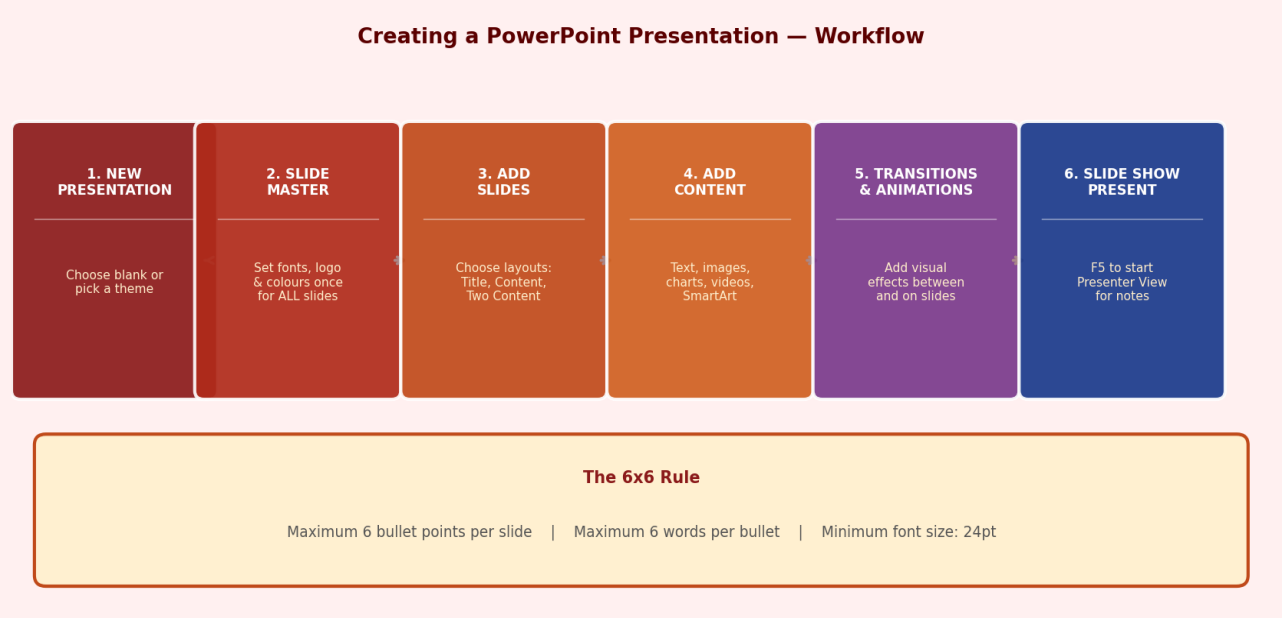


Figure 8.2 — PowerPoint Presentation Creation Workflow

Action	How To
Start from beginning	F5 or Slide Show tab > From Beginning.
Start from current slide	Shift+F5 or Slide Show > From Current Slide.
Next slide	Click, spacebar, or right arrow.
Previous slide	Backspace or left arrow.
End show	Esc key.
Presenter View	Slide Show > Use Presenter View — see notes and next slide on your screen.
Laser Pointer	Hold Ctrl and click during presentation to show laser dot.

8.11 Saving and Exporting

- Save as PPTX: Ctrl+S — editable PowerPoint format.

- Save as PDF: File > Export > Create PDF/XPS — for sharing a non-editable version.
- Save as Video: File > Export > Create a Video — useful for self-running presentations.
- PowerPoint Show (.ppsx): Opens directly in Slide Show mode without editing.

Chapter Review Questions

A. Short Answer

1. What is Microsoft PowerPoint used for? Name three settings where presentations are used.
2. What is the 6x6 rule in presentation design?
3. What is the difference between a transition and an animation?
4. What is the Slide Master and why is it useful?
5. How do you start a presentation from the first slide?

B. Fill in the Blanks

1. PowerPoint files are saved with the _____ extension.
2. The _____ panel on the left shows thumbnail previews of all slides.
3. _____ are effects that happen when moving from one slide to the next.
4. The _____ tab contains all the animation effects for individual objects.
5. Press _____ to end a slide show.

C. True or False

1. A Slide Master change affects only the currently selected slide. (True / False)
2. Presenter View shows speaker notes on the presenter's screen. (True / False)
3. PowerPoint can export presentations as video files. (True / False)
4. Exit animations make objects appear on the slide. (True / False)
5. The Notes panel at the bottom is visible to the audience during the show. (True / False)

D. Key Terms Glossary

Term	Definition
Presentation	A collection of slides used to visually communicate information.
Slide Layout	A pre-designed arrangement of placeholder areas on a slide.
Theme	A coordinated set of colours, fonts, and background applied to all slides.
Slide Master	A master template that controls the design of all slides.
Transition	A visual effect when moving from one slide to the next.
Animation	An effect applied to an individual object on a slide.
Presenter View	A display mode showing notes and next slide on the presenter's screen.
Placeholder	A pre-defined area on a slide layout for typing text or inserting content.

CHAPTER 9
Microsoft Excel

Learning Objectives

By the end of this chapter, students will be able to:

- Navigate the Excel spreadsheet interface
- Enter and format data in cells
- Write basic formulas and use common functions
- Sort and filter data
- Create charts from data
- Format a spreadsheet for printing

9.1 What is Microsoft Excel?

Microsoft Excel is a spreadsheet application used to organise, calculate, and analyse data. It is one of the most powerful and widely used business tools in the world. Excel is used for budgets, timetables, mark sheets, scientific data, financial analysis, and much more.

Excel files are saved as .xlsx. They are made up of workbooks (the file) and worksheets (individual sheets/tabs within the file).

9.2 The Excel Interface

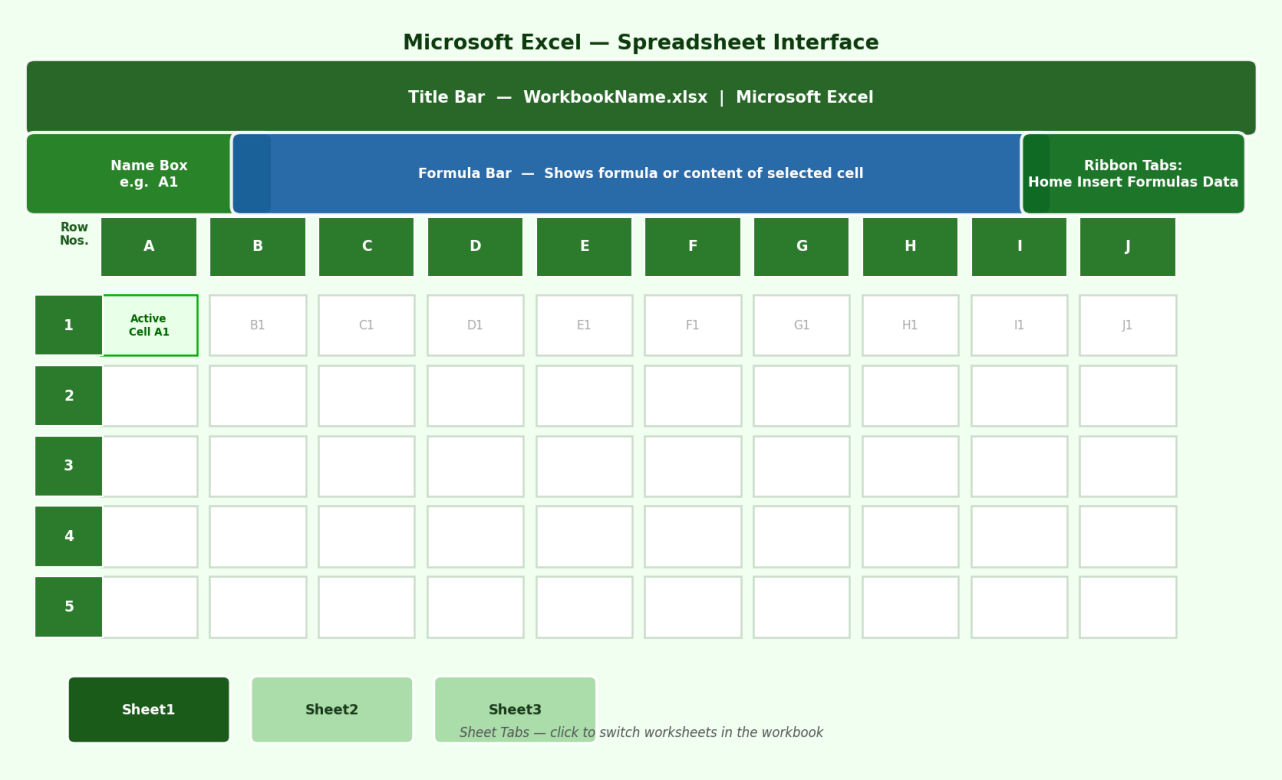


Figure 9.1 — Microsoft Excel Spreadsheet Interface

Element	Description
Title Bar	Shows the file name and Excel application name.
Name Box	Shows the address of the selected cell (e.g., A1, B3).
Formula Bar	Shows the content or formula of the selected cell.
Column Headers	Letters A, B, C... across the top — identify columns.
Row Headers	Numbers 1, 2, 3... down the left side — identify rows.
Active Cell	The currently selected cell — highlighted with a dark border.
Sheet Tabs	Bottom of screen — Sheet1, Sheet2, etc. Click to switch sheets.

9.3 Understanding Cells, Rows, and Columns

A cell is identified by its column letter followed by its row number. This is called a cell reference or cell address.

- Cell A1 = Column A, Row 1 (top-left cell).
- Cell C5 = Column C, Row 5.
- A range of cells: A1:C5 means all cells from A1 to C5.
- Entire column: Click the column letter (e.g., A) to select all of column A.
- Entire row: Click the row number (e.g., 5) to select all of row 5.

9.4 Entering Data

Data Type	Description
Text (Labels)	Words, names, headings. Aligned left by default. Example: "Name", "Subject".
Numbers (Values)	Numeric data used in calculations. Aligned right by default. Example: 85, 1200.50.
Dates	Excel recognises dates (e.g., 01/01/2024) and can perform date calculations.
Formulas	Start with = sign. Example: =A1+B1, =SUM(A1:A10).

- Type in a cell and press Enter (move down) or Tab (move right).
- Edit a cell: Press F2 or double-click the cell.
- Clear a cell: Press Delete key (removes content but keeps formatting).

9.5 Formatting Cells

9.5.1 Text Formatting

Select cells and use the Home tab for basic text formatting: Font, Size, Bold (Ctrl+B), Italic (Ctrl+I), Underline, Font Colour, and Fill Colour (background colour).

9.5.2 Number Formats

Format	Example
General	12345 (no special formatting)
Number	12,345.00 (with thousands separator and decimals)
Currency	$1,234.50 or £1,234.50
Percentage	85%
Date	01/01/2024 or 1 January 2024
Text	Treats number as text — no calculations

9.5.3 Cell Alignment and Merge

- Alignment: Home > Alignment group — Left, Centre, Right, Top, Middle, Bottom.
- Wrap Text: Home > Wrap Text — makes text visible on multiple lines in one cell.
- Merge & Centre: Select cells > Home > Merge & Centre — joins cells into one. Used for headings.

9.5.4 Borders and Fill

- Borders: Home > Borders dropdown — All Borders, Thick Box Border, etc.
- Fill Colour: Home > Fill Color (paint bucket) — adds background colour to cells.

9.6 Formulas and Functions

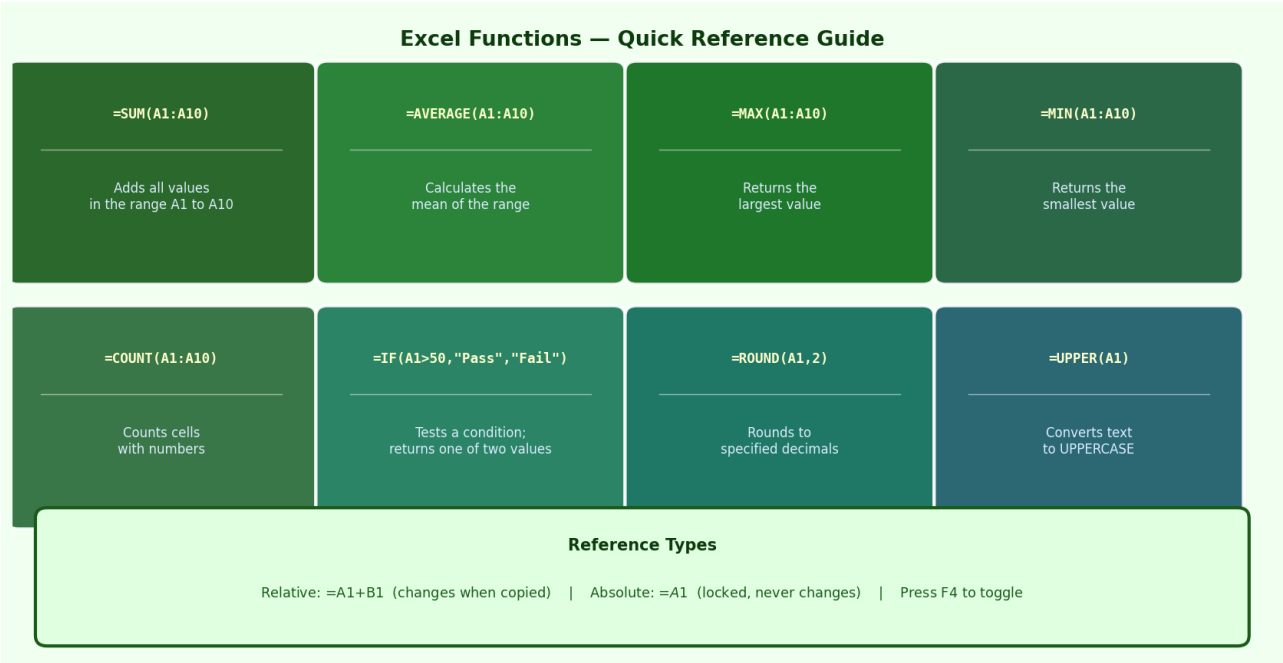


Figure 9.2 — Excel Functions Quick Reference

9.6.1 Basic Arithmetic Formulas

Formula	What it Does
=A1+B1	Adds the values in A1 and B1.
=A1-B1	Subtracts B1 from A1.
=A1*B1	Multiplies A1 by B1.
=A1/B1	Divides A1 by B1.
=(A1+B1)/2	Average of two cells using brackets for order of operations.

9.6.2 Common Functions

Function	Description and Example
=SUM(A1:A10)	Adds all values in cells A1 through A10.
=AVERAGE(A1:A10)	Calculates the arithmetic mean of A1:A10.
=MAX(A1:A10)	Returns the largest value in the range.
=MIN(A1:A10)	Returns the smallest value in the range.
=COUNT(A1:A10)	Counts how many cells contain numbers.
=IF(A1>50,"Pass","Fail")	If A1 is greater than 50, shows Pass; otherwise shows Fail.
=ROUND(A1,2)	Rounds the value in A1 to 2 decimal places.

Function	Description and Example
=UPPER(A1)	Converts text in A1 to UPPERCASE.
=LEN(A1)	Returns the number of characters in A1.

9.6.3 Absolute vs Relative References

Reference Types — Essential to Understand

Relative Reference: =A1+B1 — when copied down, becomes =A2+B2, =A3+B3, etc.

Absolute Reference: =A1 — the $ sign locks the reference. It does not change when copied.

Mixed Reference: =$A1 (column locked) or =A$1 (row locked). Press F4 to cycle through types.

9.7 Sorting and Filtering Data

Sorting

- Select any cell in the data > Data > Sort A to Z (ascending) or Z to A (descending).
- Custom Sort: Data > Sort — choose column to sort by, then add secondary sort criteria.

Filtering

- Select data > Data > Filter — adds dropdown arrows to each column header.
- Click a dropdown > choose values to show or hide.
- Clear filter: Data > Clear, or click dropdown > Select All.

9.8 Creating Charts

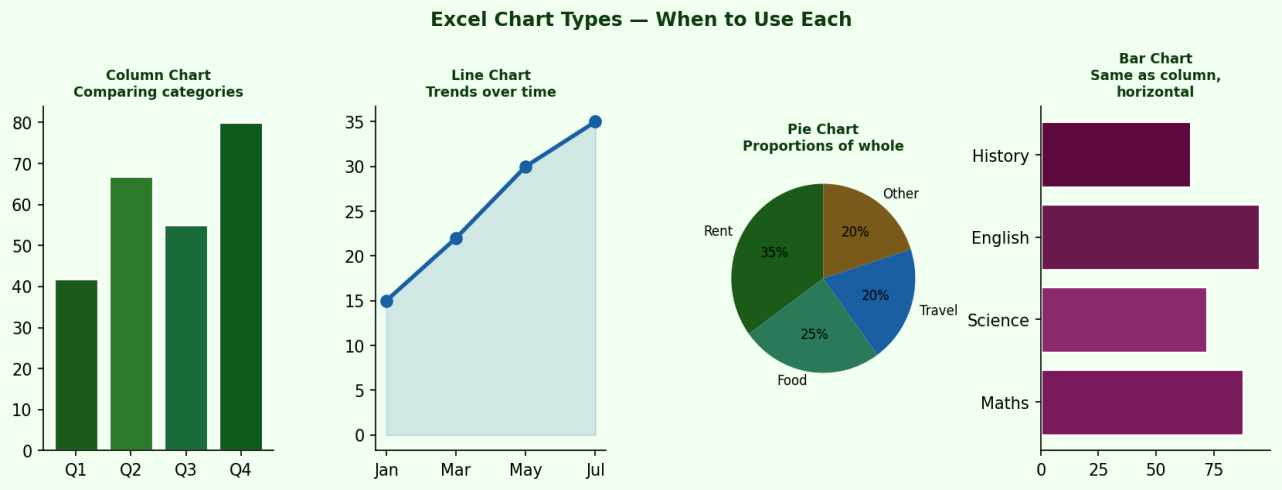


Figure 9.3 — Excel Chart Types and When to Use Each

Charts visually represent data, making it easier to see trends and comparisons.

- Select the data you want to chart (including headings).
- Insert > Charts > choose chart type (Column, Bar, Line, Pie, etc.).
- Chart Elements button (+ icon): Toggle Title, Data Labels, Legend, Gridlines.

Chart Type	Best Used For
Column / Bar Chart	Comparing values across categories (e.g., sales by region).
Line Chart	Showing trends over time (e.g., monthly temperatures).
Pie Chart	Showing proportions of a whole (e.g., budget breakdown).
Scatter Chart	Showing relationship between two numerical variables.

9.9 Printing a Spreadsheet

- File > Print (Ctrl+P) — preview appears on the right.
- Page Layout tab > Orientation (Portrait/Landscape), Size, Print Area, Margins.
- Scale to Fit: Page Layout > Scale to Fit — fit sheet to one page wide or tall.
- Print Titles: Page Layout > Print Titles — repeat row/column headers on each printed page.
- Freeze Panes: View > Freeze Panes — keeps headers visible when scrolling large sheets.

Chapter Review Questions

A. Short Answer

1. What is the difference between a workbook and a worksheet in Excel?
2. What is a cell reference? Give two examples.
3. What is the difference between a formula and a function?
4. Explain the difference between a relative and an absolute cell reference.
5. Name four types of charts and describe when to use each.

B. Write the Formula

Task	Formula
Add all values from B2 to B15	
Find the highest mark in C2:C30	
Calculate the average of D1:D10	
If E5 > 40 show "Pass" else "Fail"	
Round the value in F3 to 1 decimal place	

C. True or False

1. The Name Box shows the content of the selected cell. (True / False)
2. A1 is an absolute cell reference. (True / False)
3. A pie chart is best for showing trends over time. (True / False)
4. Wrap Text makes text visible on multiple lines within one cell. (True / False)
5. Merge & Centre joins multiple selected cells into a single cell. (True / False)

D. Key Terms Glossary

Term	Definition
Workbook	An Excel file — can contain multiple worksheets.
Cell Reference	The address of a cell (column letter + row number, e.g., B3).
Formula	A user-written calculation starting with = (e.g., =A1+B1).
Function	A pre-built formula (e.g., =SUM(), =AVERAGE(), =IF()).
Absolute Reference	A cell reference locked with $ so it does not change when copied.
Sort	Arranging data in ascending or descending order.
Filter	Showing only rows that meet specified criteria.
Chart	A visual representation of spreadsheet data.

CHAPTER 10
Internet Basics

Learning Objectives

By the end of this chapter, students will be able to:

- Define the Internet and explain how it works
- Understand key internet concepts: IP address, DNS, URL, HTTP
- Use a web browser effectively
- Search for information online efficiently
- Understand types of internet connections
- Describe common internet services

10.1 What is the Internet?

The Internet is a global network of billions of computers and devices connected together, allowing them to share information and communicate. It was originally developed in the 1960s by the US military (ARPANET) and opened to the public in the early 1990s.

The Internet works by breaking data into small pieces called packets, sending them through various routes, and reassembling them at the destination. This is called packet switching.

Internet vs World Wide Web — Not the Same Thing!

The Internet: The physical global network of cables, routers, and connections.

The World Wide Web (WWW): A service that runs ON the Internet — a collection of web pages accessed through a browser. Email, online gaming, and video streaming also use the Internet but are not part of the Web.

10.2 How the Internet Works

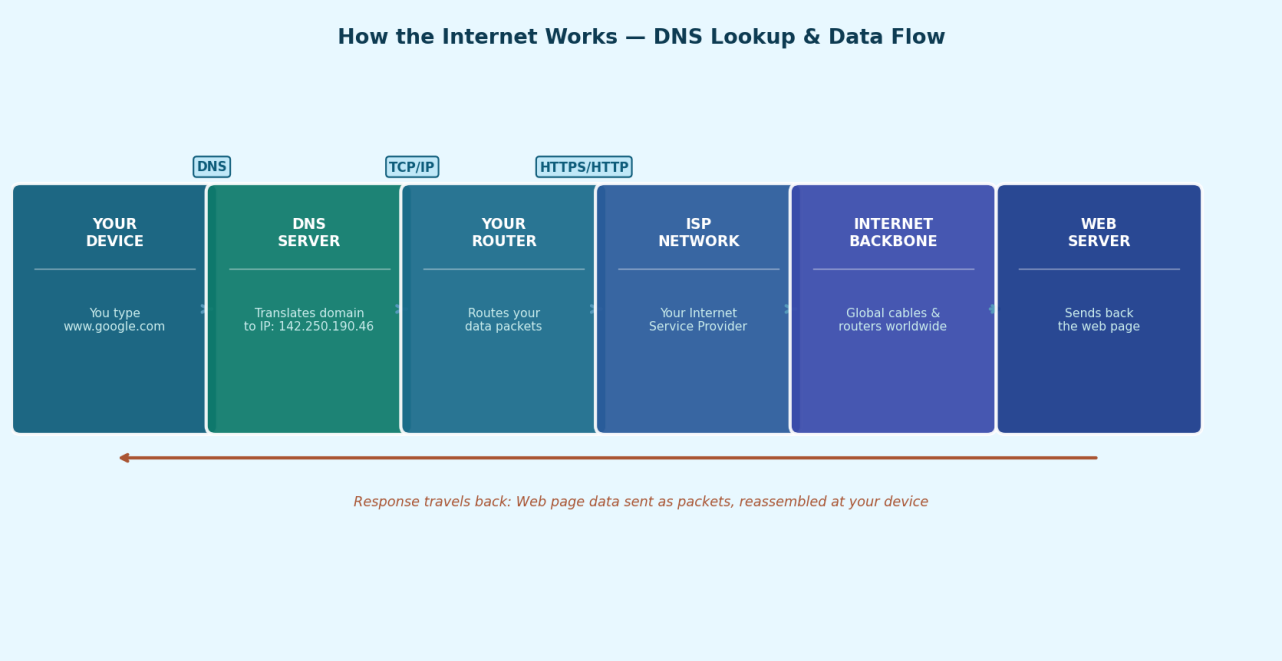


Figure 10.1 — How the Internet Works: DNS Lookup and Data Flow

10.2.1 IP Addresses

Every device on the internet has a unique IP address (Internet Protocol address) — like a postal address for computers.

- IPv4: Four groups of numbers (e.g., 192.168.1.1) — over 4 billion addresses.
- IPv6: Newer format with 8 groups of hexadecimal (e.g., 2001:0db8::1) — nearly unlimited addresses.
- Your home router has a public IP address. Devices in your home have private IP addresses.

10.2.2 DNS (Domain Name System)

Humans use domain names (e.g., www.google.com) instead of IP addresses. The DNS translates domain names into IP addresses automatically — like a phone book for the internet.

10.2.3 Protocols

Protocol	Purpose
HTTP	HyperText Transfer Protocol — transfers web pages from server to browser.
HTTPS	Secure HTTP — encrypts data between browser and server. Look for padlock icon.
FTP	File Transfer Protocol — transfers files between computers.
SMTP / POP3 / IMAP	Email protocols — sending and receiving messages.
TCP/IP	The foundational protocol suite of the internet — controls data packet routing.

10.3 Understanding URLs

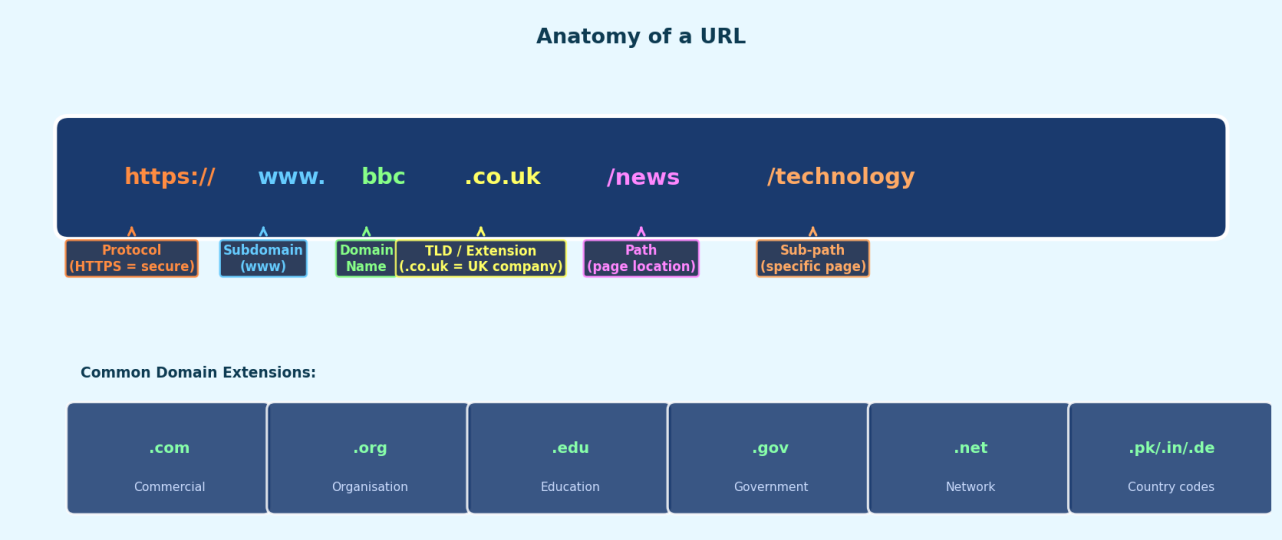


Figure 10.2 — Anatomy of a URL: Every Part Explained

Extension	Meaning
.com	Commercial — most common, used worldwide.
.org	Organisation — charities and non-profits.
.edu	Educational institution.
.gov	Government website.
.net	Network / internet organisations.
.co.uk	UK-based company.
.pk / .in / .de	Country-code TLDs for Pakistan, India, Germany, etc.

10.4 Web Browsers

A web browser is software used to access and display websites. It sends requests to web servers and renders (displays) the HTML/CSS/JavaScript it receives.

Browser	Developer
Google Chrome	Google — most widely used (~65% market share).
Mozilla Firefox	Mozilla Foundation — open source, privacy-focused.
Microsoft Edge	Microsoft — built into Windows, fast and modern.
Safari	Apple — default on Macs and iPhones.
Opera	Opera Software — has built-in VPN and ad blocker.

Feature	Description
Address Bar	Type a URL or search query here.
Tabs	Open multiple websites simultaneously in separate tabs.
Bookmarks / Favourites	Save links to frequently visited sites.
History	List of previously visited pages.
Incognito / Private Mode	Browse without saving history, cookies, or form data.
Extensions	Add-ons that add features (e.g., ad blocker, password manager).

10.5 Searching the Internet

Technique	Example
Use specific keywords	Type "photosynthesis definition" instead of "how do plants work"
Use quotes for exact phrases	"climate change causes" — finds that exact phrase
Exclude words with minus	jaguar -car — finds jaguar animal, not the car
Search specific site	site:bbc.co.uk climate change
Search by file type	photosynthesis filetype:pdf
Search by date range	Use Tools > Any time > Past year for recent results

10.6 Types of Internet Connections

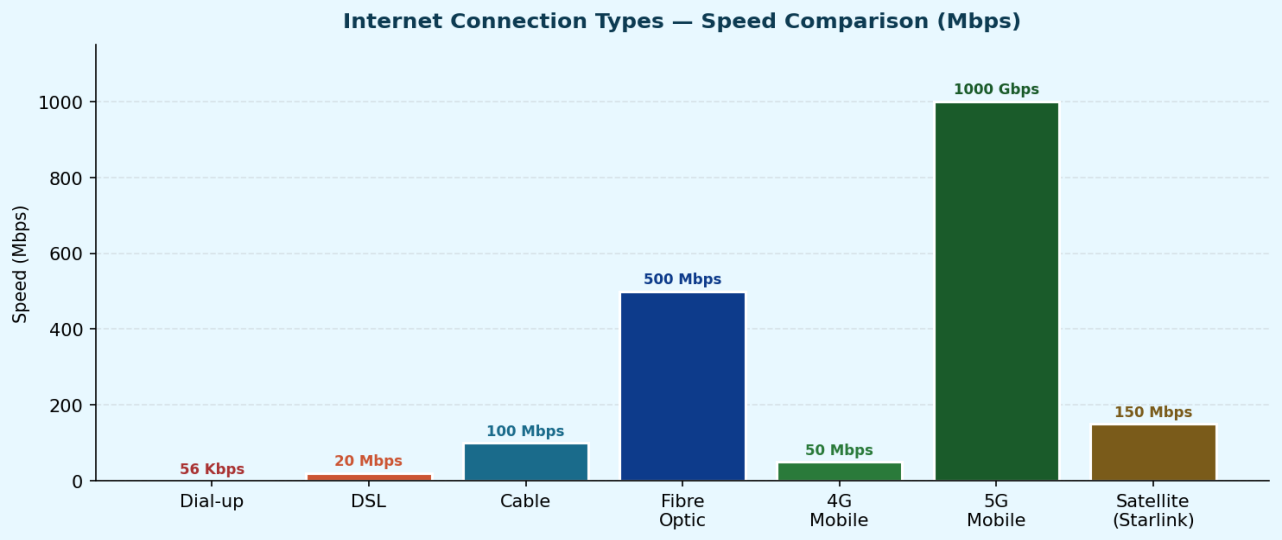


Figure 10.3 — Internet Connection Types: Speed Comparison

Connection Type	Description
Dial-up	Uses phone line. Very slow (56 Kbps). Mostly historical.
DSL (ADSL)	Digital Subscriber Line — uses phone line but much faster than dial-up.
Cable Broadband	Uses TV cable infrastructure. Fast and common in homes.
Fibre Optic	Uses glass cables and light pulses. Very fast (100 Mbps – 1 Gbps+).
Wi-Fi	Wireless connection within a local area (home, cafe, school).
4G / 5G Mobile	Cellular mobile internet. 5G offers very high speeds wirelessly.
Satellite	Internet via orbiting satellites. Available in remote areas (e.g., Starlink).

10.7 Common Internet Services

Service	Description
World Wide Web	Accessing websites through a browser.
Email	Sending and receiving electronic messages.
Streaming	Watching video (YouTube, Netflix) or listening to music in real time.
Social Media	Sharing content and connecting with others (Facebook, Instagram, X).
Cloud Services	Storing and accessing files online (Google Drive, Dropbox).
VoIP Calls	Voice or video calls over the internet (WhatsApp, Zoom, Skype).
Online Banking	Managing finances through a bank's secure

Service	Description
	website or app.

10.7a Virtual Private Networks (VPNs)

A Virtual Private Network (VPN) is a security tool that creates an encrypted, private tunnel between your device and a VPN server before your traffic reaches the wider internet. It is an important internet service that every digital citizen should understand.

How a VPN Works	What a VPN Protects Against
1. Your device connects to a VPN server (encrypted). 2. VPN server makes requests on your behalf. 3. Websites see the VPN server's IP, not yours. 4. All traffic between you and VPN is encrypted.	· Snooping on public Wi-Fi (cafes, airports, hotels). · Your ISP seeing your browsing history. · Geographic restrictions on content. · Basic tracking by websites (IP-based).

VPN in the Context of Internet Services

Service	Description
VPN	Routes all traffic through an encrypted server, hiding your IP address and protecting your data on public networks. Popular services: NordVPN, ProtonVPN, Mullvad.

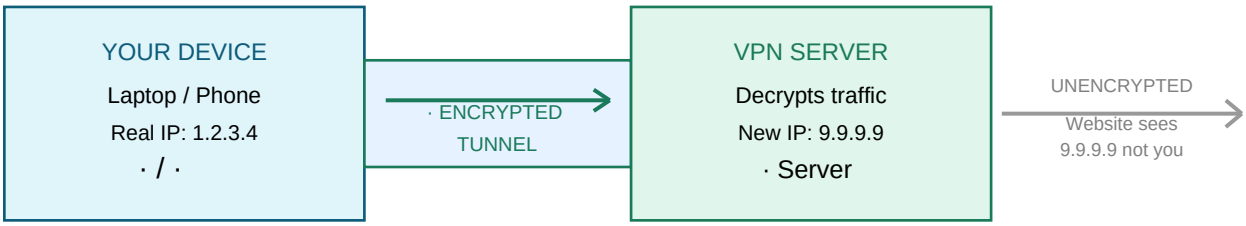
Important · VPN Limitations

A VPN does NOT make you completely anonymous online. Your VPN provider can still see your traffic. Always choose a trusted, no-logs VPN provider and understand that VPNs are a privacy tool, not a shield for illegal activity.

Key Terms · Added

Term	Definition
VPN	Virtual Private Network · encrypts internet traffic and masks your IP address.

How a VPN Tunnel Works · Visual Diagram



What This Means For Your Privacy:

- Your ISP can NOT see which websites you visit (only encrypted tunnel to VPN)
- Public Wi-Fi eavesdroppers see only encrypted data · useless to them
- Websites see the VPN server's IP, not your real location
- Your VPN provider CAN still see your traffic · choose a trusted no-logs provider

Chapter Review Questions

A. Short Answer

1. What is the Internet? How is it different from the World Wide Web?
2. What is an IP address? What is the role of DNS?
3. What is a URL? Break down this URL and explain each part: https://www.nasa.gov/science
4. What is HTTPS and why is it more secure than HTTP?
5. Name four types of internet connections and briefly describe each.

B. Fill in the Blanks

1. The Internet was originally developed as a military network called _____.
2. _____ translates domain names like google.com into IP addresses.
3. A web browser sends requests using _____ or _____ protocol.
4. The _____ provides your home or business with internet access.
5. Private browsing mode does not save _____, cookies, or form data.

C. True or False

1. The Internet and the World Wide Web are exactly the same thing. (True / False)
2. HTTPS is more secure than HTTP because it encrypts data. (True / False)
3. IPv6 provides more addresses than IPv4. (True / False)
4. Fibre optic internet is generally slower than dial-up. (True / False)
5. A URL is the unique address of a webpage on the internet. (True / False)

D. Key Terms Glossary

Term	Definition
Internet	A global network of billions of connected computers and devices.
WWW	World Wide Web — a system of web pages accessed through a browser.
IP Address	A unique numerical address identifying a device on a network.
DNS	Domain Name System — translates domain names to IP addresses.
URL	Uniform Resource Locator — the address of a web page.
Browser	Software for accessing and displaying websites (e.g., Chrome, Firefox).
ISP	Internet Service Provider — company providing internet access.
Bandwidth	The maximum data transfer speed of an internet connection.

Additional Review Questions · Virtual Private Networks (§10.7a)

A. Short Answer

6. What is a VPN? Explain how it works in simple terms.

7. Why should you use a VPN on public Wi-Fi? What threat does it protect against?

8. What is one important limitation of a VPN that users often misunderstand?

9. Name two commercial VPN services. What should you look for when choosing a VPN?

B. Fill in the Blanks

1. VPN stands for _____ _____ _____.

2. A VPN creates an encrypted _____ between your device and the VPN server.

3. When using a VPN, websites see the _____ IP address, not yours.

4. Using a VPN does NOT make you fully _____ online.

5. Three reputable VPN services are _____, _____, and _____.

C. True or False

A VPN encrypts all traffic between your device and the VPN server. (True / False)

A VPN makes you completely anonymous online. (True / False)

You should always use a VPN on trusted home Wi-Fi. (True / False)

A VPN hides your IP address from websites you visit. (True / False)

Your VPN provider can see your internet traffic. (True / False)

Key Terms · VPN (§10.7a)

Term	Definition
VPN	Virtual Private Network · encrypts traffic and masks your IP address.
Tunnel	The encrypted connection between your device and the VPN server.
No-logs policy	A VPN provider's commitment not to record your browsing activity.
IP Masking	Hiding your real IP address by routing traffic through a VPN server.

True / False Answer Key		
1.	TRUE	That is the core function of VPN
2.	FALSE	VPN provider still sees your traffic
3.	FALSE	Home router is already trusted
4.	TRUE	Websites see VPN server IP instead
5.	TRUE	Hence 'no-logs' policy matters

Fill in the Blanks · Answer Key	
1.	Virtual Private Network
2.	encrypted tunnel
3.	VPN server's
	nonymous

CHAPTER 11
Email and Online Communication

Learning Objectives

By the end of this chapter, students will be able to:

- Understand how email works and identify parts of an email address
- Compose, send, reply to, and forward emails professionally
- Organise an inbox and manage email safely
- Describe different online communication tools
- Understand netiquette — polite communication online

11.1 What is Email?

Email (Electronic Mail) is a method of sending and receiving text messages and files digitally over the internet. Email is one of the oldest and most widely used internet services. It is faster, cheaper, and more flexible than postal mail.

Feature	Description
Speed	Delivered in seconds anywhere in the world.
Cost	Free to use (just requires internet connection).
Attachments	You can send files, photos, and documents.
Access	Can be read from any device with internet access.
Storage	Emails can be stored indefinitely for reference.

11.2 How Email Works

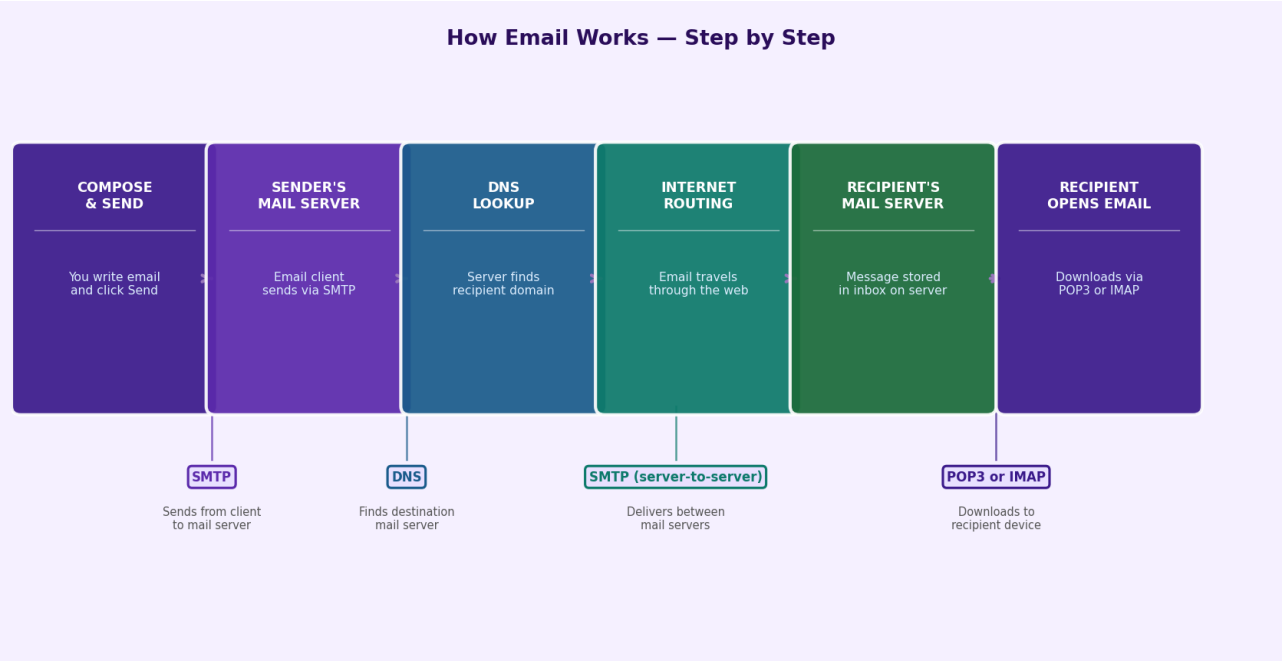


Figure 11.1 — How Email Works: From Compose to Inbox

Protocol	Role
SMTP	Simple Mail Transfer Protocol — used to SEND emails.
POP3	Post Office Protocol v3 — downloads emails to device and removes from server.
IMAP	Internet Message Access Protocol — syncs emails across devices without removing from server.

11.3 Parts of an Email Address

Email Address Format: username @ domain . extension

Example: mehrajwani771@school.edu.in

mehrajwani771 = The username (unique to this account)

school = The domain (the email service provider or organisation)

.edu.in = The extension (educational institution in India)

11.4 The Email Interface

Section	Description
Inbox	Where new incoming emails are received.
Sent	Copies of emails you have sent.
Drafts	Emails you have started writing but not yet sent.
Spam / Junk	Suspected unwanted emails filtered automatically.
Trash / Deleted	Emails you have deleted — can be recovered or permanently deleted.
Folders / Labels	Organise emails into custom categories.

11.5 Composing an Email

Field	Description
To:	The main recipient(s) email address. Required.
CC (Carbon Copy):	Other recipients who receive a copy — all recipients can see these addresses.
BCC (Blind Carbon Copy):	Hidden recipients — others cannot see these addresses. Used for mass emails.
Subject:	A brief, clear description of the email's topic. Always fill this in.
Body:	The main content of your message.
Attachment:	Files attached to the email. Click the paperclip icon to attach.
Signature:	Pre-set text added at the bottom (name, title, contact info).

Best Practices for Writing Emails

- Use a clear, specific subject line (e.g., "Assignment Submission - Chapter 3", not just "Hi").
- Start with a greeting (Dear Mr. Ali, Hello Sarah).
- Keep the message clear and concise — one main point per email.
- Use proper grammar, spelling, and punctuation.

- Sign off professionally (Regards, Sincerely, Thank you).
- Only attach files that are necessary. Compress large files.

11.6 Managing Your Inbox

Action	Description
Reply	Respond to the sender only.
Reply All	Respond to the sender AND all other recipients. Use carefully.
Forward	Send a received email to someone else.
Delete	Remove an email. Moves to Trash.
Flag / Star	Mark important emails for easy retrieval.
Move to Folder	Organise emails into categories.
Unsubscribe	Remove yourself from mailing lists to reduce inbox clutter.

11.7 Email Safety

Phishing Warning — Stay Alert

Phishing emails pretend to be from banks, delivery companies, or websites to trick you into entering your password or card details.

Signs of phishing: urgent language ("Your account will be closed!"), spelling errors, suspicious sender address, links that look wrong.

- Never open attachments from unknown senders — may contain viruses.
- Do not click links in suspicious emails — phishing attacks steal your information.
- Never share your email password with anyone.
- Use a strong password and enable two-factor authentication (2FA).
- Check the sender's address carefully — fake addresses mimic real ones.

11.8 Other Online Communication Tools

Tool / Platform	Description
Instant Messaging (IM)	Real-time text chat. Examples: WhatsApp, Telegram, Signal.
Video Conferencing	Real-time audio and video communication. Examples: Zoom, Google Meet, Teams.
Social Media	Platforms for sharing posts, photos, and interacting publicly. Examples: Instagram, X.
Forums	Communities where people post questions and discuss topics. Examples: Reddit.

Tool / Platform	Description
Blogs	Regularly updated web pages sharing articles or opinions.
VoIP (Voice over IP)	Phone calls made over the internet instead of phone lines.

11.9 Netiquette

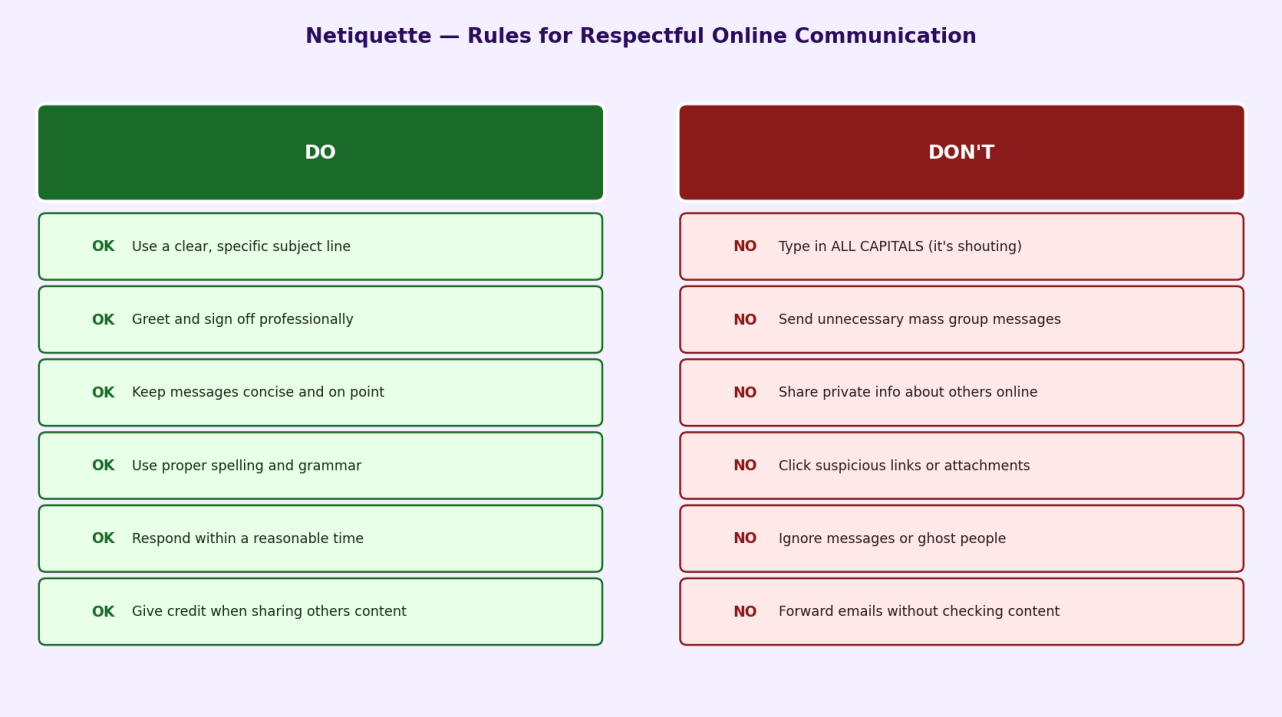


Figure 11.2 — Netiquette: Online Communication Do's and Don'ts

Chapter Review Questions

A. Short Answer

1. What is email? Name three advantages over postal mail.
2. What is the difference between CC and BCC?
3. Explain how email works from sending to receiving (step by step).
4. What is phishing? Name three signs of a phishing email.
5. What is netiquette? Give four examples of good netiquette.

B. Fill in the Blanks

1. The protocol used to SEND emails is called _____.
2. _____ protocol syncs emails across all devices without deleting from the server.
3. The _____ field in an email hides recipients from each other.
4. Email addresses follow the format _____ @ _____.
5. Clicking on links in suspicious emails is called falling for a _____ attack.

C. True or False

1. 'Reply All' sends a response only to the original sender. (True / False)
2. POP3 downloads emails to your device and removes them from the server. (True / False)
3. Typing in ALL CAPS in an online message is considered polite. (True / False)
4. IMAP is recommended when accessing email on multiple devices. (True / False)
5. You should always open attachments from unknown senders. (True / False)

D. Key Terms Glossary

Term	Definition
Email	Electronic mail — digital messages sent and received over the internet.
SMTP	Protocol used to send emails between servers.
IMAP	Protocol that syncs emails across multiple devices.
CC	Carbon Copy — visible recipients who receive a copy of the email.
BCC	Blind Carbon Copy — recipients hidden from other recipients.
Phishing	A scam email designed to steal passwords or personal information.
Spam	Unwanted, bulk email sent to many addresses.
Netiquette	Rules for polite and respectful communication online.

CHAPTER 12
Cyber Safety and Digital Etiquette

Learning Objectives

By the end of this chapter, students will be able to:

- Identify common cyber threats: viruses, malware, phishing, cyberbullying
- Create strong passwords and practise safe login habits
- Understand the importance of privacy online
- Behave responsibly and respectfully in digital environments
- Know how to respond to cyberbullying and online harassment

12.1 Why Cyber Safety Matters

As we spend more of our lives online — for study, communication, and entertainment — it is increasingly important to stay safe. Cyber threats can steal your personal data, damage your devices, harm your reputation, or put you in danger. Being a responsible digital citizen means knowing the risks and taking steps to protect yourself and others.

12.2 Common Cyber Threats

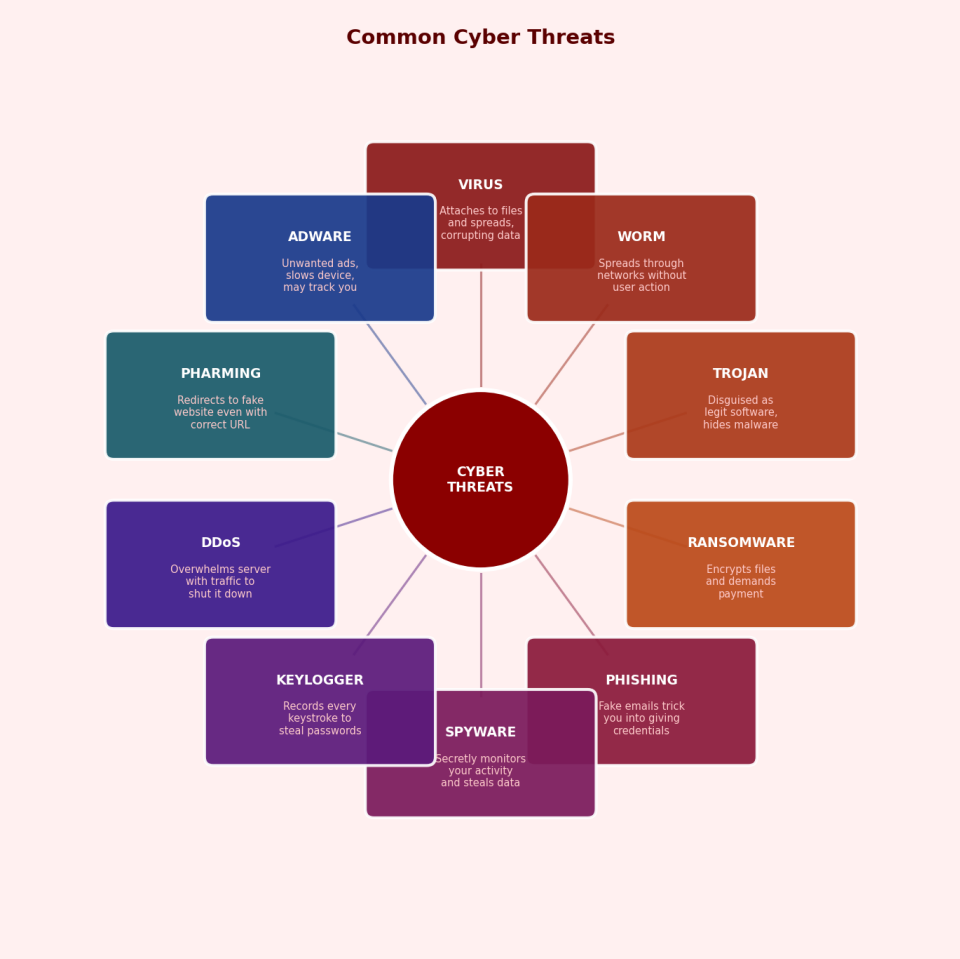


Figure 12.1 — Common Cyber Threats: Types and Descriptions

Threat	Description
Virus	A malicious program that attaches to files and spreads, corrupting or deleting data.
Worm	Similar to a virus but spreads across networks automatically without user action.
Trojan Horse	Disguises itself as legitimate software but secretly performs harmful actions.
Ransomware	Encrypts your files and demands payment to restore access.
Spyware	Secretly monitors your activity and sends data to a third party.
Adware	Displays unwanted adverts and may slow down your device.

Threat	Description
Phishing	Fake emails, messages, or websites that trick you into giving personal information.
Keylogger	Records every keystroke to steal passwords and private information.
DDoS Attack	Overwhelms a server with traffic to shut down a website or service.

12.3 Creating Strong Passwords

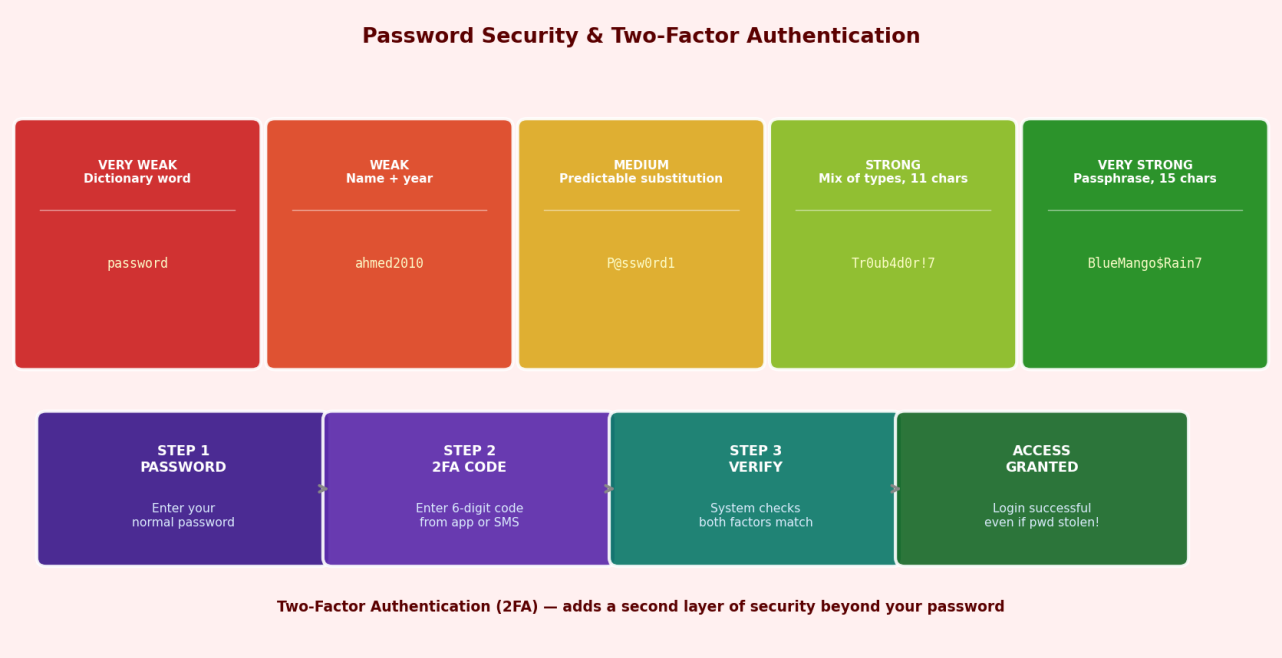


Figure 12.2 — Password Strength Scale and Two-Factor Authentication

Rules for a Strong Password

- At least 12 characters long.
- Combination of UPPERCASE, lowercase, numbers, and symbols.
- Not a real word, your name, birthday, or "password123".
- Different password for every account.
- Use a passphrase: A string of random words (e.g., BlueMango$Rain7Tree).

Weak Password	Strong Password
Password	G!r9@Lp2#kT7
12345678	BlueMango$Rain7Tree
ahmed2010	aH!m3d&X0z#99qW
mehrajwani	mehraj#Y0u_N0T!really

Password Managers & Two-Factor Authentication (2FA)

A password manager stores all your passwords securely in an encrypted vault. You only need to remember one master password. Examples: Bitwarden, LastPass, 1Password.

2FA adds an extra layer of security. Even if someone has your password, they also need a second proof of identity — usually a code sent to your phone or generated by an app.

- Enable 2FA on all important accounts: email, social media, banking.
- Methods: SMS code, authenticator app (Google Authenticator), physical security key.

12.4 Safe Browsing Habits

- Check for HTTPS and the padlock icon before entering personal information on a website.
- Do not download software from unknown or untrusted websites.
- Do not click pop-up ads — they may download malware.
- Use a reputable antivirus program and keep it updated.
- Keep your OS and browser updated — updates patch security vulnerabilities.
- Use a VPN (Virtual Private Network) on public Wi-Fi to encrypt your traffic.
- Log out of accounts when using shared or public computers.

12.5 Protecting Your Privacy Online

- Never share your full name, address, phone number, or school details with strangers online.
- Review and limit app permissions — does that game really need your location?
- Use privacy settings on social media to control who sees your content.
- Be careful about what you post — once online, it is very hard to fully remove.

The Digital Footprint

A digital footprint is the trail of data you leave behind every time you use the internet — search history, social media posts, comments, photos, and purchases. This information can be seen by employers, schools, and strangers.

Always think before you post.

12.6 Cyberbullying

Cyberbullying is the use of digital technology to harass, threaten, humiliate, or harm another person. It can happen via social media, messaging apps, online games, or email.

Forms of Cyberbullying

Form	Description
Harassment	Sending repeated mean or threatening messages.
Exclusion	Deliberately leaving someone out of online groups.
Impersonation	Creating a fake profile pretending to be someone else.
Outing	Sharing someone's private information or photos without consent.
Trolling	Posting inflammatory content to provoke emotional responses.

How to Respond to Cyberbullying

- Do not respond to the bully — it often escalates things.
- Take screenshots as evidence.
- Block and report the person on the platform.
- Tell a trusted adult: parent, teacher, or school counsellor.
- Contact the platform's safety team if the content is dangerous.

12.7 Screen Time and Digital Wellbeing

Too much screen time can affect your sleep, mental health, and physical wellbeing. Healthy digital habits include:

- Set daily screen time limits and stick to them.
- Avoid screens at least one hour before bedtime.
- Take regular breaks: every 20 minutes, look at something 20 feet away for 20 seconds (20-20-20 rule).
- Balance online and offline activities: sports, reading, social time.
- Use apps like Screen Time (iOS) or Digital Wellbeing (Android) to monitor usage.

12.8 Copyright and Intellectual Property

Copyright protects original creative works — books, music, software, images, and videos. Using someone else's work without permission is illegal and unethical.

- Cite sources when using information from websites in your work.
- Use Creative Commons licensed images for school projects (check licence terms).
- Do not download pirated software, music, or films.
- If creating content, you automatically hold copyright over your own original work.

12.2a Social Engineering

While technical attacks like viruses and ransomware exploit weaknesses in software, social engineering attacks exploit weaknesses in people. A social engineer manipulates a victim into revealing confidential information or performing an action that helps the attacker · no hacking required. Social engineering is responsible for the majority of successful cyberattacks worldwide.

Types of Social Engineering Attacks

Attack Type	How It Works
Phishing	Fake emails pretending to be trusted organisations to steal credentials. (Covered in detail in §11.7 and §12.2)
Vishing (Voice Phishing)	Attacker calls by phone pretending to be bank, IT support, or government. Pressures victim to reveal passwords or transfer money.
Smishing (SMS Phishing)	Fake text messages with malicious links. Common format: 'Your parcel could not be delivered · click here to reschedule.'
Pretexting	Attacker creates a fabricated scenario (pretext) to gain trust. Example: 'I'm from IT · I need your password to fix your account.'
Baiting	Leaving infected USB drives in car parks or offices. Curiosity leads victims to plug them in, installing malware automatically.
Tailgating (Piggybacking)	Physically following an authorised person through a secure door without using a badge. Exploits politeness · 'Can you hold the door?'
Quid Pro Quo	Offering something in exchange for information. Example: fake IT helpdesk offering to fix a problem in exchange for login credentials.
Watering Hole	Infecting a website frequently visited by the target group. When employees visit, malware is silently installed on their device.

The Golden Rule of Social Engineering Defence

Verify before you trust. Legitimate organisations will NEVER:

- Ask for your password by phone, email, or chat
- Pressure you to act immediately without time to think
- Ask you to bypass security procedures 'just this once'

If in doubt, hang up / ignore the message and call back on an official number.

Key Terms · Added

Term	Definition
Social Engineering	Manipulating people psychologically to gain access or information.
Vishing	Voice phishing · social engineering via phone calls.
Smishing	SMS phishing · social engineering via text messages.
Pretexting	Creating a false scenario to manipulate a victim into compliance.
Tailgating	Physically following authorised personnel through a secure door.

Chapter Review Questions

A. Short Answer

1. What is ransomware and how does it work?
2. List five rules for creating a strong password.
3. What is two-factor authentication? Why is it recommended?
4. What is cyberbullying? Give four examples of cyberbullying behaviour.
5. What should you do if you are being cyberbullied?

B. Fill in the Blanks

1. A _____ is malicious software that encrypts your files and demands payment.
2. A _____ stores all your passwords securely in an encrypted vault.
3. The trail of data you leave behind online is called a digital _____.
4. The 20-20-20 rule recommends taking a break every _____ minutes.
5. Original creative works are protected by _____ law.

C. True or False

1. Sharing your home address with online strangers is safe. (True / False)
2. HTTPS websites are more secure than HTTP websites. (True / False)
3. You should always respond to cyberbullying messages immediately. (True / False)
4. A password manager helps you use strong, unique passwords for every account. (True / False)
5. A worm spreads across networks without any user action. (True / False)

D. Key Terms Glossary

Term	Definition
Malware	Any malicious software designed to harm or exploit a computer or user.
Phishing	A cyberattack using fake emails to steal credentials or personal data.
Ransomware	Malware that encrypts data and demands payment for the decryption key.
2FA	Two-Factor Authentication — requires two forms of ID to log in.
Cyberbullying	Using technology to harass, threaten, or humiliate another person.
Digital Footprint	The data trail created by your online activities.
VPN	Virtual Private Network — encrypts internet traffic for privacy and security.
Copyright	Legal protection of original creative works from unauthorised use.

CHAPTER 13

Logical Thinking and Problem Solving

Learning Objectives

By the end of this chapter, students will be able to:

- Understand what computational thinking means
- Apply decomposition to break problems into smaller parts
- Use pattern recognition to identify similarities
- Create abstractions to simplify problems
- Follow a structured problem-solving process
- Apply logical thinking to everyday situations

13.1 What is Logical Thinking?

Logical thinking means reasoning in a clear, step-by-step way to solve a problem or reach a conclusion. It involves identifying facts, spotting patterns, and making decisions based on evidence rather than guesswork.

Computers are logical machines — they follow instructions exactly. To program a computer effectively, you must think logically and clearly. This skill is useful not just in computing but in everyday life: deciding what to do, solving puzzles, planning tasks.

13.2 Computational Thinking

Computational thinking is a problem-solving approach used by computer scientists. It has four key pillars that together provide a complete framework for tackling any complex problem.

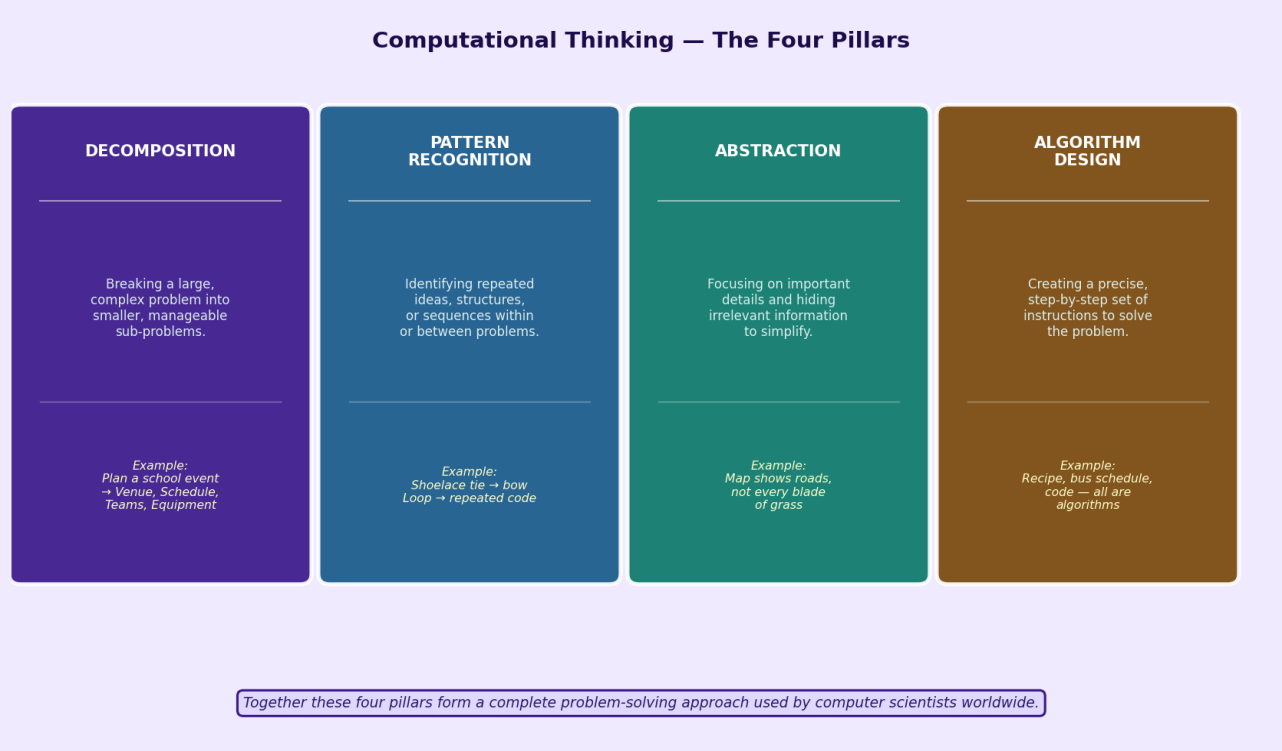


Figure 13.1 — The Four Pillars of Computational Thinking

Pillar	Description
Decomposition	Breaking a large, complex problem into smaller, more manageable sub-problems.
Pattern Recognition	Identifying similarities, trends, or repeating patterns within or between problems.
Abstraction	Focusing on the important details and ignoring irrelevant information.
Algorithm Design	Creating a clear, step-by-step solution to solve the problem or sub-problem.

13.2.1 Decomposition

When you face a large problem, it is overwhelming to try to solve it all at once. Decomposition means splitting it into smaller pieces, each of which is easier to understand and solve.

Example: Planning a School Event

Big problem: Organise a school sports day. Decompose it into:

1. Book the venue
2. Create a schedule of events
3. Invite participants and form teams
4. Arrange equipment
5. Plan refreshments
6. Set up safety procedures

Each sub-task is now much easier to manage independently.

13.2.2 Pattern Recognition

Patterns are repeated ideas, structures, or sequences. Recognising patterns helps you solve problems faster because a solution that worked before may work again.

- Example: If you can tie a shoelace, you can tie a bow on a gift box — same pattern.
- In programming: A loop is used when you recognise a pattern of repeated actions.
- In data: You recognise a pattern that temperatures are higher in summer every year.

13.2.3 Abstraction

Abstraction means simplifying a problem by hiding unnecessary details and only focusing on what matters.

- A map is an abstraction of the real world — it shows roads and landmarks, not every blade of grass.
- A school timetable is an abstraction — it shows subject names and times, not how each lesson is structured.
- In programming: Using a function called printScore() is an abstraction — you do not need to know how it works, just that it prints the score.

13.2.4 Algorithm Design

An algorithm is a precise, step-by-step set of instructions for solving a problem. Good algorithms are:

Property	Description
Finite	Must have a clear start and end — cannot run forever.
Unambiguous	Each step is clear and cannot be misinterpreted.
Effective	They solve the problem correctly for all valid inputs.
Efficient	They solve the problem without unnecessary steps.

13.3 Problem-Solving Process

A structured approach to solving problems ensures nothing is missed. Follow these eight steps:

Step	Action
1. Understand	Read carefully. What is given? What is needed?
2. Decompose	Break the problem into smaller parts.
3. Identify Patterns	Have you seen something similar before?
4. Abstraction	What details are important? What can be ignored?
5. Plan (Algorithm)	Write out the steps needed to solve the problem.
6. Implement	Code it or carry it out.
7. Test	Does the solution work for all cases, including edge cases?
8. Refine	If it does not work perfectly, adjust and try again.

13.4 Logic and Boolean Thinking

Boolean logic uses only two values: TRUE and FALSE. Computers use Boolean logic in all decision-making.

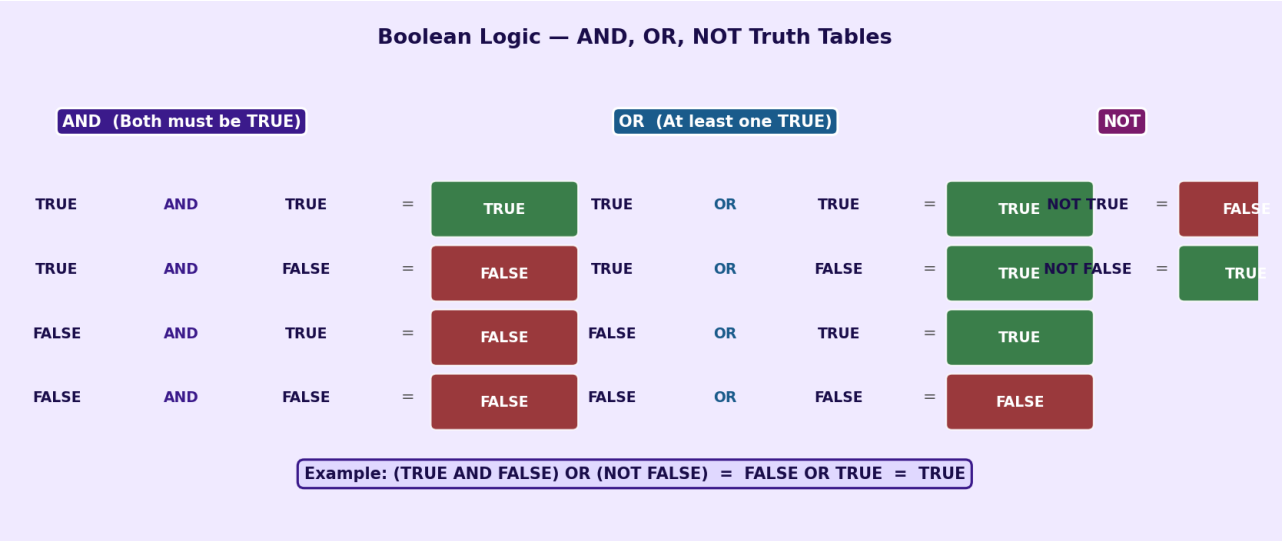


Figure 13.2 — Boolean Logic: AND, OR, NOT Truth Tables

Operator	Meaning and Example
AND	Both conditions must be true. Example: "Is it raining AND cold?" — only TRUE if both are true.
OR	At least one condition must be true. Example: "Is it raining OR cold?" — TRUE if either is true.

Operator	Meaning and Example
NOT	Reverses the truth value. Example: "NOT raining" — TRUE if it is NOT raining.

13.5 Sequence, Selection, and Iteration

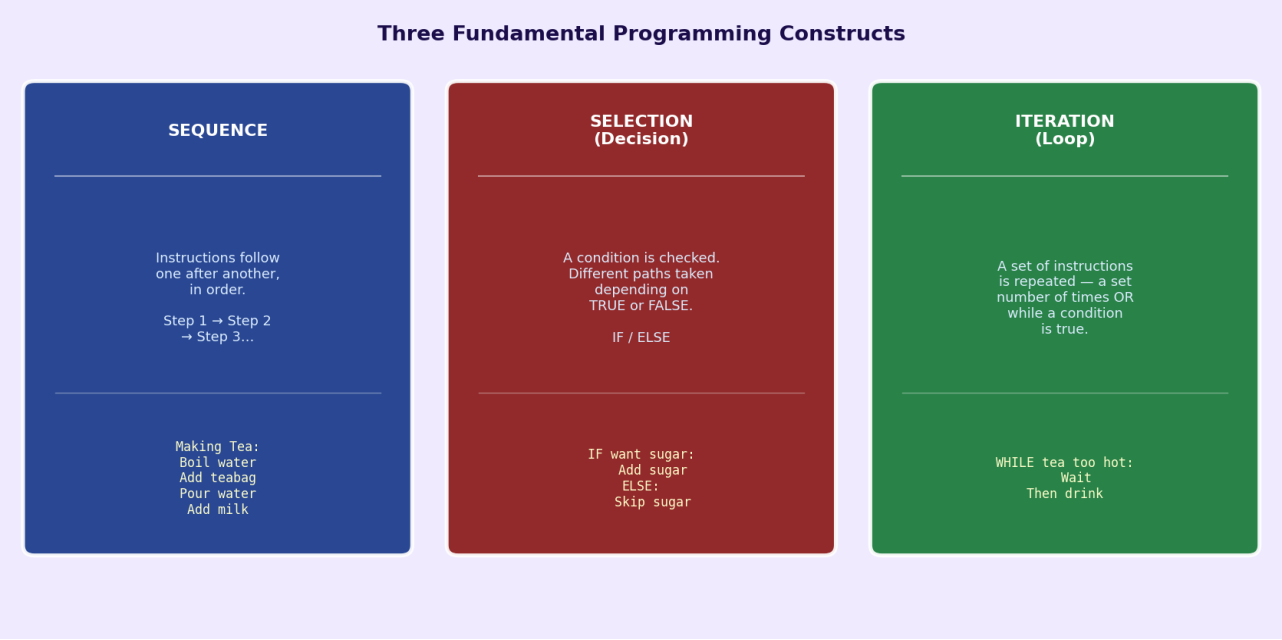


Figure 13.3 — The Three Fundamental Programming Constructs

Construct	Description
Sequence	Instructions are followed one after another, in order. Step 1, then Step 2, then Step 3...
Selection (Decision)	A condition is checked and different paths are followed depending on TRUE or FALSE. (IF/ELSE)
Iteration (Loop)	A set of instructions is repeated — either a set number of times (FOR loop) or while a condition is true (WHILE loop).

Everyday Example: Making Tea

SEQUENCE: Boil water → Add teabag → Pour water → Add milk → Remove teabag.

SELECTION: IF you want sugar, add sugar. ELSE don't add sugar.

ITERATION: WHILE tea is too hot, wait. Then drink.

Chapter Review Questions

A. Short Answer

1. What is computational thinking? Name and explain its four pillars.
2. What is decomposition? Give a real-life example.
3. Explain abstraction. How does a map illustrate abstraction?
4. What is Boolean logic? Give examples using AND, OR, and NOT.
5. Explain the three fundamental constructs in all algorithms.

B. True or False

1. Decomposition means solving all parts of a problem at the same time. (True / False)
2. An algorithm must have a clear start and end. (True / False)
3. TRUE AND FALSE evaluates to TRUE. (True / False)
4. Iteration means repeating a set of instructions. (True / False)
5. Abstraction focuses on the important details and ignores irrelevant ones. (True / False)

C. Apply Your Knowledge

1. Decompose the problem: 'Build a school website' into at least 5 sub-tasks.
2. Write a step-by-step algorithm (at least 6 steps) for making a cheese sandwich, including one selection and one sequence.
3. Evaluate: Is (TRUE AND FALSE) OR (NOT FALSE) equal to TRUE or FALSE? Show your working.

D. Key Terms Glossary

Term	Definition
Computational Thinking	A problem-solving approach using decomposition, patterns, abstraction, and algorithms.
Decomposition	Breaking a complex problem into smaller, manageable sub-problems.
Pattern Recognition	Identifying repeated ideas or structures in a problem.
Abstraction	Focusing on essential details while ignoring irrelevant ones.
Algorithm	A precise, step-by-step set of instructions to solve a problem.
Boolean Logic	Logic that uses only TRUE and FALSE values.
Sequence	Instructions carried out one after another in a fixed order.
Iteration	Repeating a set of instructions a number of times or while a condition is met.

CHAPTER 14
Algorithms

Learning Objectives

By the end of this chapter, students will be able to:

- Define an algorithm and identify its key properties
- Write algorithms in pseudocode
- Trace through algorithms and predict outputs
- Understand common searching and sorting algorithms
- Evaluate algorithms for efficiency
- Express algorithms in different forms

14.1 What is an Algorithm?

An algorithm is a precise, finite, step-by-step set of instructions designed to solve a specific problem or accomplish a task. Algorithms exist everywhere in life — a recipe, a bus timetable, a traffic signal sequence — but in computing, algorithms are the heart of every program ever written.

Property	Description
Finite	Must have a clear start and end — cannot run forever.
Definite / Unambiguous	Every step must be precise — no room for interpretation.
Input	Has zero or more inputs (data given to the algorithm).
Output	Produces one or more results.
Effective	Each step is basic enough to be carried out and actually solves the problem.
Efficient	Solves the problem in a reasonable number of steps, not wastefully.

14.2 Ways to Express Algorithms

14.2.1 Natural Language

Writing the steps in plain English. Easy to understand but can be ambiguous.

Example: Finding the Largest of Two Numbers

1. Get number A from the user.
2. Get number B from the user.
3. If A is greater than B, then the largest is A.
4. Otherwise, the largest is B.
5. Display the largest number.

14.2.2 Pseudocode

Pseudocode is a simplified, code-like notation that describes an algorithm using structured language. It is not actual code — it cannot be run — but it closely resembles the structure of real programming code. It is used to plan before writing actual code.

Pseudocode: Find Largest of Two Numbers

```
INPUT A

INPUT B

IF A > B THEN

    OUTPUT "A is largest"
```

```
ELSE

    OUTPUT "B is largest"

ENDIF
```

14.3 Pseudocode Conventions

Construct	Pseudocode Syntax
Input	INPUT variableName
Output	OUTPUT message or variable
Assignment	variable <- value (or variable = value)
Selection (IF)	IF condition THEN ... ELSE ... ENDIF
Count-controlled loop	FOR i <- 1 TO 10 ... ENDFOR
Condition loop	WHILE condition DO ... ENDWHILE
Repeat until	REPEAT ... UNTIL condition

14.4 Tracing an Algorithm

Tracing means manually working through an algorithm step by step, tracking the value of each variable. This is used to check an algorithm is correct before coding it.

Trace Table Example — Sum Numbers 1 to 5

```
Algorithm:  total <- 0  |  FOR i <- 1 TO 5  |  total <- total + i  |
ENDFOR  |  OUTPUT total
```

Trace: i=1: total=1 → i=2: total=3 → i=3: total=6 → i=4: total=10 → i=5: total=15 → Output: 15

14.5 Common Algorithms

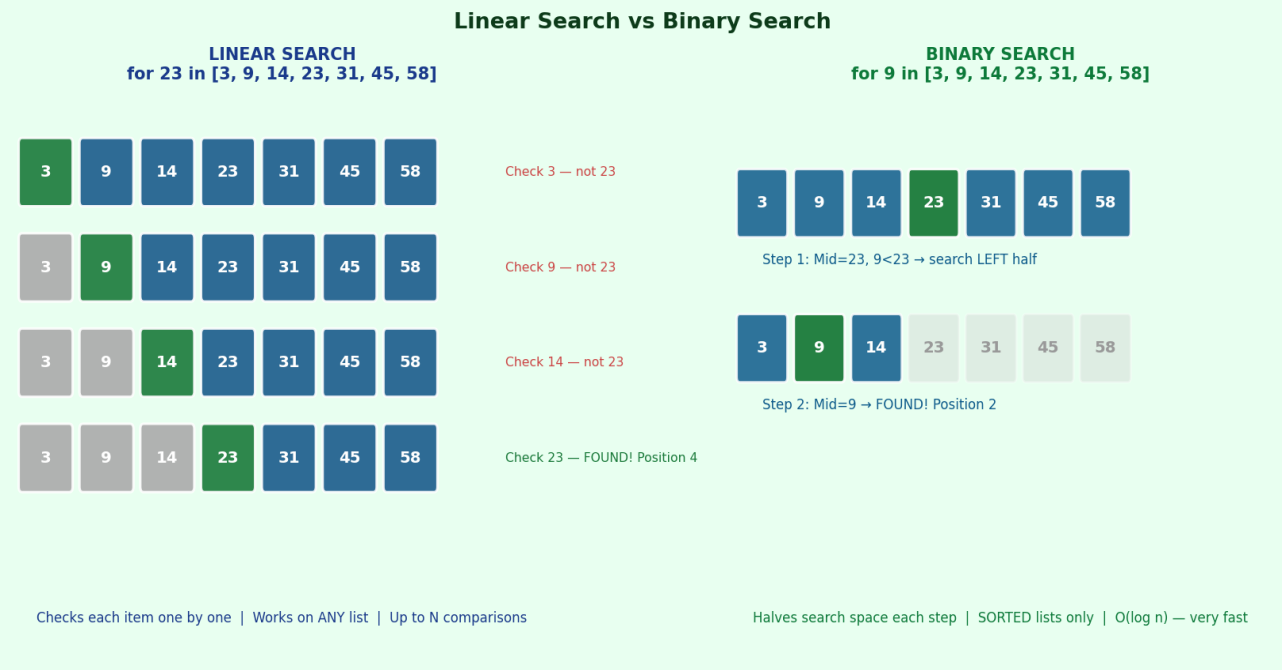


Figure 14.1 — Linear Search vs Binary Search: Step-by-Step Comparison

14.5.1 Linear Search

Linear search looks at each item one by one from the beginning until the target is found or the list ends.

- Best case: item is first — 1 comparison.
- Worst case: item is last or missing — n comparisons.
- Works on unsorted and sorted lists.

14.5.2 Binary Search

Binary search works only on sorted lists. It checks the middle item, then eliminates half the list with each comparison — making it far more efficient for large sorted lists.

- Far more efficient than linear search for large sorted lists.
- Each step halves the search space — very fast.
- Requires the list to be sorted first.

14.5.3 Bubble Sort

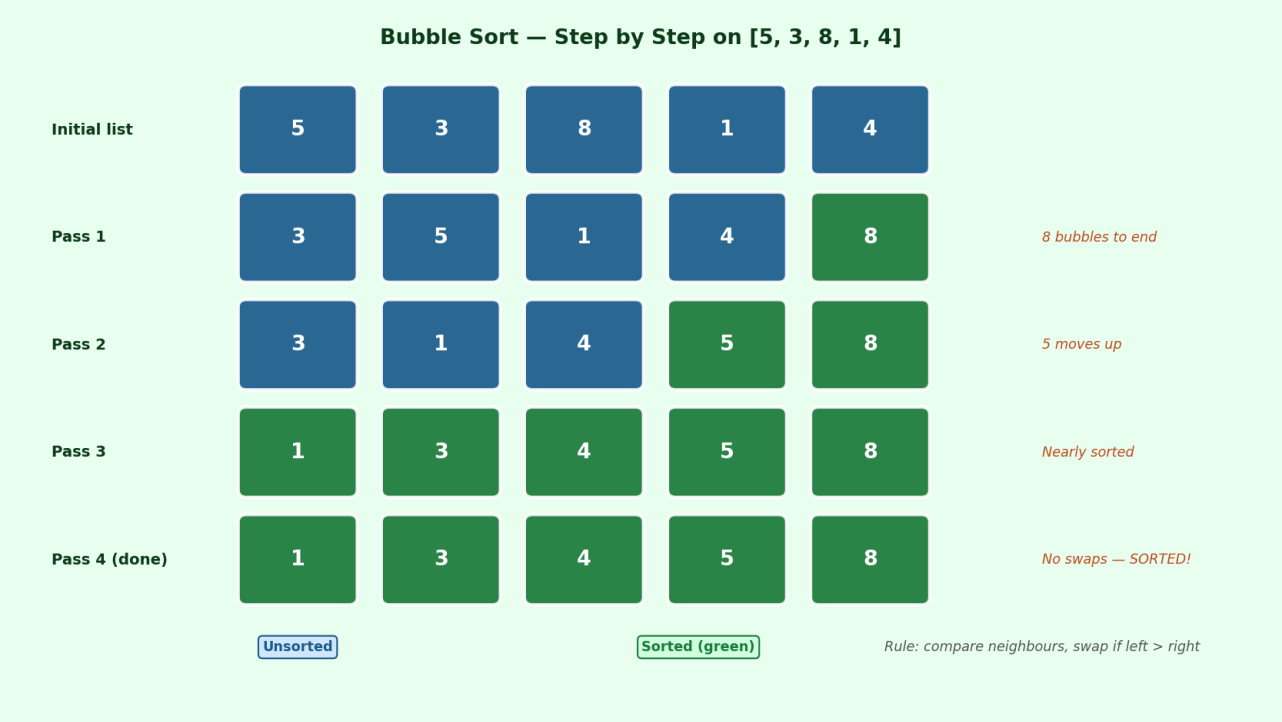


Figure 14.2 — Bubble Sort: Step-by-Step on [5, 3, 8, 1, 4]

Bubble sort repeatedly steps through a list, compares adjacent items, and swaps them if they are in the wrong order. Larger values "bubble up" to the end with each pass.

14.5.4 Comparing Sorting Algorithms

Algorithm	Description
Bubble Sort	Simple to understand, slow for large lists. Good for teaching.
Insertion Sort	Efficient for small or nearly sorted lists.
Merge Sort	Efficient for large lists but uses more memory.
Quick Sort	Very fast on average, used in real-world applications.

14.6 Algorithm Efficiency

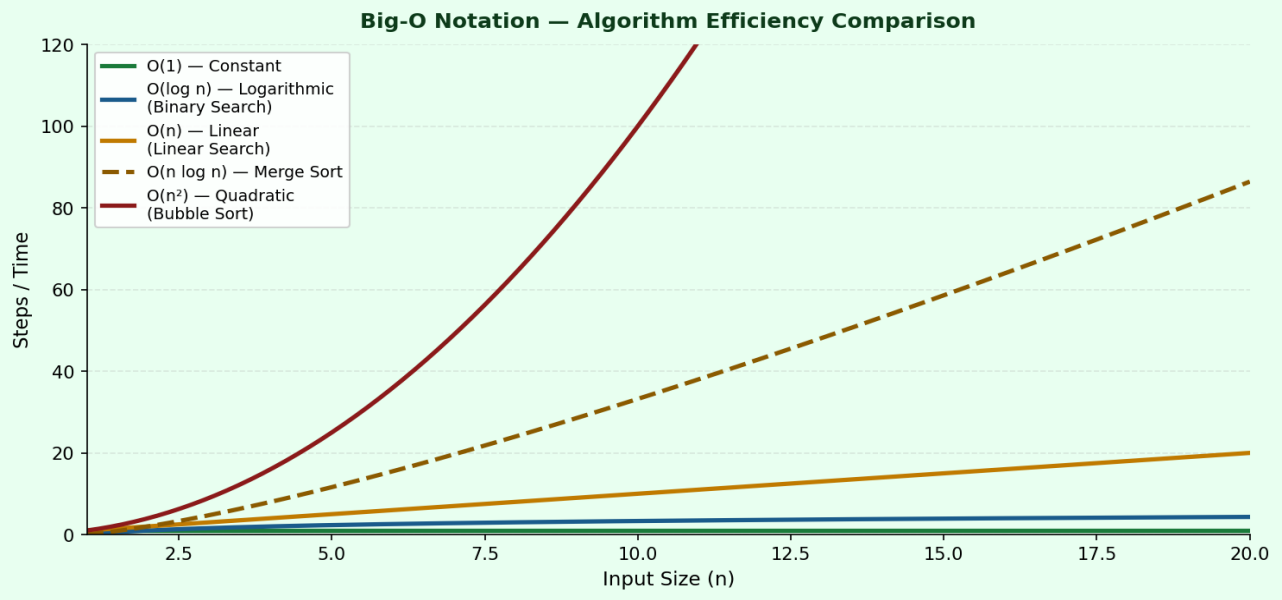


Figure 14.3 — Big-O Notation: How Efficiency Changes with Input Size

Notation	Meaning
O(1)	Constant time — speed does not change with input size.
O(log n)	Logarithmic — very efficient (e.g., binary search). Halves the problem each step.
O(n)	Linear time — speed grows proportionally with input (e.g., linear search).
O(n log n)	Log-linear — efficient for large inputs (e.g., merge sort).
O(n²)	Quadratic — slow for large inputs (e.g., bubble sort).

Chapter Review Questions

A. Short Answer

1. What is an algorithm? List five properties of a good algorithm.
2. What is pseudocode? Why is it used before writing actual code?
3. Explain the difference between linear search and binary search.
4. Trace through bubble sort on the list [4, 2, 7, 1, 5]. Show all passes.
5. What does O(log n) efficiency mean? Which search algorithm uses it?

B. Write the Pseudocode

1. Write pseudocode to calculate the average of five numbers entered by the user.
2. Write pseudocode that checks if a number is positive, negative, or zero.
3. Write pseudocode to count from 1 to 10 and output each number.

C. True or False

1. An algorithm must always produce an output. (True / False)
2. Binary search can be used on an unsorted list. (True / False)
3. Bubble sort is generally more efficient than merge sort for large lists. (True / False)
4. Pseudocode is actual code that can be executed by a computer. (True / False)
5. Tracing an algorithm means working through it manually to check it. (True / False)

D. Key Terms Glossary

Term	Definition
Algorithm	A precise, step-by-step set of instructions to solve a problem.
Pseudocode	A simplified code-like notation used to plan algorithms.
Trace Table	A table used to manually track variable values as an algorithm runs.
Linear Search	A search algorithm that checks each item one by one.
Binary Search	An efficient search algorithm for sorted lists that halves the search space each step.
Bubble Sort	A sorting algorithm that repeatedly swaps adjacent out-of-order items.
Efficiency	How quickly and resource-efficiently an algorithm solves a problem.
Big-O Notation	A notation used to describe the efficiency of an algorithm.

CHAPTER 15
Flowcharts

Learning Objectives

By the end of this chapter, students will be able to:

- Understand what a flowchart is and why it is used
- Identify and use standard flowchart symbols correctly
- Draw flowcharts for sequence, selection, and iteration
- Convert between pseudocode and flowcharts
- Trace through a flowchart to determine output

15.1 What is a Flowchart?

A flowchart is a visual diagram that uses standard symbols connected by arrows to represent the steps and decisions in a process or algorithm. Flowcharts make it easy to visualise the logic of a program before coding it, and are used to communicate algorithms to others clearly.

Flowcharts are used in computing, engineering, business process design, science, and education.

15.2 Standard Flowchart Symbols

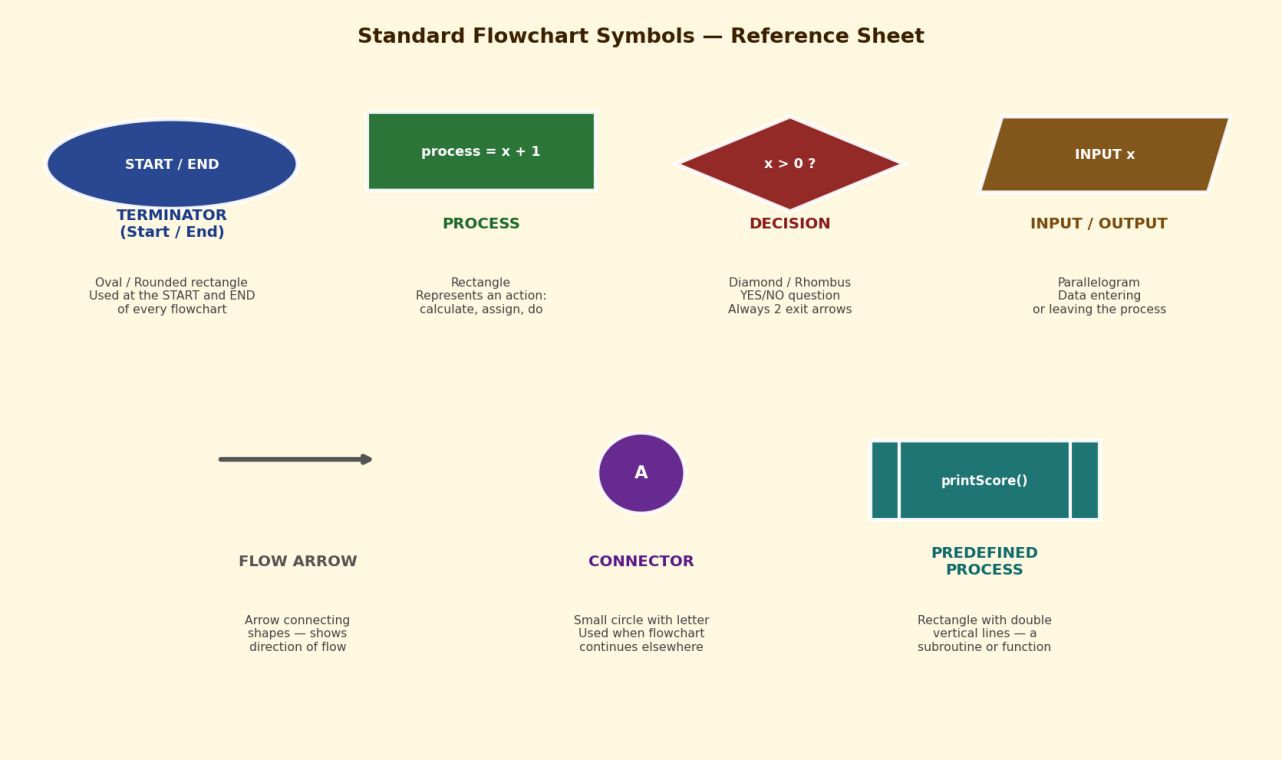


Figure 15.1 — Standard Flowchart Symbols Reference Sheet

Symbol Name	Shape and Use
Terminator (Start/End)	Oval / Rounded rectangle. Used at the START and END of every flowchart. Every flowchart must have exactly one START and at least one END.
Process	Rectangle. Represents an action or instruction — calculating, assigning, or doing something. Example: total = total + score
Decision	Diamond / Rhombus. Represents a YES/NO or TRUE/FALSE question. Always has two exit arrows labelled Yes/No or True/False.
Input / Output	Parallelogram. Represents data entering (INPUT) or leaving (OUTPUT) the process.
Flow Arrows	Arrows. Connect symbols and show the direction of flow.
Connector	Small circle with a letter. Used when the flowchart continues on another page to avoid

Symbol Name	Shape and Use
	crossing arrows.
Predefined Process	Rectangle with double vertical lines. Represents a sub-routine or function defined elsewhere.

15.3 Rules for Drawing Flowcharts

- Every flowchart must start with a START terminator.
- Every flowchart must end with an END terminator (may be multiple END symbols).
- All arrows must flow in the same general direction (usually top to bottom).
- Decision diamonds always have exactly two exit paths (Yes/No).
- Arrows must not cross each other if possible.
- Each symbol should have one entry point (except START).
- Use consistent, neat shapes of the same size.

15.4 Flowchart Examples

Example 1: Sequence — Calculate Area of a Rectangle

This is a pure sequence — no decisions or loops. Steps are followed one after another: START → INPUT length → INPUT width → area = length × width [Process] → OUTPUT area → END.

Example 2: Selection — Check if Number is Positive or Negative

The decision diamond has two exit paths: Yes (number is positive → OUTPUT "Positive") and No (number is not positive → OUTPUT "Negative or Zero"). Both paths lead to END.

Example 3: Iteration — Print Numbers 1 to 5

The loop arrow goes back up to the decision, repeating the process until counter exceeds 5. START → counter = 1 → Is counter ≤ 5? → Yes: OUTPUT counter, counter = counter + 1, loop back → No: END.

Example 4: Combined — Login System (Full Flowchart)

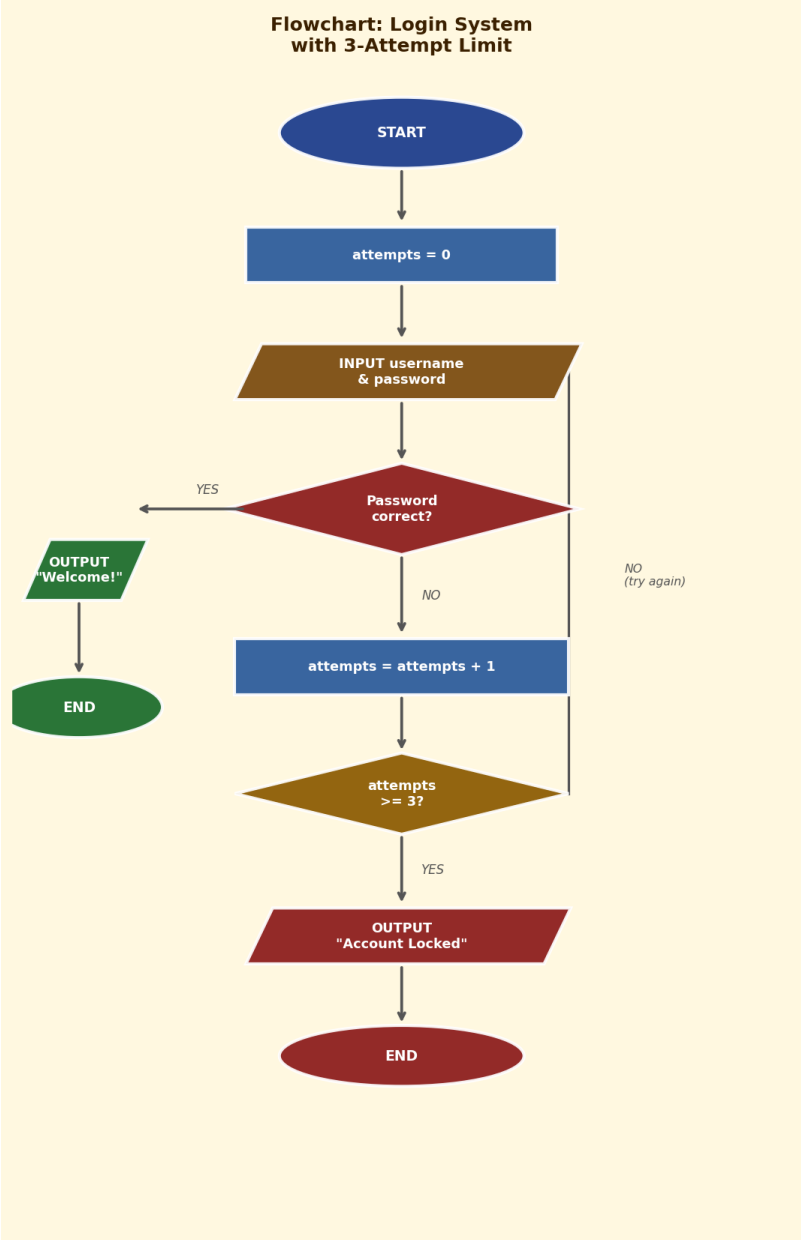


Figure 15.2 — Complete Flowchart: Login System with 3-Attempt Limit

15.5 Converting Between Pseudocode and Flowchart

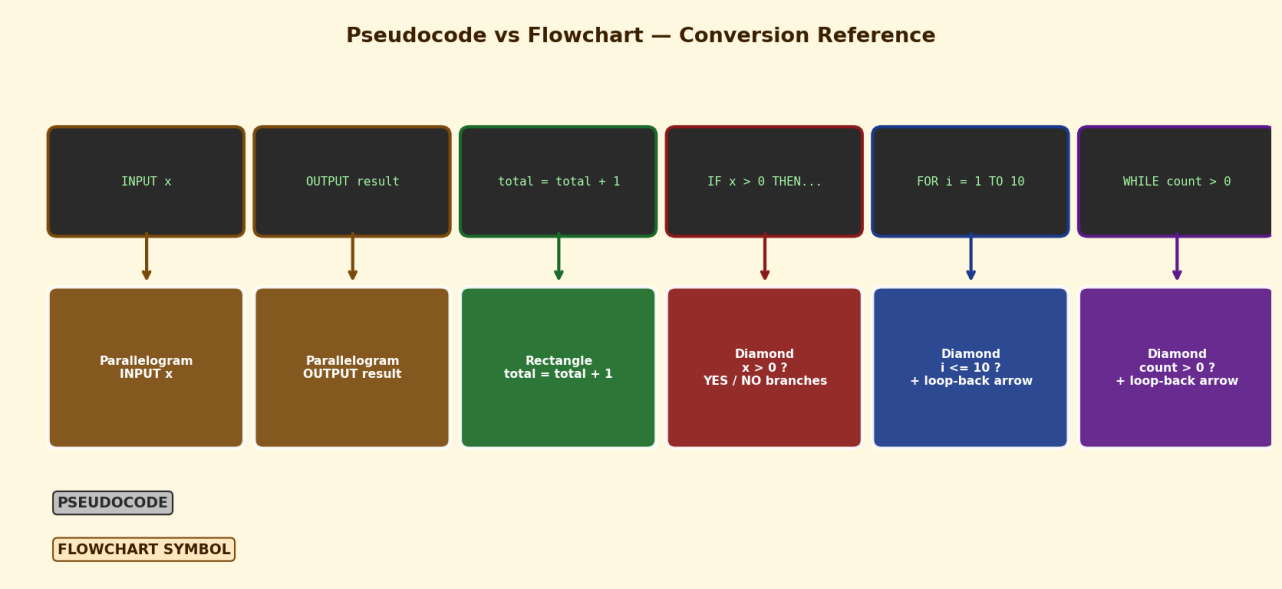


Figure 15.3 — Pseudocode to Flowchart Conversion Reference

Pseudocode Construct	Flowchart Equivalent
INPUT x	Parallelogram with "INPUT x"
OUTPUT result	Parallelogram with "OUTPUT result"
total = total + 1	Rectangle with "total = total + 1"
IF x > 0 THEN ... ELSE ...	Diamond with "x > 0?" and two branches (Yes/No)
FOR i = 1 TO 10	Diamond with "i <= 10?" and loop-back arrow
WHILE count > 0	Diamond with "count > 0?" and loop-back arrow

---

15.6 Tracing a Flowchart

Tracing a flowchart means following its arrows step by step for a given input to determine the output. Use a trace table to record variable values at each step.

Trace Example: Add numbers while sum < 20

Flowchart: START → sum=0 → INPUT number → sum = sum + number → Is sum < 20? YES→loop NO→OUTPUT sum → END

Input: 7, 8, 6 → Trace: sum=0 → sum=7 (7<20, loop) → sum=15 (15<20, loop) → sum=21 (21≥20, stop) → Output: 21

15.7 Flowchart Tools

Tool	Description
Paper and Pencil	Most important for exams — practice drawing by hand with a ruler.
Microsoft Word / PowerPoint	Insert > Shapes — flowchart shapes built in. Easy for school work.

Tool	Description
Diagrams.net (draw.io)	Free online tool, drag-and-drop flowchart shapes. Very popular.
Lucidchart	Professional online diagram tool with collaboration features.
Microsoft Visio	Professional diagramming software used in industry.

Chapter Review Questions

A. Short Answer

1. What is a flowchart? Why are flowcharts useful?
2. Name and describe five standard flowchart symbols.
3. What shape is used for a decision? How many exit paths must it have?
4. What is the difference between a sequence, selection, and iteration flowchart?
5. Convert this pseudocode into a flowchart description: INPUT age → IF age >= 18 THEN OUTPUT "Adult" ELSE OUTPUT "Minor" ENDIF

B. Draw the Flowchart (on paper or using a tool)

1. Draw a flowchart that inputs two numbers and outputs the larger one.
2. Draw a flowchart that outputs the multiplication table of a number entered by the user (×1 to ×10).
3. Draw a flowchart for a password check: user gets 3 attempts; if correct, show "Welcome"; if all 3 fail, show "Locked".

C. True or False

1. An oval shape is used for processing calculations in a flowchart. (True / False)
2. A decision symbol has exactly two exit arrows. (True / False)
3. Every flowchart must begin with a START terminator. (True / False)
4. A parallelogram represents input or output in a flowchart. (True / False)
5. Flowcharts and pseudocode can represent the same algorithm. (True / False)

D. Key Terms Glossary

Term	Definition
Flowchart	A visual diagram using standard symbols to represent an algorithm.
Terminator	Oval shape marking the START or END of a flowchart.
Process Box	Rectangle representing an action or calculation step.
Decision Diamond	Diamond shape representing a YES/NO question in a flowchart.
Parallelogram	Shape used for INPUT or OUTPUT operations.
Flow Arrow	Arrow connecting shapes to show the direction of execution.
Iteration	A loop — a section of a flowchart that repeats based on a condition.
Trace	Manually following an algorithm step by step for a given input.

CHAPTER 16
Introduction to Coding (Scratch)

Learning Objectives

By the end of this chapter, students will be able to:

- Understand what Scratch is and why it is used for learning coding
- Navigate the Scratch interface
- Create sprites and use backdrops
- Use motion, looks, sound, and control blocks
- Build simple animations, stories, and games in Scratch
- Understand the link between Scratch blocks and real programming concepts

16.1 What is Scratch?

Scratch is a free, visual block-based programming language developed by the MIT Media Lab. Instead of typing code, beginners drag and snap together coloured blocks that represent programming instructions. This makes coding accessible to young learners without needing to worry about syntax errors.

Scratch is used worldwide in schools to introduce programming concepts including sequences, loops, conditionals, variables, and events. Projects include animations, interactive stories, quizzes, and games. Scratch is available at scratch.mit.edu — free to use in any browser.

16.2 The Scratch Interface

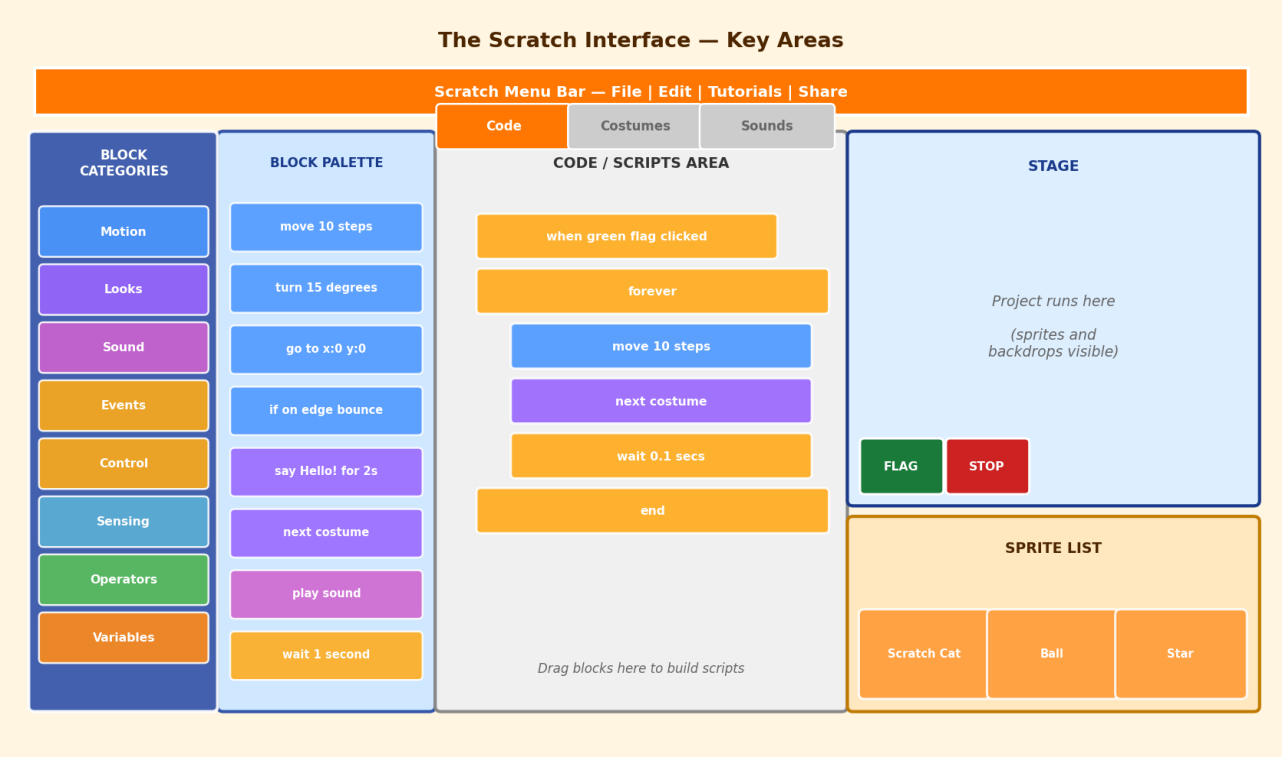


Figure 16.1 — The Scratch Interface: Key Areas and Panels

Area	Description
Stage (top right)	The screen where your project runs. Shows sprites and backgrounds.
Sprite List (bottom right)	All sprites in your project. Click a sprite to select and edit it.
Block Categories (left)	Coloured categories: Motion, Looks, Sound, Events, Control, Sensing, Operators, Variables, My Blocks.
Block Palette (centre-left)	Shows available blocks for the selected category.
Scripts / Code Area (centre)	Where you drag and assemble blocks to create scripts.
Costume Tab	Edit how a sprite looks. Each sprite can have multiple costumes.
Sounds Tab	Add and record sounds for your sprites.

Area	Description
Green Flag / Red Stop	Green flag runs the project. Red stop sign stops it.

16.3 Key Concepts in Scratch

16.3.1 Sprites

A sprite is a character or object in your Scratch project. The default sprite is the Scratch Cat. You can:

- Choose from the Scratch library (hundreds of characters and objects).
- Upload your own image.
- Draw your own using the built-in paint editor.
- Use a photo from your camera.

16.3.2 Backdrops

A backdrop is the background image on the Stage. Each project can have multiple backdrops — switching between them creates scene changes. Choose from library, draw, or upload.

16.3.3 Costumes

A sprite can have multiple costumes — different images that represent different states (e.g., walking frames). Switching between costumes quickly creates animation.

16.4 Block Categories



Figure 16.2 — Scratch Block Categories: What Each Colour Does

Category (Colour)	What it Does
Motion (Blue)	Move, turn, glide, and set position/direction of a sprite.
Looks (Purple)	Change costumes, size, colour effects.

Category (Colour)	What it Does
	Show/hide sprites. Say or think text.
Sound (Pink)	Play sounds and music. Set volume and pitch.
Events (Yellow)	Triggers — when green flag clicked, when key pressed, when sprite clicked.
Control (Orange)	Wait, repeat, forever, if/else, stop, create clone. Loops and conditionals.
Sensing (Light Blue)	Detect if sprite is touching something, ask the user a question, use mouse/keyboard.
Operators (Green)	Maths: +, -, *, /. Comparisons: >, <, =. Logic: AND, OR, NOT.
Variables (Dark Orange)	Create, set, change, and show/hide variables. Create lists.
My Blocks (Red)	Create custom blocks (functions/procedures).

16.5 Building Your First Projects

Project 1: Make the Cat Move

Script: Cat moves 10 steps when arrow key pressed

```
when [right arrow] key pressed  →  [Events: yellow]

move 10 steps  →  [Motion: blue]

Repeat for all four arrow keys to move in all directions.
```

Project 2: Simple Walking Animation

Script: Walking animation with costume switching

```
when green flag clicked  →  [Events]

forever  →  [Control: orange loop]

    next costume  →  [Looks]

    move 5 steps  →  [Motion]

    wait 0.1 seconds  →  [Control]

    if on edge, bounce  →  [Motion]
```

Project 3: Simple Quiz Game

Script: Ask a question and check the answer

```
when green flag clicked

ask "What is 5 + 3?" and wait  →  [Sensing]
```

```
if (answer = 8) then  →  [Control + Operators]
    say "Correct!" for 2 seconds  →  [Looks]
Else
    say "Wrong! The answer is 8." for 2 seconds
```

16.6 Programming Concepts in Scratch

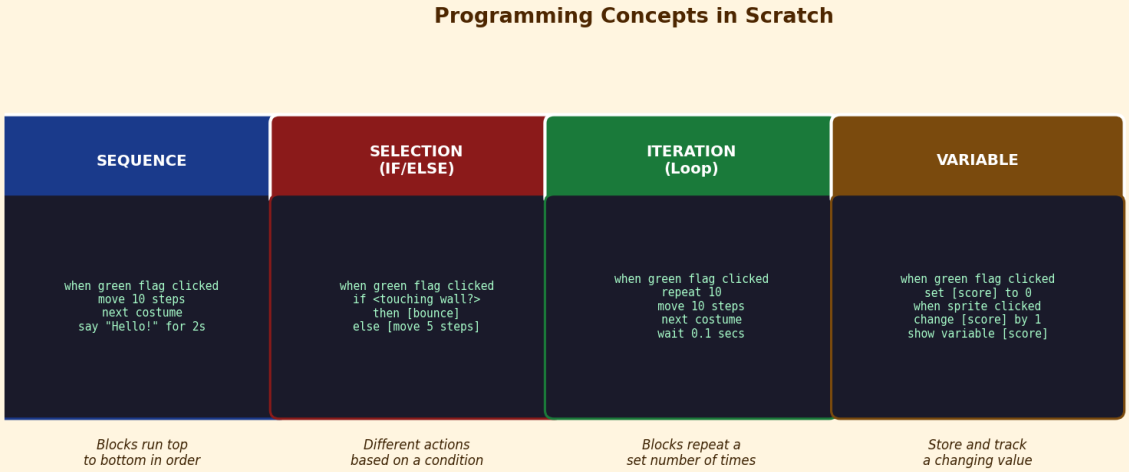


Figure 16.3 — Programming Concepts: How They Appear in Scratch

Programming Concept	Scratch Block Example
Sequence	A stack of blocks — each runs in order, top to bottom.
Selection (IF)	'if <condition> then ... else ...' block in Control.
Iteration — Count Loop	'repeat 10' block — runs the inside 10 times.
Iteration — Infinite Loop	'forever' block — loops forever until stopped.
Iteration — Condition Loop	'repeat until <condition>' block.
Variable	Create a variable in Variables category. Set, change, show it.
Event Handling	'when green flag clicked', 'when key pressed' blocks — Events.
Input	'ask ... and wait' in Sensing. Answer stored in the 'answer' variable.
Output	'say' or 'think' block in Looks. Or use 'show variable'.
Procedure / Function	'define' block in My Blocks — create a reusable block.

16.7 Sharing and Saving Projects

- Save: File > Save Now (Ctrl+S). Projects auto-save when logged in.
- Save to computer: File > Save to your computer — saves as an .sb3 file.
- Load from computer: File > Load from your computer.
- Share online: Click the Share button — project becomes publicly visible on the Scratch website.
- Remix: You can remix (copy and modify) other people's public projects.

Chapter Review Questions

A. Short Answer

1. What is Scratch and who developed it?
2. What is a sprite? What is a backdrop? How are they different?
3. Which Scratch block category contains loops and IF statements?
4. What is the difference between a 'repeat' block and a 'forever' block?
5. How do you create a variable in Scratch? Give an example of how you would use it.

B. Match the Scratch Block to the Programming Concept

Scratch Block	Programming Concept
if <condition> then ... else ...	
repeat 10	
set score to 0	
ask "What is your name?" and wait	
when green flag clicked	

C. True or False

1. In Scratch, you write code by typing commands on a keyboard. (True / False)
2. A sprite can have more than one costume. (True / False)
3. The 'forever' block runs its contents exactly 10 times. (True / False)
4. Variables are created in the Variables block category. (True / False)
5. Scratch projects can be saved as .sb3 files on your computer. (True / False)

D. Key Terms Glossary

Term	Definition
Scratch	A free, visual, block-based programming language for beginners from MIT.
Sprite	A character or object in a Scratch project.
Backdrop	The background image displayed on the Stage.
Costume	A visual state of a sprite — multiple costumes enable animation.
Script	A set of assembled code blocks in Scratch that performs a task.
Event	A trigger that starts a script (e.g., when green flag clicked).
Variable	A named storage location that holds a value that can change.
Clone	A duplicate copy of a sprite created during the running of a project.

CHAPTER 17
Basic Python Programming

Learning Objectives

By the end of this chapter, students will be able to:

- Understand why Python is popular and where it is used
- Write and run basic Python programs
- Use variables, data types, and operators
- Use input() and print() for interaction
- Write if/elif/else statements
- Use for and while loops
- Define and call functions
- Understand lists and basic list operations

17.1 What is Python?

Python is a high-level, general-purpose programming language created by Guido van Rossum, first released in 1991. It is famous for its clean, readable syntax that resembles plain English, making it one of the best languages for beginners.

Why Python?	Where Python is Used
Easy to read and write	Web development (Django, Flask)
Free and open source	Data science and machine learning
Huge community and libraries	Artificial intelligence and deep learning
Works on all platforms	Automation and scripting
Versatile — from beginner to expert	Scientific research and education

Setting Up Python

- Download: python.org — download the latest version for your OS.
- IDLE: Python's built-in editor — simple and good for beginners.
- Thonny: Beginner-friendly Python IDE with built-in debugger.
- Online: replit.com or onlinegdb.com — run Python in your browser, no installation needed.
- VS Code: Professional editor with Python extension — recommended for intermediate learners.

17.2 Your First Python Program

```
# This is a comment -- Python ignores it
print('Hello, World!')

# Output: Hello, World!
```

Figure 17.1 — Every Python journey begins with Hello, World!

print() is a built-in function that outputs text to the screen. Text must be enclosed in quotes (single or double). Comments start with # and are ignored by Python — use them to explain your code.

17.3 Variables and Data Types

A variable is a named container that stores a value. In Python, you create a variable simply by assigning it a value — no need to declare its type first.

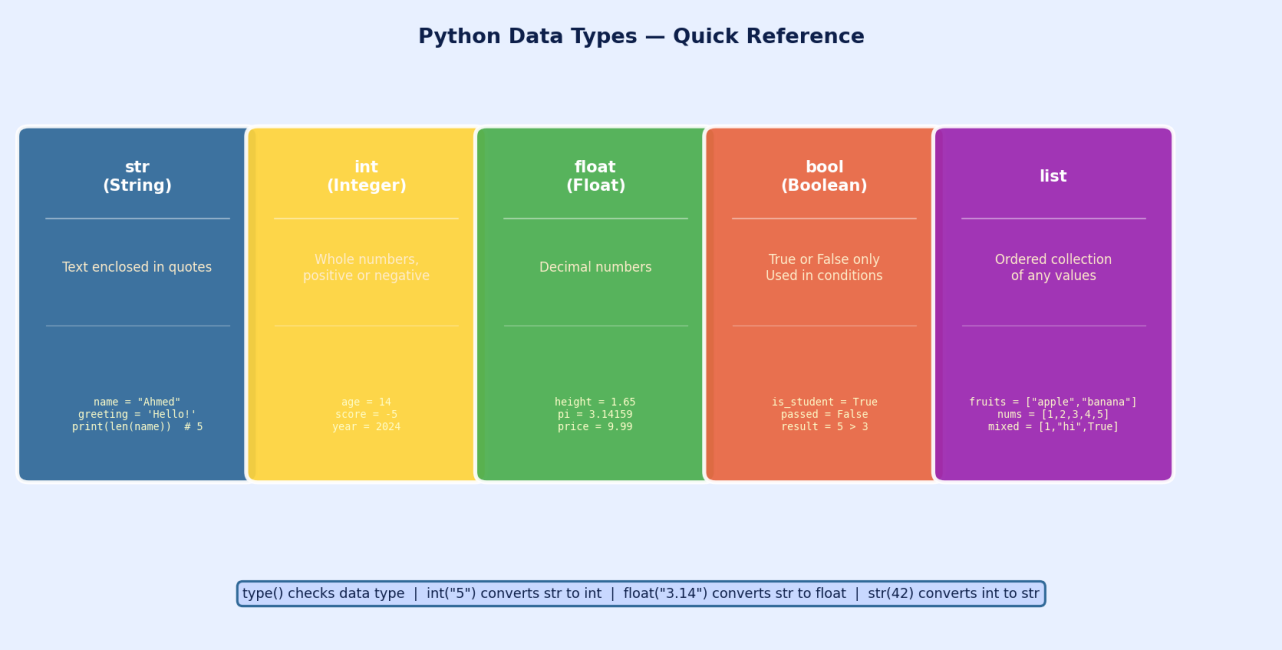


Figure 17.2 — Python Data Types: str, int, float, bool, and list

```
name       = 'Ahmed'     # str   (String — text)
age        = 14          # int   (Integer — whole number)
height     = 1.65        # float (Float — decimal number)
is_student = True        # bool  (Boolean — True or False)
fruits     = ['apple', 'banana', 'cherry']  # list

# Check type with type()
print(type(age))      # Output: <class 'int'>
print(type(name))     # Output: <class 'str'>
```

17.4 Operators

Operator Type	Operators and Examples
Arithmetic	+ (add) - (subtract) * (multiply) / (divide) // (floor divide) % (modulo) ** (power) Example: 10 % 3 = 1 | 2 ** 4 = 16 | 7 // 2 = 3
Comparison	== (equal) != (not equal) > (greater) < (less) >= (greater or equal) <= (less or equal) Always returns True or False
Assignment	= (assign) += (add and assign) -= (subtract and assign) *= /= Example: x += 3 is the same as x = x + 3
Logical	and or not Example: (5 > 3) and (2 < 4) → True

17.5 Input and Output

```
# Getting input from user
name = input('What is your name? ')
```

```
age  = int(input('How old are you? '))    # convert str to int

# Printing output — three methods:
print('Hello,', name)
print('You are', age, 'years old.')
print(f'Hello, {name}! You are {age} years old.')  # f-string (best
method)
```

Important: input() always returns a STRING

To use input as a number, convert it: `age = int(input(...))` or `price = float(input(...))`

17.6 Selection — If Statements



Figure 17.3 — Python Control Structures: if/elif/else and for/while loops

```
mark = int(input('Enter your mark: '))

if mark >= 90:
    print('Grade: A')
elif mark >= 75:
    print('Grade: B')
elif mark >= 60:
    print('Grade: C')
elif mark >= 50:
    print('Grade: D')
else:
    print('Grade: F - Please try harder!')

# Python uses INDENTATION (4 spaces) to define code blocks
```

17.7 Iteration — Loops

For Loop (Count-Controlled)

```
# Print numbers 1 to 5
for i in range(1, 6):
    print(i)

# range(1, 6)      generates: 1, 2, 3, 4, 5
# range(5)         generates: 0, 1, 2, 3, 4
# range(0, 10, 2)  generates: 0, 2, 4, 6, 8  (step = 2)
```

While Loop (Condition-Controlled)

```
# Count down from 5
counter = 5
while counter > 0:
    print(counter)
    counter -= 1   # same as: counter = counter - 1
print('Blastoff!')

# Loop runs WHILE counter > 0. Stops when counter reaches 0.
```

17.8 Functions

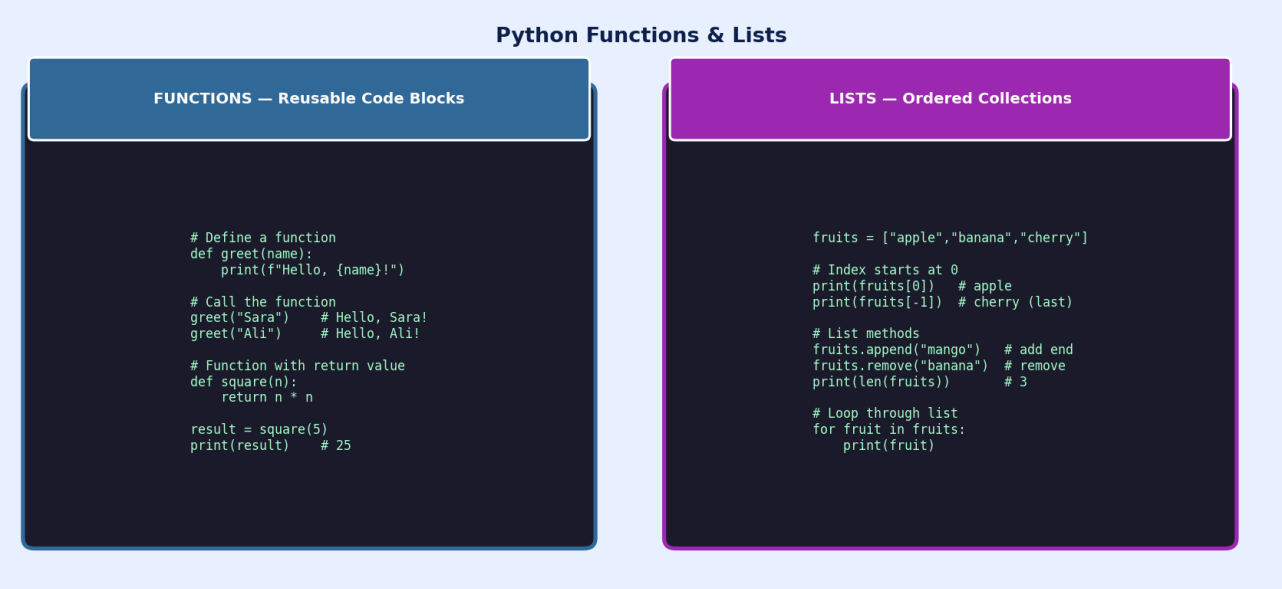


Figure 17.4 — Python Functions and Lists: Defining, Calling, and List Operations

A function is a reusable block of code that performs a specific task. Using functions avoids repeating code and makes programs easier to read and maintain.

```
# Define a function
def greet(name):
    print(f'Hello, {name}!')

# Call the function
greet('Sara')    # Output: Hello, Sara!
greet('Ali')     # Output: Hello, Ali!
```

```
# Function with a return value
def square(number):
    return number * number

result = square(5)
print(result)   # Output: 25
```

17.9 Lists

A list stores multiple values in a single variable, in order. Lists can contain any data type and can be changed (they are mutable).

```
fruits = ['apple', 'banana', 'cherry']

# Accessing items — index starts at 0
print(fruits[0])    # apple
print(fruits[-1])   # cherry (last item)

# List methods
fruits.append('mango')   # adds to end
fruits.remove('banana')  # removes by value
print(len(fruits))       # number of items: 3

# Loop through a list
for fruit in fruits:
    print(fruit)
```

17.10 A Complete Python Program

Mini Grade Calculator — putting it all together

```
# Grade Calculator Program

def get_grade(mark):
    if mark >= 90: return "A"
    elif mark >= 75: return "B"
    elif mark >= 60: return "C"
    elif mark >= 50: return "D"
    else: return "F"

name  = input("Enter student name: ")
marks = []

for subject in range(3):
    m = int(input(f"Enter mark for subject {subject + 1}: "))
    marks.append(m)

average = sum(marks) / len(marks)
grade   = get_grade(average)

print(f"Student: {name}")
print(f"Average: {average:.1f}")
print(f"Grade:   {grade}")
```

17.9a Dictionaries

A dictionary stores data as key-value pairs. Unlike a list which uses numbered index positions, a dictionary lets you label each value with a meaningful key.
Dictionaries are one of Python's most important and widely used data structures · used in APIs, databases, configuration files, and almost every real Python program.

```
# Create a dictionary
student = {
    'name':  'Ahmed',
    'age':   14,
    'grade': 'A'
}

# Access values by key
print(student['name'])      # Ahmed
print(student['grade'])     # A

# Add or update a key
student['subject'] = 'Maths'
student['age'] = 15         # updates existing key

# Delete a key
del student['subject']

# Check if a key exists
if 'name' in student:
    print('Name found!')

# Loop through key-value pairs
for key, value in student.items():
    print(key, ':', value)
```

Common Dictionary Methods

Method / Operation	Description & Example
d[key]	Access value: student['name'] · 'Ahmed'
d[key] = value	Add or update: student['age'] = 15
del d[key]	Delete a key-value pair
d.get(key)	Safe access · returns None if key missing (no error)
d.keys()	Returns all keys: dict_keys(['name', 'age', 'grade'])
d.values()	Returns all values: dict_values(['Ahmed', 14, 'A'])
d.items()	Returns key-value pairs for looping
len(d)	Number of key-value pairs in the dictionary
key in d	True/False · check if key exists

Lists vs Dictionaries · When to Use Which

Use a list when order matters and items are identified by position: scores = [85, 90, 78]
Use a dictionary when items are identified by a meaningful label: student = {'name':'Ahmed'}
Rule of thumb: if you find yourself writing list[0], list[1], list[2] with meaning, use a dict.

17.11 Error Handling · try / except

When a Python program encounters an unexpected situation · such as a user typing text when a number is expected · it raises an exception and crashes. Error handling lets you catch these exceptions and respond gracefully instead of crashing.

Basic Syntax:

```
try:
    age = int(input('Enter your age: '))
    print('You are', age, 'years old.')
except ValueError:
    print('Please enter a valid whole number.')
```

The try block contains the code that might cause an error. If an exception occurs, Python jumps to the except block instead of crashing. Execution continues normally after the except block.

Common Exception Types:

Exception	When It Occurs
ValueError	Wrong data type · e.g. int('hello')
ZeroDivisionError	Division by zero · e.g. 10 / 0
IndexError	List index out of range · e.g. lst[99] on a short list
FileNotFoundError	Trying to open a file that does not exist
TypeError	Wrong type for an operation · e.g. 'hello' + 5

Extended Example with Multiple Exceptions:

```
try:
    num   = int(input('Enter a number: '))
    result = 100 / num
    print('Result:', result)
except ValueError:
    print('That is not a valid number.')
except ZeroDivisionError:
    print('Cannot divide by zero.')
finally:
    print('Program finished.')  # always runs
```

· Common Python Mistakes · Beginners Should Watch Out For These

1. Forgetting int() or float() after input() · input() always returns a string.
2. Off-by-one in range() · range(1,6) gives 1,2,3,4,5 (not 6).
3. Incorrect indentation · Python uses 4 spaces; mixing tabs and spaces causes errors.
4. Using = instead of == in conditions · = assigns, == compares.
5. Mutable default arguments · avoid using lists as default function parameters.

Chapter Review Questions

A. Short Answer

1. What is Python? Name three areas where it is used professionally.
2. What is the difference between int, float, str, and bool?
3. What does input() return? How do you convert it to a number?
4. What is the difference between a for loop and a while loop?
5. What is a function? Why are functions useful?

B. What is the Output?

Code	Output
print(2 ** 3)	
print(10 % 3)	
x = 5 x += 3 print(x)	
for i in range(2, 8, 2): print(i)	
fruits = ['apple', 'banana', 'cherry'] print(fruits[-1])	

C. Write the Python Code

1. Write a program that asks the user for two numbers and prints their sum, difference, and product.
2. Write a function called is_even(n) that returns True if n is even and False if it is odd.
3. Write a program that prints the multiplication table of a number entered by the user.

D. Key Terms Glossary

Term	Definition
Python	A high-level, readable programming language used in many fields.
Variable	A named storage location holding a value.
Data Type	The kind of value a variable holds (int, str, float, bool, list).
print()	Built-in function to display output on screen.
input()	Built-in function to receive text input from the user.
Function	A reusable block of code that performs a specific task.
List	An ordered collection of values stored in one variable.
Indentation	Spaces at the start of a line that define code blocks in Python.

Additional Review Questions · Error Handling (§17.11)

A. Short Answer

6. What is an exception in Python? Give two examples of exceptions that can occur.

7. What is the purpose of a try/except block? How does it prevent a program from crashing?

8. What does the finally clause do? When would you use it?

9. Write a try/except block that safely converts user input to a float and handles invalid input.

10. Name three common Python mistakes beginners make. How would you fix each?

B. True or False

A ValueError occurs when you try to convert 'hello' to an integer using int(). (True / False)

The except block only runs if an error occurs in the try block. (True / False)

You can have multiple except clauses for different error types. (True / False)

The finally block only runs if no exception occurred. (True / False)

ZeroDivisionError occurs when you divide a number by zero. (True / False)

Key Terms Glossary · Error Handling (§17.11)

Term	Definition
Exception	An error that occurs at runtime, interrupting normal program execution.
try / except	A Python block that attempts code (try) and catches errors (except) gracefully.
ValueError	Exception raised when a value is the wrong type · e.g. int('hello').
ZeroDivisionError	Exception raised when dividing a number by zero.
IndexError	Exception raised when accessing a list index that does not exist.
FileNotFoundError	Exception raised when a file the program tries to open does not exist.
TypeError	Exception raised when an operation is applied to the wrong data type.
finally	A clause that always executes after try/except, regardless of errors.
Error Handling	Using try/except to catch and respond to exceptions rather than crashing.

CHAPTER 18
Artificial Intelligence

Learning Objectives

By the end of this chapter, students will be able to:

- Define Artificial Intelligence and explain its key concepts
- Describe different types of AI and machine learning
- Identify real-world applications of AI in everyday life
- Understand how AI learns from data
- Discuss the benefits and limitations of current AI systems

18.1 What is Artificial Intelligence?

Artificial Intelligence (AI) is the branch of computer science concerned with creating systems that can perform tasks that would normally require human intelligence. These tasks include understanding language, recognising images, making decisions, solving problems, and learning from experience.

The term "Artificial Intelligence" was coined by computer scientist John McCarthy in 1956. Today, AI is one of the most rapidly advancing fields in technology.

Key Distinction: Intelligence vs Artificial Intelligence

Intelligence: The ability to learn, reason, solve problems, and adapt.

Artificial Intelligence: Machines that simulate aspects of human intelligence using algorithms and data.

18.2 Types of AI

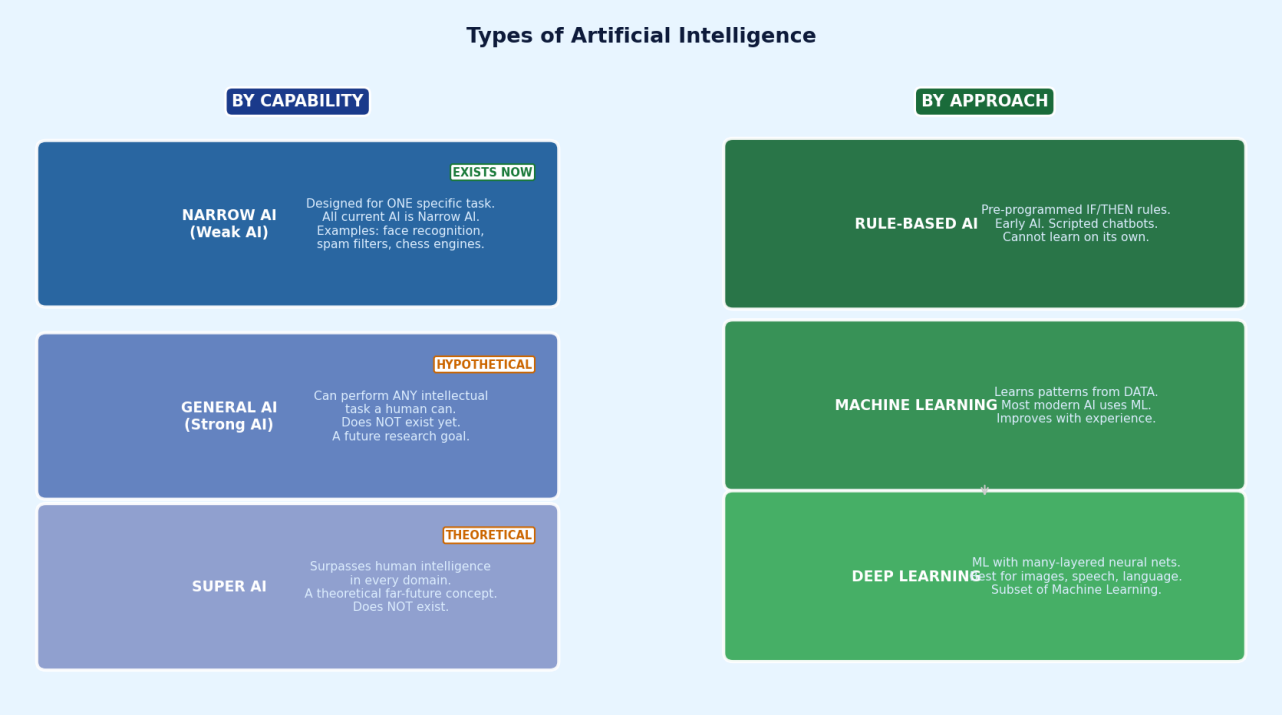


Figure 18.1 — Types of AI: By Capability and By Approach

18.2.1 Based on Capability

Type	Description
Narrow AI (Weak AI)	AI designed for one specific task. Cannot do anything outside its training. Examples: face recognition, spam filters, chess engines. ALL current AI is Narrow AI.
General AI (Strong AI)	Hypothetical AI that can perform any intellectual task a human can. Does NOT exist yet — it is a future goal.
Super AI	Hypothetical AI that surpasses human intelligence in every domain. Does NOT exist — a theoretical concept for the far future.

18.2.2 Based on Approach

Approach	Description
Rule-Based AI	Uses pre-programmed IF/THEN rules. Early AI systems. Example: simple chatbots with scripted responses.
Machine Learning (ML)	Systems that learn from data without being explicitly programmed. Most modern AI uses ML.
Deep Learning	A subset of ML using multi-layered artificial neural networks. Excellent at images, speech, and language.

18.3 Machine Learning

Machine Learning (ML) is the most important technique in modern AI. Instead of writing rules manually, we feed an ML system data and it learns the rules by itself.

18.3.1 How Machines Learn

Step	Action
1. Collect Data	Gather and prepare large amounts of labelled training examples.
2. Choose Algorithm	Select the learning algorithm suited to the problem.
3. Train the Model	The algorithm finds patterns in the training data.
4. Test the Model	Evaluate on new, unseen data to check accuracy.
5. Evaluate & Improve	Adjust and retrain until performance is good enough.
6. Deploy	Release the model in a real application.

18.3.2 Types of Machine Learning

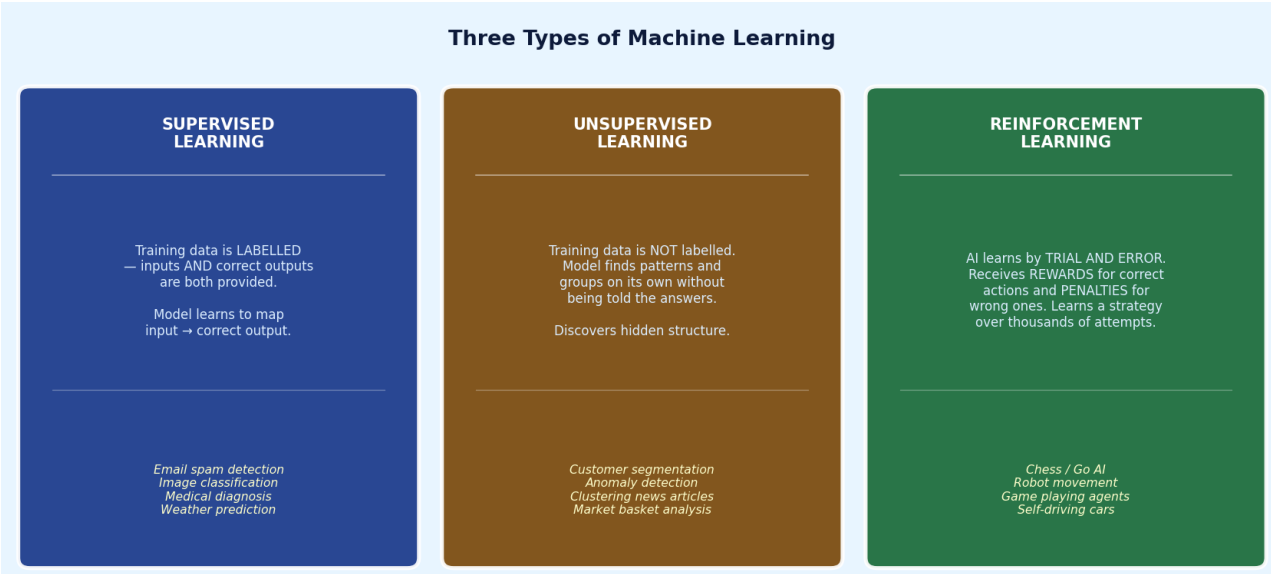


Figure 18.2 — Three Types of Machine Learning

Type	Description and Examples
Supervised Learning	Training data is labelled — inputs and correct outputs are both given. The model learns to map inputs to outputs. Examples: email spam detection (spam/not spam), image classification (cat/dog).
Unsupervised Learning	Training data is NOT labelled. The model finds patterns and groups on its own. Examples: customer segmentation, anomaly detection, clustering similar news articles.

Type	Description and Examples
Reinforcement Learning	The AI learns by trial and error — it receives rewards for correct actions and penalties for wrong ones. Examples: game-playing AI (chess, Go), robot movement training.

18.3.3 Neural Networks and Deep Learning

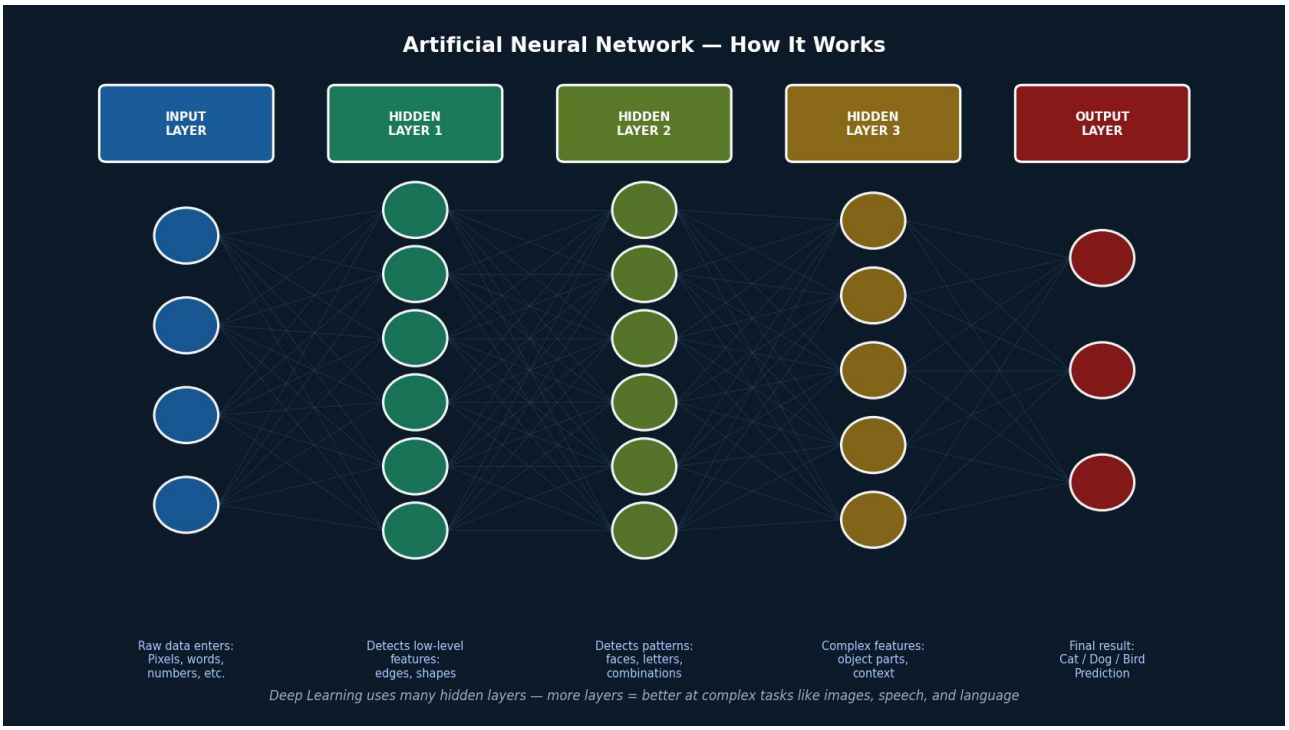


Figure 18.3 — Artificial Neural Network: Input → Hidden Layers → Output

A neural network is a computing model inspired by the human brain. It is made up of layers of nodes (neurons) connected by weighted links. Data flows through input layers, hidden layers, and output layers.

- Deep Learning: Neural networks with many hidden layers — extremely powerful for complex tasks.
- Convolutional Neural Networks (CNNs): Specialised for image recognition.
- Recurrent Neural Networks (RNNs): Specialised for sequences and time-series data (speech, text).
- Large Language Models (LLMs): Very large neural networks trained on vast text. Examples: GPT-4, Claude, Gemini.

18.4 AI Applications in Real Life



Figure 18.4 — AI Applications Across Every Field of Modern Life

Field	AI Application
Healthcare	Medical image analysis (detecting cancer in X-rays), drug discovery, patient diagnosis support, wearable health monitors.
Transportation	Self-driving cars (Tesla, Waymo), GPS route optimisation, traffic management, flight autopilot.
Education	Personalised learning platforms, automated essay grading, AI tutors, plagiarism detection.
Entertainment	Recommendation engines (Netflix, Spotify, YouTube), game AI opponents, generative art and music.
Finance	Fraud detection, algorithmic stock trading, credit scoring, chatbot customer service.
Language	Machine translation (Google Translate), voice assistants (Siri, Alexa), speech-to-text.
Security	Cybersecurity threat detection, biometric authentication (fingerprint, face ID).

18.5 Natural Language Processing (NLP)

NLP is the branch of AI that enables computers to understand, interpret, and generate human language.

- Machine Translation: Google Translate converts text between 100+ languages.
- Sentiment Analysis: Determines whether a review is positive, negative, or neutral.
- Chatbots and Virtual Assistants: Answer questions in natural language (Siri, Alexa, Claude, ChatGPT).
- Speech Recognition: Converts spoken words to text (Siri, Google Voice, dictation).

18.6 Computer Vision

Computer Vision enables computers to interpret and understand images and video.

- Facial Recognition: Identifying or verifying a person from a photo or video.
- Object Detection: Identifying objects in images (used in self-driving cars).
- Medical Imaging: Detecting tumours, fractures, and diseases in X-rays and MRIs.
- Quality Control: Detecting defects in products on manufacturing lines.

18.7 Limitations of AI

Limitation	Explanation
Narrow Scope	Current AI only works in the specific area it was trained on.
Needs Large Data	ML models need huge amounts of high-quality training data.
Bias	If training data is biased, the AI will be biased (e.g., unfair hiring algorithms).
Explainability	Deep learning models are "black boxes" — hard to explain why they made a decision.
No Common Sense	AI lacks real-world understanding and general reasoning.
Energy Consumption	Training large AI models requires vast amounts of electricity.
Fragility	AI can fail in unexpected ways when it encounters data outside its training.

Chapter Review Questions

A. Short Answer

1. What is Artificial Intelligence? Give a definition in your own words.
2. What is the difference between Narrow AI and General AI?
3. Explain the difference between supervised, unsupervised, and reinforcement learning.
4. What is a neural network? What is deep learning?
5. Name five real-world applications of AI in different fields.

B. Fill in the Blanks

1. All current AI systems are examples of _____ AI.
2. _____ learning trains models using labelled examples.
3. NLP stands for Natural Language _____.
4. Computer Vision allows computers to understand _____ and video.
5. AI training data that is _____ can lead to unfair or discriminatory decisions.

C. True or False

1. General AI exists and is used in most smartphones today. (True / False)
2. Reinforcement learning uses a reward system to train AI. (True / False)
3. Deep learning models are always easy to explain and interpret. (True / False)
4. Machine Learning systems learn from data rather than explicit rules. (True / False)
5. CNNs are particularly good at processing image data. (True / False)

D. Key Terms Glossary

Term	Definition
Artificial Intelligence	Computer systems that simulate human-like intelligence.
Machine Learning	AI that learns from data without being explicitly programmed.
Deep Learning	ML using multi-layered neural networks for complex tasks.
Neural Network	A computing model inspired by the human brain's structure.
NLP	Natural Language Processing — AI for understanding human language.
Computer Vision	AI for interpreting and analysing images and video.
Supervised Learning	ML trained on labelled data (input + correct output).
Reinforcement Learning	ML trained through reward and penalty feedback.

CHAPTER 19
Ethics of AI

Learning Objectives

By the end of this chapter, students will be able to:

- Understand what AI ethics means and why it matters
- Identify key ethical issues: bias, privacy, fairness, transparency
- Discuss the social impact of AI on employment and society
- Understand the concept of responsible AI
- Evaluate AI decisions from an ethical perspective
- Articulate personal views on the future of AI

19.1 Why Ethics Matters in AI

Artificial Intelligence is becoming part of nearly every aspect of modern life — from the videos recommended to you, to decisions about who gets a bank loan, to medical diagnosis. As AI becomes more powerful and widespread, the decisions it makes have real consequences for real people.

AI ethics is the field that asks: How should AI be designed, deployed, and governed to ensure it is safe, fair, and beneficial for everyone? This is not just a question for engineers — it concerns all of us.

Why Ethics Cannot Be Ignored

AI systems have already been found to:

- **Wrongly identify innocent people as criminals (facial recognition bias)**
- **Deny loans or jobs to people unfairly (biased training data)**
- **Spread misinformation at scale (deepfake videos)**
- **Make life-or-death decisions in autonomous weapons**

Getting AI ethics right is urgent and important.

19.2 Key Principles of Ethical AI

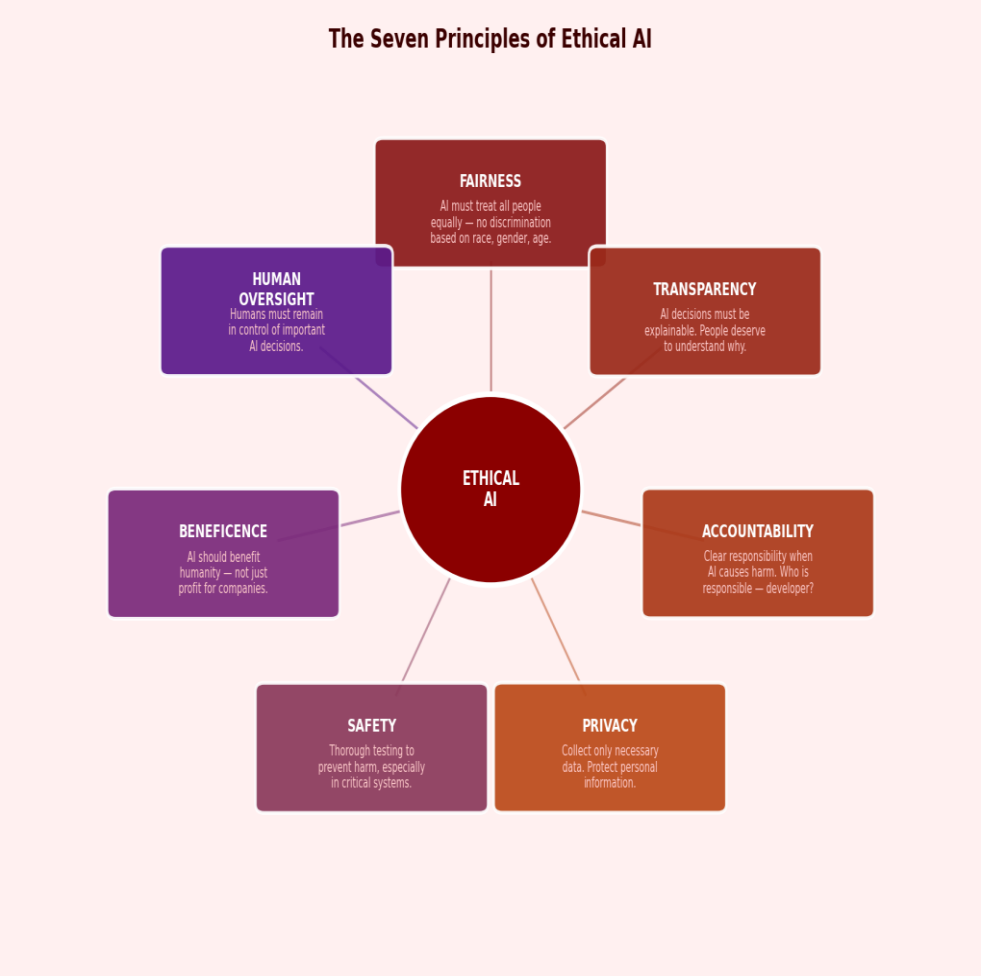


Figure 19.1 — The Seven Principles of Ethical AI

Principle	Description
Fairness	AI systems should treat all people equally, regardless of race, gender, age, religion, or background. They should not discriminate.
Transparency	AI decisions should be explainable. People should be able to understand why an AI made a particular decision — especially in high-stakes areas.
Accountability	When an AI system causes harm, there must be clear responsibility. Who is accountable — the developer, the company, the user?
Privacy	AI systems should collect and use only the data necessary for their purpose, and protect personal information.
Safety	AI must be tested thoroughly to ensure it does not cause harm, especially in critical systems like healthcare, transport, and military.
Beneficence	AI should be designed to benefit humanity, not just profit for companies.
Human Oversight	Humans must remain in control of important AI decisions, especially where life or rights are at stake.

19.3 AI Bias

AI bias occurs when an AI system produces unfair or discriminatory results. Bias usually enters AI through training data — if the data reflects historical inequalities, the AI will learn and perpetuate those inequalities.

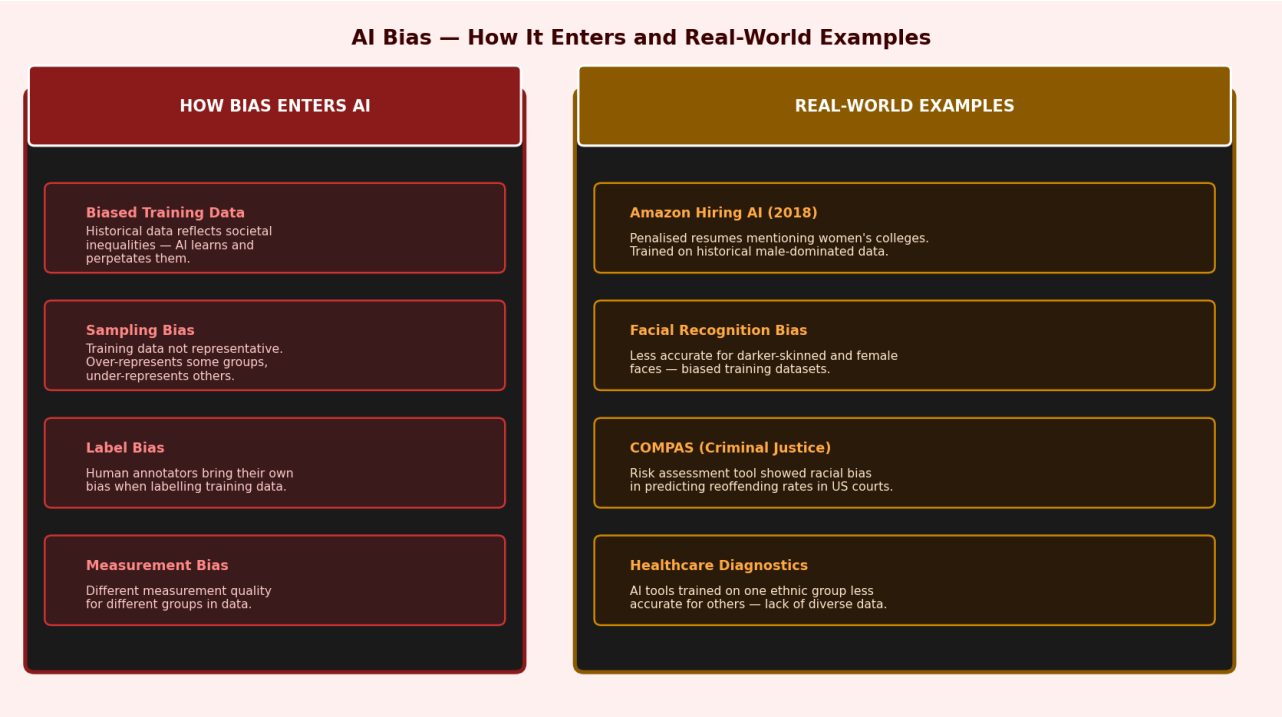


Figure 19.2 — AI Bias: How It Enters and Real-World Examples

How to Reduce AI Bias

- Use diverse, representative training data.
- Audit AI systems regularly for discrimination.
- Include diverse teams in AI development.
- Test AI on edge cases and minority groups.
- Make the decision-making process transparent and explainable.

19.4 Privacy and Surveillance

AI requires huge amounts of data — much of it personal. Ethical concerns around privacy include:

- Data Collection: Apps and devices collect location, messages, purchases, and browsing history — often without users fully understanding what is shared.
- Facial Recognition Surveillance: AI-powered CCTV cameras can identify individuals in public without their knowledge or consent — raising serious civil liberties concerns.
- Profiling: AI can build detailed profiles of individuals to predict or influence behaviour.
- Data Breaches: If personal data used to train AI is hacked, people's privacy is violated.

GDPR — General Data Protection Regulation (EU)

The EU's GDPR gives people rights over their personal data, including: the right to know what data is collected, the right to have it deleted, and the right not to be subject to fully automated decisions that significantly affect them.

19.5 Deepfakes and Misinformation

A deepfake is a highly realistic AI-generated image or video that makes it appear someone said or did something they did not. Deepfake technology uses deep learning (especially Generative Adversarial Networks — GANs) to produce convincing fake media.

Risks of Deepfakes

- Political misinformation: Fake videos of politicians saying things they never said.
- Non-consensual intimate images: A serious form of online abuse and harassment.
- Fraud: Fake audio of a CEO's voice used to trick companies into transferring money.
- Eroding trust: When nothing can be trusted as genuine, public discourse suffers.

Detecting Deepfakes

- AI detection tools are being developed to identify deepfakes.
- Look for: unnatural blinking, lighting inconsistencies, blurry edges around the face.
- Verify sources — only trust media from reliable, verified organisations.

19.6 AI and Employment

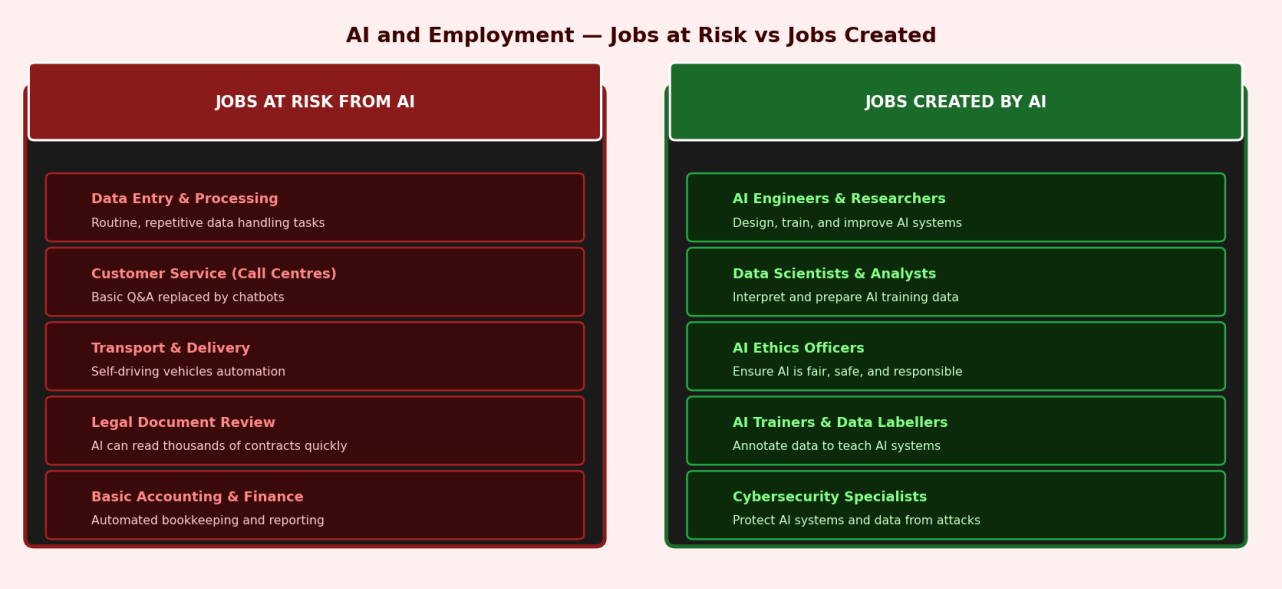


Figure 19.3 — AI and Employment: Jobs at Risk vs New Jobs Created

The Key Skill for the Future

Humans working alongside AI — using creativity, emotional intelligence, and critical thinking that machines cannot replicate.

19.7 Autonomous Weapons and AI in Warfare

Lethal Autonomous Weapons (LAWs) — sometimes called "killer robots" — are weapons that can identify and attack targets without human input. This raises profound ethical questions:

- Can a machine make a morally justified decision to take a human life?
- Who is responsible if an autonomous weapon kills a civilian by mistake?
- Should there be an international ban on fully autonomous lethal weapons?

Many AI researchers, ethicists, and human rights organisations argue that life-and-death decisions must always involve meaningful human control.

19.8 Responsible AI Development

Major organisations and governments are developing frameworks and principles for responsible AI. Key commitments include:

- Human-centred design: AI should serve human needs and respect human rights.
- Inclusive development: AI should not exclude or disadvantage marginalised groups.
- Environmental responsibility: Large AI models have significant carbon footprints — sustainability matters.
- International cooperation: AI governance cannot be left to one country or company.

Organisation	AI Ethics Initiative
European Union	AI Act (2024) — first comprehensive AI regulation law in the world.
UNESCO	Recommendation on the Ethics of AI — global

19.8a Global AI Governance (Beyond the EU)

While the EU AI Act is the first comprehensive legal framework, many other nations and organisations have developed significant AI governance initiatives. Understanding the global landscape is essential for anyone working with AI at an international level.

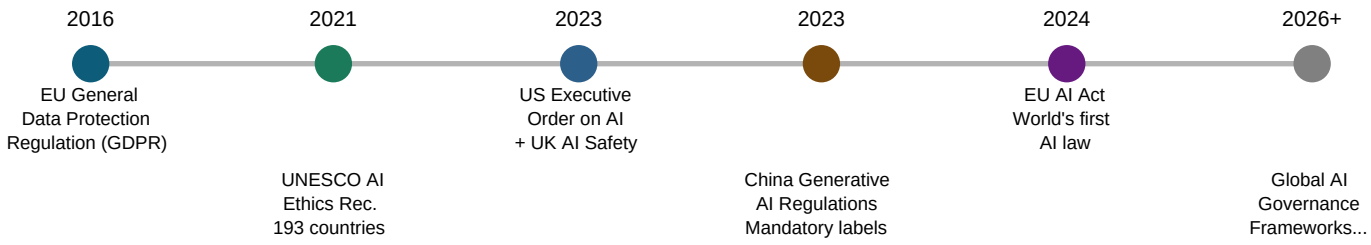
Organisation / Region	AI Governance Initiative
United Kingdom	AI Safety Institute (2023) · world's first government AI safety body. Focuses on frontier AI risks and coordinates international testing.
United States	Executive Order on AI (2023) · requires safety testing for powerful AI, protects privacy, and promotes responsible innovation. NIST AI Risk Management Framework provides voluntary guidance.
China	Generative AI Regulations (2023) · requires AI-generated content to be labelled, prevents AI from undermining state authority, and mandates data security reviews.
Canada	Voluntary Code of Conduct on Generative AI (2023) · industry commitments on safety, transparency, and human rights protections.
UNESCO	Recommendation on the Ethics of AI (2021) · adopted by all 193 UNESCO member states. Covers human rights, transparency, and environmental sustainability.
G7 / GPAI	Hiroshima AI Process (2023) and Global Partnership on AI · international coordination on responsible AI across leading democracies.

Key Insight · AI Governance is Global

No single country or region controls AI development. Effective governance requires international cooperation to set shared standards on safety, transparency, and human rights · just as aviation and nuclear safety require global frameworks.

As you study AI ethics, be aware that different countries may permit or restrict AI uses differently. A system legal in one country may be banned in another. This complexity makes understanding multiple governance frameworks increasingly important.

Global AI Governance · Key Milestones Timeline



Takeaway: AI governance is evolving rapidly. As of 2024, the EU AI Act is the most comprehensive law. Other regions are following with their own frameworks.

Organisation	AI Ethics Initiative
	framework adopted by 193 countries.
Google	AI Principles — guidelines for responsible AI development.
Microsoft	Responsible AI Standard — six principles: fairness, reliability, privacy, inclusiveness, transparency, accountability.
Anthropic (Claude's creator)	Constitutional AI — training AI to be helpful, harmless, and honest.

19.9 Should We Be Worried About AI?

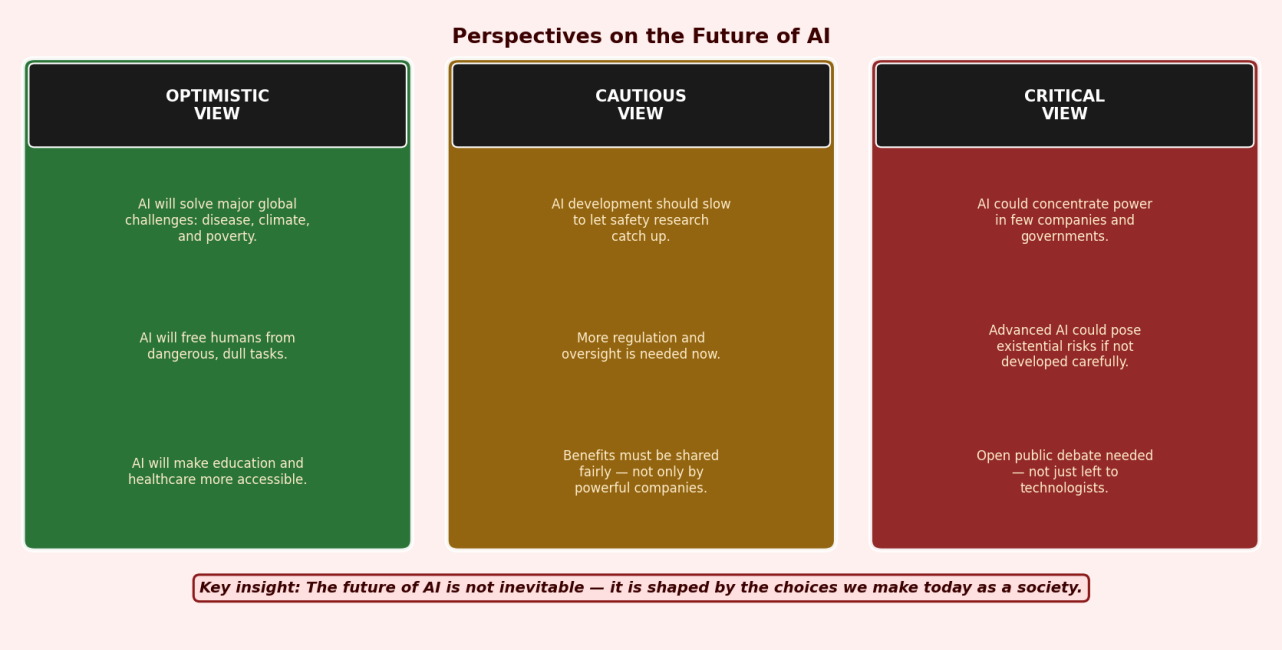


Figure 19.4 — Three Perspectives on the Future of AI

Chapter Review Questions

A. Short Answer

1. What is AI ethics? Why does it matter?
2. What is AI bias? Give two real-world examples and explain their harm.
3. What is a deepfake? What are three risks it poses to society?
4. How is AI changing employment? Name three jobs at risk and three new jobs being created.
5. What are the key principles of responsible AI? Name and explain four.

B. Discussion Questions

1. A hospital uses AI to decide which patients receive organ transplants. What ethical issues does this raise? Who should have the final decision?
2. Should facial recognition cameras be used in public spaces by governments? Argue both for and against.
3. Is it ethical for companies to use AI to screen job applicants? What safeguards should be required?

C. True or False

1. AI bias only occurs if the programmers are personally biased. (True / False)
2. Deepfakes are created using deep learning technology. (True / False)
3. The EU AI Act is the first comprehensive AI regulation law in the world. (True / False)
4. AI is expected to create no new jobs — only destroy existing ones. (True / False)
5. Human oversight of AI decisions is especially important in high-stakes areas. (True / False)

D. Key Terms Glossary

Term	Definition
AI Ethics	The study of moral principles guiding how AI is designed, used, and governed.
AI Bias	Unfair or discriminatory outputs from an AI system, usually caused by biased training data.
Deepfake	AI-generated realistic fake image or video making it appear someone did or said something false.
Transparency	The ability to explain and understand how an AI system reaches its decisions.
Accountability	Being responsible for the consequences of an AI system's actions or decisions.
GDPR	EU regulation giving individuals rights over their personal data.
Autonomous Weapons	Weapons that can identify and attack targets without human control.
Responsible AI	AI developed and deployed with fairness, safety, privacy, and human welfare in mind.